

Mathematics via Lattice Models

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Part I

Independent Spins / Ising Chain

1 Microstates, Hamiltonians, partition functions

Exercise 1.1 (Configuration spaces, Hamiltonians, Boltzmann weights, and the partition function). Let S be a finite set (the local state space) and Λ a finite set (the sites). The configuration space is $\Omega_\Lambda = S^\Lambda$, a microstate is a point $\sigma \in \Omega_\Lambda$, and $\sigma_i = \sigma(i)$ is its value at $i \in \Lambda$. A Hamiltonian is a function $H : \Omega_\Lambda \rightarrow \mathbb{R}$. For $\beta > 0$ the Boltzmann weight of σ is $e^{-\beta H(\sigma)}$ and the partition function is

$$Z_\Lambda(\beta) = \sum_{\sigma \in \Omega_\Lambda} e^{-\beta H(\sigma)}.$$

Physically $\beta = 1/(k_B T)$ with T the absolute temperature and k_B Boltzmann's constant; β is called the inverse temperature.

- (a) Suppose $H(\sigma) = \sum_{i \in \Lambda} \phi_i(\sigma_i)$ for functions $\phi_i : S \rightarrow \mathbb{R}$. Prove that $Z_\Lambda = \prod_{i \in \Lambda} z_i$ with $z_i = \sum_{s \in S} e^{-\beta \phi_i(s)}$.
- (b) (Independent spins in a field.) Take $S = \{\pm 1\}$, a field $h \in \mathbb{R}$, and $H(\sigma) = -h \sum_{i \in \Lambda} \sigma_i$. Prove $Z_\Lambda = (2 \cosh \beta h)^{|\Lambda|}$.
- (c) (Zeeman Hamiltonian for spin s .) Let $s \in \frac{1}{2}\mathbb{N}$, $S = \{-s, -s+1, \dots, s\}$ (the $2s+1$ levels), and for a magnetic induction $B \in \mathbb{R}$, a Landé factor $g > 0$, and the Bohr magneton $\mu_B > 0$ put $H(\sigma) = -g\mu_B B \sum_{i \in \Lambda} \sigma_i$. With $x = \beta g\mu_B B$ prove

$$Z_\Lambda = \left(\frac{\sinh((2s+1)x/2)}{\sinh(x/2)} \right)^{|\Lambda|},$$

the right side being interpreted as $(2s+1)^{|\Lambda|}$ at $x = 0$. Prove that Z_Λ extends to an entire function of x and that $\log Z_\Lambda$ is real analytic in $x \in \mathbb{R}$.

Exercise 1.2 (The Ising Hamiltonian on a finite graph and its $\mathbb{Z}_2$ symmetry). Let $\Gamma = (V, E)$ be a finite graph with unordered edges $ij \in E$, and let $\Omega_V = \{\pm 1\}^V$. For a coupling $J > 0$ and a field $h \in \mathbb{R}$ the Ising Hamiltonian is

$$H(\sigma) = -J \sum_{ij \in E} \sigma_i \sigma_j - h \sum_{i \in V} \sigma_i.$$

Put $K = \beta J$ (the reduced coupling) and write

$$Z_\Gamma(K, \beta h) = \sum_{\sigma \in \Omega_V} \exp\left(K \sum_{ij \in E} \sigma_i \sigma_j + \beta h \sum_{i \in V} \sigma_i\right).$$

The group $\mathbb{Z}_2 = \{\pm 1\}$ acts on Ω_V by $\sigma \mapsto -\sigma$.

- (a) Prove that H is $\mathbb{Z}_2$ -invariant if and only if $h = 0$ or $V = \emptyset$, and that $Z_\Gamma(K, \beta h) = Z_\Gamma(K, -\beta h)$ for all h .
- (b) For $\sigma \in \Omega_V$ let $u(\sigma)$ be the number of edges ij with $\sigma_i \neq \sigma_j$ and $n_-(\sigma)$ the number of sites with $\sigma_i = -1$. Prove

$$Z_\Gamma(K, \beta h) = e^{K|E| + \beta h|V|} \sum_{\sigma \in \Omega_V} x^{u(\sigma)} z^{n_-(\sigma)}, \quad x = e^{-2K}, \quad z = e^{-2\beta h},$$

- so that $e^{-K|E|-\beta h|V|}Z_\Gamma$ is a polynomial in (x, z) with nonnegative integer coefficients, of degree at most $|E|$ in x and exactly $|V|$ in z . Determine its constant term and its coefficient of $z^{|V|}$.
- (c) Prove that $Z_\Gamma(K, \beta h) > 0$ for all real $(K, \beta h)$ and that Z_Γ extends to an entire function of $(K, \beta h) \in \mathbb{C}^2$.

Exercise 1.3 (The Curie–Weiss partition function as a binomial sum). Let Γ be the complete graph on n sites and take the mean-field Hamiltonian

$$H_n(\sigma) = -\frac{J}{n} \sum_{1 \leq i < j \leq n} \sigma_i \sigma_j - h \sum_{i=1}^n \sigma_i.$$

Put $M(\sigma) = \sum_i \sigma_i$.

- (a) Prove $\sum_{i < j} \sigma_i \sigma_j = \frac{1}{2}(M(\sigma)^2 - n)$ and deduce

$$Z_n = \sum_{k=0}^n \binom{n}{k} \exp\left(\frac{K}{2n}(n-2k)^2 - \frac{K}{2} + \beta h(n-2k)\right), \quad K = \beta J.$$

- (b) Let $s(m) = -\frac{1+m}{2} \log \frac{1+m}{2} - \frac{1-m}{2} \log \frac{1-m}{2}$ for $m \in [-1, 1]$. Prove the bounds $\frac{1}{n+1} e^{ns(1-2k/n)} \leq \binom{n}{k} \leq e^{ns(1-2k/n)}$ for $0 \leq k \leq n$, and deduce

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log Z_n = \sup_{m \in [-1, 1]} \left(\frac{K}{2} m^2 + \beta h m + s(m) \right).$$

- (c) Prove that for $K \leq 1$ and $h = 0$ the supremum is attained only at $m = 0$, and that for $K > 1$ and $h = 0$ it is attained exactly at $\pm m_K$ with $m_K > 0$ the positive solution of $m = \tanh(Km)$.

2 Gibbs measures

Exercise 2.1 (Gibbs measures, expectations, and the Markov property of the Ising model). Let Ω be a finite set, $H : \Omega \rightarrow \mathbb{R}$, and $\beta > 0$. The Gibbs measure is the probability measure on Ω with

$$\mu(\sigma) = \frac{e^{-\beta H(\sigma)}}{Z}, \quad Z = \sum_{\sigma \in \Omega} e^{-\beta H(\sigma)}.$$

For $A : \Omega \rightarrow \mathbb{C}$ the expectation is $\langle A \rangle = \mathbb{E}_\mu A = \sum_\sigma A(\sigma) \mu(\sigma)$, and for $A, B : \Omega \rightarrow \mathbb{C}$ the truncated correlation is $\langle A; B \rangle = \langle AB \rangle - \langle A \rangle \langle B \rangle$. For a subset $\Delta \subset V$ of the vertex set of a finite graph $\Gamma = (V, E)$ and $\eta \in \{\pm 1\}^{V \setminus \Delta}$, define the Ising Hamiltonian on Δ with boundary condition η ,

$$H_\Delta^\eta(\sigma) = -J \sum_{\substack{ij \in E \\ i, j \in \Delta}} \sigma_i \sigma_j - h \sum_{i \in \Delta} \sigma_i - J \sum_{\substack{ij \in E \\ i \in \Delta, j \notin \Delta}} \sigma_i \eta_j, \quad \sigma \in \{\pm 1\}^\Delta,$$

with Gibbs measure μ_Δ^η and partition function Z_Δ^η . Let μ be the Gibbs measure of the Ising Hamiltonian H of the section Microstates, Hamiltonians, partition functions on all of Γ .

- (a) Prove that under μ the conditional law of σ_i given $(\sigma_j)_{j \neq i}$ is

$$\mu(\sigma_i = \pm 1 \mid (\sigma_j)_{j \neq i}) = \frac{e^{\pm \beta(J \sum_{j \sim i} \sigma_j + h)}}{2 \cosh(\beta(J \sum_{j \sim i} \sigma_j + h))},$$

where $j \sim i$ means $ij \in E$; in particular it depends on $(\sigma_j)_{j \neq i}$ only through the neighbours of i .

- (b) (Finite-volume consistency.) Prove that for every $\Delta \subset V$ and every $\eta \in \{\pm 1\}^{V \setminus \Delta}$ with $\mu(\sigma|_{V \setminus \Delta} = \eta) > 0$,

$$\mu(\sigma|_{\Delta} = \tau \mid \sigma|_{V \setminus \Delta} = \eta) = \mu_{\Delta}^{\eta}(\tau), \quad \tau \in \{\pm 1\}^{\Delta},$$

and that μ_{Δ}^{η} depends on η only through its restriction to the set of sites of $V \setminus \Delta$ adjacent to Δ .

- (c) Prove that for $\Delta' \subset \Delta \subset V$ and $\eta \in \{\pm 1\}^{V \setminus \Delta}$, $\mu_{\Delta'}^{\eta}(\sigma|_{\Delta'} = \tau \mid \sigma|_{\Delta \setminus \Delta'} = \zeta) = \mu_{\Delta'}^{\zeta \eta}(\tau)$, where $\zeta \eta$ denotes the configuration on $V \setminus \Delta'$ equal to ζ on $\Delta \setminus \Delta'$ and to η on $V \setminus \Delta$.

Exercise 2.2 (Gibbs entropy, relative entropy, and the variational principle). For a probability measure ν on a finite set Ω the Shannon–Gibbs entropy is

$$S(\nu) = - \sum_{\sigma \in \Omega} \nu(\sigma) \log \nu(\sigma),$$

with $0 \log 0 = 0$, and for probability measures ν, μ with $\mu(\sigma) > 0$ for all σ the relative entropy is $D(\nu \parallel \mu) = \sum_{\sigma} \nu(\sigma) \log(\nu(\sigma)/\mu(\sigma))$.

- (a) Prove $0 \leq S(\nu) \leq \log |\Omega|$, with $S(\nu) = 0$ if and only if ν is a point mass and $S(\nu) = \log |\Omega|$ if and only if ν is uniform.
- (b) (Gibbs inequality.) Using $\log t \leq t - 1$ for $t > 0$ with equality only at $t = 1$, prove $D(\nu \parallel \mu) \geq 0$ with equality if and only if $\nu = \mu$.
- (c) (Variational principle.) Let $H : \Omega \rightarrow \mathbb{R}$, $\beta > 0$, and let μ_{β} be the Gibbs measure. Prove that for every probability measure ν on Ω ,

$$S(\nu) - \beta \mathbb{E}_{\nu} H = \log Z - D(\nu \parallel \mu_{\beta}) \leq \log Z,$$

so that μ_{β} is the unique maximizer of $\nu \mapsto S(\nu) - \beta \mathbb{E}_{\nu} H$ over probability measures on Ω , and $S(\mu_{\beta}) = \log Z + \beta \langle H \rangle$.

- (d) Prove that μ_{β} is the unique maximizer of $S(\nu)$ over probability measures ν with $\mathbb{E}_{\nu} H = \langle H \rangle_{\mu_{\beta}}$.

Exercise 2.3 (Symmetries of Gibbs measures and vanishing of odd correlations). Let μ be the Gibbs measure of a Hamiltonian H on a finite set Ω and let $g : \Omega \rightarrow \Omega$ be a bijection with $H \circ g = H$.

- (a) Prove that μ is g -invariant and that $\langle A \circ g \rangle = \langle A \rangle$ for every $A : \Omega \rightarrow \mathbb{C}$.
- (b) For the Ising Hamiltonian on a finite graph $\Gamma = (V, E)$ at $h = 0$ and $A \subset V$ put $\sigma_A = \prod_{i \in A} \sigma_i$. Prove $\langle \sigma_A \rangle = 0$ whenever $|A|$ is odd; in particular $\langle \sigma_i \rangle = 0$ for every $i \in V$ and every $\beta > 0$.
- (c) Prove that μ is invariant under every automorphism of Γ (for all h), and that for a vertex-transitive graph $\langle \sigma_i \rangle$ is independent of i .

3 $\log Z$, cumulants, entropy, free energy

Exercise 3.1 (The logarithm of the partition function as a cumulant generating function). Let X be a real random variable on a finite probability space (Ω, ν) . Its cumulant generating function is $\kappa_X(t) = \log \mathbb{E}_{\nu} e^{tX}$, $t \in \mathbb{R}$, and its cumulants are the coefficients $\kappa_n(X)$ in the expansion $\kappa_X(t) = \sum_{n \geq 1} \kappa_n(X) t^n / n!$ near $t = 0$. Let $H : \Omega \rightarrow \mathbb{R}$ be a Hamiltonian on a finite set, μ_{β} its Gibbs measure, and $Z(\beta)$ its partition function. Put $F(\beta) = -\beta^{-1} \log Z(\beta)$, the free energy.

- (a) Prove that κ_X is real analytic and convex, that $\kappa_1(X) = \mathbb{E}_{\nu} X$, $\kappa_2(X) = \text{Var}_{\nu} X$, and $\kappa_3(X) = \mathbb{E}_{\nu} (X - \mathbb{E}_{\nu} X)^3$, and that κ_X is strictly convex unless X is ν -almost surely constant.

- (b) Prove that $\log Z(\beta + t) - \log Z(\beta) = \kappa_{-H}(t)$ computed in μ_β . Deduce

$$\frac{\partial}{\partial \beta} \log Z = -\langle H \rangle, \quad \frac{\partial^2}{\partial \beta^2} \log Z = \text{Var}(H) = \langle H; H \rangle \geq 0,$$

so that the internal energy $U(\beta) = \langle H \rangle_{\mu_\beta}$ is nonincreasing in β , and strictly decreasing unless H is constant.

- (c) Let $\Omega = \{\pm 1\}^\Lambda$, $H = H_0 - hM$ with $M(\sigma) = \sum_{i \in \Lambda} \sigma_i$ and H_0 independent of h , and regard $\log Z$ as a function of the variables $(\beta, \beta h)$. Prove

$$\frac{\partial \log Z}{\partial(\beta h)} = \langle M \rangle, \quad \frac{\partial^2 \log Z}{\partial(\beta h)^2} = \text{Var}(M), \quad \frac{\partial^2 \log Z}{\partial \beta \partial(\beta h)} = -\langle H_0; M \rangle,$$

and that the Hessian of $\log Z$ with respect to $(\beta, \beta h)$ is the covariance matrix of $(-H_0, M)$ under μ_β . Deduce that $\log Z$ is a convex function of $(\beta, \beta h) \in \mathbb{R}^2$ and that $\beta F = -\log Z$ is concave.

- (d) (Fluctuation–response.) Define the magnetization per site $m = |\Lambda|^{-1} \langle M \rangle$, the susceptibility $\chi = \partial m / \partial h$ at fixed β , and the specific heat per site $c = |\Lambda|^{-1} \partial U / \partial T$ at fixed h , where $T = 1/(k_B \beta)$. Prove

$$\chi = \frac{\beta \text{Var}(M)}{|\Lambda|} \geq 0, \quad c = \frac{k_B \beta^2 \text{Var}(H)}{|\Lambda|} \geq 0.$$

Exercise 3.2 (Free energy, thermodynamic entropy, and the Legendre transform). Let $H : \Omega \rightarrow \mathbb{R}$ on a finite set Ω , with Gibbs measures μ_β and partition function $Z(\beta)$, $\beta > 0$. Use the free energy F of the preceding exercise; the internal energy is $U(\beta) = \langle H \rangle_{\mu_\beta}$, and the thermodynamic entropy is $S_{\text{th}}(\beta) = k_B S(\mu_\beta)$ with S the Shannon–Gibbs entropy of the section Gibbs measures. Put $u_- = \min H$, $u_+ = \max H$, and for $u \in [u_-, u_+]$ let $s(u) = \sup\{S(\nu) : \nu \text{ a probability measure on } \Omega, \mathbb{E}_\nu H = u\}$.

- (a) Prove $S_{\text{th}} = k_B(\log Z + \beta U)$ and $F = U - TS_{\text{th}}$ with $T = 1/(k_B \beta)$; prove $\partial(\beta F) / \partial \beta = U$ and $\partial F / \partial T = -S_{\text{th}}$.
(b) Prove that s is concave on $[u_-, u_+]$, that $s(u_\pm) = \log |H^{-1}(u_\pm)|$, and that $\beta \mapsto U(\beta)$ is a real-analytic bijection of $\mathbb{R}$ onto (u_-, u_+) when H is nonconstant (extend μ_β to all $\beta \in \mathbb{R}$).
(c) (Legendre transform.) Prove, for all $\beta \in \mathbb{R}$,

$$\log Z(\beta) = \sup_{u \in [u_-, u_+]} (s(u) - \beta u),$$

and that for $u \in (u_-, u_+)$, with $\beta(u)$ the unique solution of $U(\beta) = u$,

$$s(u) = S(\mu_{\beta(u)}) = \inf_{\beta \in \mathbb{R}} (\log Z(\beta) + \beta u), \quad s'(u) = \beta(u).$$

- (d) Prove that $\beta \mapsto S_{\text{th}}(\beta)$ is nonincreasing and that $\lim_{\beta \rightarrow \infty} S_{\text{th}}(\beta) = k_B \log |H^{-1}(u_-)|$ (the residual entropy of the ground states).

Exercise 3.3 (Independent spins: Curie’s law, the Brillouin function, and the Schottky anomaly). Let $s \in \frac{1}{2}\mathbb{N}$ and consider the Zeeman Hamiltonian $H = -g\mu_B B \sum_{i \in \Lambda} \sigma_i$ on $\{-s, \dots, s\}^\Lambda$ of the section Microstates, Hamiltonians, partition functions, with $x = \beta g\mu_B B$. The Brillouin function of spin s is

$$B_s(y) = \frac{2s+1}{2s} \coth\left(\frac{2s+1}{2s} y\right) - \frac{1}{2s} \coth\left(\frac{y}{2s}\right), \quad y \in \mathbb{R} \setminus \{0\}, \quad B_s(0) = 0.$$

- (a) Prove that B_s is real analytic and odd, strictly increasing, with $B_s(y) \rightarrow 1$ as $y \rightarrow \infty$ and $B_s(y) = \frac{s+1}{3s} y - \frac{(s+1)(2s^2+2s+1)}{90s^3} y^3 + O(y^5)$ as $y \rightarrow 0$; and that $B_{1/2}(y) = \tanh y$.
- (b) Prove that the magnetic moment per site is

$$\frac{\langle g\mu_B \sum_i \sigma_i \rangle}{|\Lambda|} = g\mu_B s B_s(sx),$$

so that the ratio of the moment to its saturation value $g\mu_B s$ is $B_s(g\mu_B s B/(k_B T))$.

- (c) (Curie's law.) Prove that the zero-field susceptibility per site is

$$\chi = \frac{(g\mu_B)^2 s(s+1)}{3k_B T},$$

both from (b) and from the fluctuation–response identity of the previous exercise using $\text{Var}(\sigma_i) = s(s+1)/3$ under the uniform measure on $\{-s, \dots, s\}$.

- (d) (Schottky anomaly.) For $s = \frac{1}{2}$ let $\Delta = g\mu_B |B| > 0$ be the level splitting. Prove that the specific heat per site is

$$c = k_B \left(\frac{\beta \Delta}{2} \right)^2 \text{sech}^2 \left(\frac{\beta \Delta}{2} \right),$$

that $c \rightarrow 0$ both as $T \rightarrow 0$ and as $T \rightarrow \infty$, and that c attains its maximum at the unique solution of $\beta \Delta \tanh(\beta \Delta/2) = 2$, namely $k_B T/\Delta = 0.4168\dots$, with maximal value $c_{\max} = 0.4392\dots k_B$.

4 Transfer matrices

Exercise 4.1 (Tensor products, finite-dimensional spaces, and intrinsic traces). A tensor product $V \otimes W$ of complex vector spaces represents bilinear maps from $V \times W$: composition with its bilinear map induces $\text{Hom}(V \otimes W, U) \simeq \text{Bil}(V, W; U)$. Write $V^* = \text{Hom}(V, \mathbb{C})$ and $\text{ev}_V : V^* \otimes V \rightarrow \mathbb{C}$ for evaluation. A vector space V is finite-dimensional if there is a map $\text{coev}_V : \mathbb{C} \rightarrow V \otimes V^*$ satisfying

$$(\text{id}_V \otimes \text{ev}_V)(\text{coev}_V \otimes \text{id}_V) = \text{id}_V, \quad (\text{ev}_V \otimes \text{id}_{V^*})(\text{id}_{V^*} \otimes \text{coev}_V) = \text{id}_{V^*}.$$

For such a space and $A \in \text{End}(V)$ define $\text{Tr } A = \text{ev}_V \tau(A \otimes \text{id}) \text{coev}_V$, where τ interchanges the factors, and $\dim V = \text{Tr } \text{id}_V$. The exterior power $\bigwedge^n V$ represents alternating n -linear maps. If $\dim V = n$, the scalar by which $\bigwedge^n A$ acts on the line $\bigwedge^n V$ is $\det A$.

- (a) Construct the tensor product as the quotient of the free vector space on $V \times W$ by the bilinearity relations. Prove uniqueness of the representing object and of coevaluation when it exists. Show that $W \otimes V^* \rightarrow \text{Hom}(V, W)$ is an isomorphism for finite-dimensional V , using the snake identities.
- (b) For a finite set S use the evaluation functions and their evaluation functionals to construct coevaluation on $\mathbb{C}^S$. Prove that an operator with kernel $A(s, t)$ has $\text{Tr } A = \sum_s A(s, s)$. Prove $\text{Tr}(AB) = \text{Tr}(BA)$ for maps $A : V \rightarrow W$, $B : W \rightarrow V$, and prove multiplicativity under tensor products and additivity under direct sums.
- (c) Split off a nonzero line using a linear functional and use induction to show that $\dim V$ is a nonnegative integer, $\bigwedge^{\dim V} V$ is a line, and the next exterior power vanishes. Prove $\det(AB) = \det A \det B$.
- (d) Define the partial trace $\text{Tr}_V : \text{End}(V \otimes W) \rightarrow \text{End}(W)$ by inserting coev_V , applying the operator to $V \otimes W$, and contracting V with V^* by evaluation and the symmetry. Prove $\text{Tr}_V(A \otimes B) = (\text{Tr } A)B$ and its cyclicity with respect to operators acting only on V .

Exercise 4.2 (Inner products and spectral projections). A Hermitian inner product is a positive definite sesquilinear pairing, conjugate-linear in its first argument. The adjoint of A is the operator A^* satisfying $\langle v, Aw \rangle = \langle A^*v, w \rangle$. An operator is self-adjoint if $A = A^*$, unitary if $A^*A = AA^* = \text{id}$, and an orthogonal projection if $P = P^* = P^2$.

- (a) On a finite-dimensional inner product space prove existence and uniqueness of adjoints and the orthogonal decomposition $V = W \oplus W^\perp$ for every subspace W .
- (b) For self-adjoint A maximize $\langle v, Av \rangle$ on the unit sphere, prove that the maximizing line is invariant, and induct on its orthogonal complement. Obtain the intrinsic decomposition $V = \bigoplus_{\lambda \in \text{Spec } A} \ker(A - \lambda)$ and $A = \sum_\lambda \lambda P_\lambda$ with mutually orthogonal spectral projections. Prove $\text{Tr } A = \sum_\lambda \lambda \dim \text{im } P_\lambda$ and $\det A = \prod_\lambda \lambda^{\dim \text{im } P_\lambda}$.
- (c) Prove that commuting self-adjoint operators preserve one another's spectral subspaces, and hence admit joint spectral projections. Define $f(A) = \sum_\lambda f(\lambda) P_\lambda$ and prove the product rule for this functional calculus.

Exercise 4.3 (The transfer matrix of the periodic Ising chain and the exact free energy). Let $\Lambda = \mathbb{Z}/N\mathbb{Z}$ with edges $\{n, n+1\}$, $N \geq 3$ (the periodic Ising chain), so that $H(\sigma) = -J \sum_{n \in \mathbb{Z}/N\mathbb{Z}} \sigma_n \sigma_{n+1} - h \sum_n \sigma_n$. Let $\mathbb{C}^{\{\pm 1\}}$ be the space of functions on $\{\pm 1\}$ with its standard inner product, and let T be the operator on it with matrix entries in the configuration basis

$$\langle \sigma | T | \sigma' \rangle = \exp \left(K \sigma \sigma' + \frac{\beta h}{2} (\sigma + \sigma') \right), \quad \sigma, \sigma' \in \{\pm 1\}.$$

- (a) Prove $Z_N = \text{Tr}(T^N)$.
- (b) Prove that T is symmetric with positive entries and $\det T = 2 \sinh 2K > 0$, hence positive definite, with eigenvalues

$$\lambda_\pm = e^K \cosh \beta h \pm \sqrt{e^{2K} \sinh^2 \beta h + e^{-2K}}, \quad \lambda_+ > \lambda_- > 0.$$

- (c) Prove $Z_N = \lambda_+^N + \lambda_-^N$ and $\lim_{N \rightarrow \infty} N^{-1} \log Z_N = \log \lambda_+$, so that the free energy per site converges to

$$f(\beta, h) = -\beta^{-1} \log \left(e^K \cosh \beta h + \sqrt{e^{2K} \sinh^2 \beta h + e^{-2K}} \right),$$

with $f(\beta, 0) = -\beta^{-1} \log(2 \cosh K)$.

- (d) Prove that at $h = 0$ the internal energy per site and the specific heat per site converge to $-J \tanh K$ and $k_B K^2 \text{sech}^2 K$, and that the limiting magnetization per site is

$$m(\beta, h) = \frac{\partial \log \lambda_+}{\partial (\beta h)} = \frac{\sinh \beta h}{\sqrt{\sinh^2 \beta h + e^{-4K}}}.$$

Exercise 4.4 (Analyticity of the chain free energy and the absence of a phase transition in one dimension). Keep the notation of the previous exercise and put $D = \{(\beta, h) : \beta > 0, h \in \mathbb{R}\}$.

- (a) Prove that $(K, \beta h) \mapsto \lambda_+$ extends to a holomorphic function without zeros on a connected open set $\mathcal{U} \subset \mathbb{C}^2$ containing $\mathbb{R}^2$, and that $|\lambda_-/\lambda_+| < 1$ on $\mathcal{U}$ after shrinking $\mathcal{U}$ to a neighbourhood of any compact subset of $\mathbb{R}^2$.
- (b) Prove that on such a neighbourhood $N^{-1} \log Z_N = \log \lambda_+ + N^{-1} \log(1 + (\lambda_-/\lambda_+)^N)$ converges uniformly to $\log \lambda_+$, and deduce that every partial derivative of $N^{-1} \log Z_N$ in $(\beta, \beta h)$ converges uniformly on compact subsets of D to the corresponding derivative of $\log \lambda_+$. Conclude that the limiting magnetization, susceptibility, internal energy, and specific heat are the derivatives of f and are real analytic on D .

- (c) Prove that $m(\beta, h) \rightarrow 0$ as $h \rightarrow 0$ for every $\beta \in (0, \infty)$, that $h \mapsto m(\beta, h)$ is real analytic and strictly increasing, and that the zero-field susceptibility is $\chi(\beta, 0) = \beta e^{2K}$, finite for every $\beta < \infty$.
- (d) Prove that for $N \geq 3$ the finite- N partition function $Z_N(K, \beta h)$ has no zero with $K \in \mathbb{R}$, $\beta h \in \mathbb{R}$, and that a zero with $K \in \mathbb{R}$ and $\beta h \in \mathbb{C}$ requires $|\lambda_+| = |\lambda_-|$, where $\lambda_{\pm}$ are the two roots of $\lambda^2 - 2e^K \cosh(\beta h)\lambda + 2\sinh 2K$ (use $\lambda_+^N = -\lambda_-^N$). Prove that $|\lambda_+| = |\lambda_-|$ forces $\operatorname{Re}(\bar{a}\sqrt{b}) = 0$ with $a = e^K \cosh \beta h$ and $b = e^{2K} \sinh^2 \beta h + e^{-2K}$.

Exercise 4.5 (Spectral decomposition of the transfer matrix, two-point functions, and the correlation length). Keep the notation of the periodic chain and let $P_{\pm}$ be the spectral projections of T onto its eigenlines, $T = \lambda_+ P_+ + \lambda_- P_-$. Let Σ be the operator on $\mathbb{C}^{\{\pm 1\}}$ of multiplication by the function $\sigma \mapsto \sigma$.

- (a) Prove, for $0 \leq r \leq N$,

$$\langle \sigma_0 \sigma_r \rangle_N = \frac{\operatorname{Tr}(\Sigma T^r \Sigma T^{N-r})}{\operatorname{Tr} T^N} = \frac{\sum_{a,b \in \{\pm 1\}} \lambda_a^r \lambda_b^{N-r} \operatorname{Tr}(P_a \Sigma P_b \Sigma)}{\lambda_+^N + \lambda_-^N}.$$

- (b) Prove that $\operatorname{Tr}(P_+ \Sigma P_+ \Sigma) = \operatorname{Tr}(P_+ \Sigma)^2$, $\operatorname{Tr}(P_+ \Sigma P_- \Sigma) = \operatorname{Tr}(P_- \Sigma P_+ \Sigma) \geq 0$, and, with r fixed and $N \rightarrow \infty$,

$$\langle \sigma_0 \sigma_r \rangle \rightarrow \operatorname{Tr}(P_+ \Sigma)^2 + \operatorname{Tr}(P_+ \Sigma P_- \Sigma) (\lambda_- / \lambda_+)^r,$$

so that $\lim_N \langle \sigma_0; \sigma_r \rangle_N = \operatorname{Tr}(P_+ \Sigma P_- \Sigma) e^{-r/\xi}$ with the correlation length

$$\xi = \frac{1}{\log(\lambda_+ / \lambda_-)}.$$

- (c) At $h = 0$ prove $\lambda_{\pm} = 2 \cosh K$, $2 \sinh K$, $\operatorname{Tr}(P_+ \Sigma) = 0$, $\operatorname{Tr}(P_+ \Sigma P_- \Sigma) = 1$, and

$$\langle \sigma_0 \sigma_r \rangle_N = \frac{t^r + t^{N-r}}{1 + t^N}, \quad t = \tanh K, \quad \xi = -\frac{1}{\log \tanh K},$$

and that $\xi \sim \frac{1}{2} e^{2K}$ as $K \rightarrow \infty$ while $\xi \rightarrow 0$ as $K \rightarrow 0$.

- (d) Prove that at $h = 0$ the susceptibility of the section $\log Z$, cumulants, entropy, free energy satisfies $\chi = \beta \sum_{r \in \mathbb{Z}/N\mathbb{Z}} \langle \sigma_0 \sigma_r \rangle_N$ and converges as $N \rightarrow \infty$ to $\beta(1+t)/(1-t) = \beta e^{2K}$.

Exercise 4.6 (Transfer matrices on a strip: positivity and the simple leading eigenvalue). Let $L, N \geq 3$ and let $\mathbb{T}_{N,L} = (\mathbb{Z}/N\mathbb{Z}) \times (\mathbb{Z}/L\mathbb{Z})$ be the $N \times L$ torus with nearest-neighbour edges. Write a configuration as a sequence of rows $\sigma^{(1)}, \dots, \sigma^{(N)} \in \{\pm 1\}^{\mathbb{Z}/L\mathbb{Z}}$ and let $\mathcal{H}_L = \mathbb{C}^{\{\pm 1\}^{\mathbb{Z}/L\mathbb{Z}}} \cong (\mathbb{C}^2)^{\otimes L}$ be the row space, the tensor factor at $j \in \mathbb{Z}/L\mathbb{Z}$ being $\mathbb{C}^{\{\pm 1\}}$. Define $U(\tau) = -J \sum_j \tau_j \tau_{j+1} - h \sum_j \tau_j$ (the row energy) and $W(\tau, \tau') = -J \sum_j \tau_j \tau'_j$ (the interaction of consecutive rows), and the transfer matrix T_L on $\mathcal{H}_L$ by

$$\langle \tau | T_L | \tau' \rangle = \exp\left(-\frac{\beta}{2} U(\tau) - \beta W(\tau, \tau') - \frac{\beta}{2} U(\tau')\right).$$

- (a) Prove that the Ising Hamiltonian on $\mathbb{T}_{N,L}$ equals $\sum_n (U(\sigma^{(n)}) + W(\sigma^{(n)}, \sigma^{(n+1)}))$ and that $Z_{\mathbb{T}_{N,L}} = \operatorname{Tr}(T_L^N)$.
- (b) Prove $T_L = V^{1/2} (T_{h=0})^{\otimes L} V^{1/2}$, where V is the positive diagonal operator of multiplication by $e^{-\beta U}$ and $T_{h=0}$ is the chain transfer matrix at $h = 0$ of the first exercise of this section. Deduce that T_L is symmetric with strictly positive entries and positive definite.

- (c) Prove that T_L commutes with the translation operator $(R\psi)(\tau) = \psi(\tau \circ (j \mapsto j+1))$ and, when $h = 0$, with the spin reversal $(F\psi)(\tau) = \psi(-\tau)$.
- (d) Let A be a symmetric kernel on $\mathbb{R}^S$ for a finite set S , with $A(s, t) > 0$ for all s, t , and let $\lambda_{\max} = \max \text{spec } A$. Prove: if v maximizes the Rayleigh quotient $\langle v, Av \rangle / \langle v, v \rangle$, then so does $|v|$ (entrywise absolute value), hence $A|v| = \lambda_{\max}|v|$ and $|v|$ has strictly positive entries; deduce that $\lambda_{\max}$ is a simple eigenvalue with an entrywise positive eigenvector, and that $\lambda_{\max} > |\lambda|$ for every other eigenvalue λ .
- (e) Conclude that the largest eigenvalue $\lambda_{\max}(L)$ of T_L is simple and $\lim_{N \rightarrow \infty} N^{-1} \log Z_{\mathbb{T}_{N,L}} = \log \lambda_{\max}(L)$.

Exercise 4.7 (Prediction: magnetization and calorimetry of independent spins and the Ising chain). Consider independent ions of spin s , with Zeeman energy $-g\mu_B B\sigma$, and the zero-field Ising chain $H = -J \sum_i \sigma_i \sigma_{i+1}$, $J > 0$. Use $\beta = 1/(k_B T)$ and $K = \beta J$.

- (a) Derive the moment per ion $M = g\mu_B s B_s(g\mu_B s B/(k_B T))$ and the zero-field susceptibility $(g\mu_B)^2 s(s+1)/(3k_B T)$. For $g = 2$ compute the saturation moments for $s = 3/2, 5/2, 7/2$ and the slopes of the normalized magnetization curves against B/T .
- (b) Derive the chain's specific heat and its peak,

$$\frac{c}{k_B} = K^2 \text{sech}^2 K, \quad k_B T_{\max} = 0.8336 \dots J, \quad c_{\max} = 0.4392 \dots k_B,$$

by proving uniqueness of the solution of $K \tanh K = 1$. Obtain the high- and low-temperature asymptotics.

- (c) From the chain entropy per spin $k_B [\log(2 \cosh K) - K \tanh K]$ prove $\int_0^\infty c(T) dT/T = k_B \log 2$. Calculate the predicted peak temperature for $J/k_B = 9 \text{ K}$, regarding this as a specified model parameter.
- (d) Identify the measurements of magnetization, saturation, and heat capacity that test these formulas. For the independent-ion calculation state the assumptions of negligible exchange and crystal-field splitting relative to $k_B T$; for the chain state the absence of interchain exchange and the need to subtract lattice heat capacity. Use analyticity of the chain free energy to show that a finite-temperature ordering anomaly would require physics beyond this Hamiltonian.

Part II

2D Ising Model

5 Thermodynamic limit

Exercise 5.1 (Boundary terms, subadditivity, and the existence of the pressure). Let $d \geq 1$. A box is a finite set $\Lambda \subset \mathbb{Z}^d$; $\Lambda_L = \{1, \dots, L\}^d$; For state limits put $\Lambda_L^c = \{-L, \dots, L\}^d$. $E(\Lambda)$ is the set of nearest-neighbour edges with both ends in Λ ; $\partial\Lambda$ is the set of sites of Λ with a nearest neighbour outside Λ ; $|\Lambda|$ is the number of sites. Let $\Omega = \{\pm 1\}^{\mathbb{Z}^d}$ with the product σ -algebra. For $\eta \in \Omega$ the Ising Hamiltonian in Λ with boundary condition η is

$$H_\Lambda^\eta(\sigma) = -J \sum_{ij \in E(\Lambda)} \sigma_i \sigma_j - h \sum_{i \in \Lambda} \sigma_i - J \sum_{\substack{i \in \Lambda, j \notin \Lambda \\ i \sim j}} \sigma_i \eta_j, \quad \sigma \in \Omega_\Lambda = \{\pm 1\}^\Lambda,$$

with partition function $Z_\Lambda^\eta(\beta, h)$ and Gibbs measure μ_Λ^η (the section Gibbs measures of Part I); the free boundary condition $\eta = \emptyset$ omits the last sum, giving $Z_\Lambda^\emptyset$. Put $K = \beta J$.

- (a) Prove $\left| \log Z_\Lambda^\eta - \log Z_\Lambda^\emptyset \right| \leq 2dK |\partial\Lambda|$ for every η , and, if $\Lambda = \Lambda_1 \sqcup \Lambda_2$,

$$e^{-K|E(\Lambda_1, \Lambda_2)|} Z_{\Lambda_1}^\emptyset Z_{\Lambda_2}^\emptyset \leq Z_\Lambda^\emptyset \leq e^{K|E(\Lambda_1, \Lambda_2)|} Z_{\Lambda_1}^\emptyset Z_{\Lambda_2}^\emptyset,$$

where $E(\Lambda_1, \Lambda_2)$ is the set of edges between Λ_1 and Λ_2 .

- (b) Prove that $\psi_k = 2^{-kd} \log Z_{\Lambda_{2^k}}^\emptyset$ satisfies $|\psi_{k+1} - \psi_k| \leq dK 2^{-k}$, hence converges to a limit $\psi(\beta, h)$, and that $|\Lambda_L|^{-1} \log Z_{\Lambda_L}^\emptyset \rightarrow \psi(\beta, h)$ as $L \rightarrow \infty$ (tile Λ_L by translates of Λ_{2^k} and a remainder of at most $d 2^k L^{d-1}$ sites, and use $\left| \log Z_\Lambda^\emptyset \right| \leq (dK + \beta|h| + \log 2) |\Lambda|$).
- (c) A sequence of boxes $\Lambda^{(n)}$ is a van Hove sequence if $|\Lambda^{(n)}| \rightarrow \infty$ and, for every r ,

$$\frac{\left| \{i \in \Lambda^{(n)} : \text{dist}(i, \mathbb{Z}^d \setminus \Lambda^{(n)}) \leq r\} \right|}{|\Lambda^{(n)}|} \rightarrow 0.$$

Prove that $|\Lambda^{(n)}|^{-1} \log Z_{\Lambda^{(n)}}^{\eta_n} \rightarrow \psi(\beta, h)$ for every van Hove sequence and every choice of boundary conditions η_n (including $\emptyset$), and for the torus $\mathbb{T}_L = (\mathbb{Z}/L\mathbb{Z})^d$ with periodic boundary conditions. The limit $\psi(\beta, h)$ is the pressure; the free energy density is $f = -\beta^{-1}\psi$.

- (d) Prove $dK \leq \psi(\beta, h) - \beta|h| \leq dK + \log 2$, that ψ is even in h , and that $\psi(\beta, h) - \beta|h| \rightarrow dK$ as $|h| \rightarrow \infty$.

Exercise 5.2 (Convexity of the pressure, magnetization, and one-sided derivatives). Keep the notation of the previous exercise and regard ψ as a function of $(\beta, \beta h)$ as in the section $\log Z$, cumulants, entropy, free energy of Part I.

- (a) Prove that ψ is convex in $(\beta, \beta h) \in (0, \infty) \times \mathbb{R}$ and continuous, and that for fixed β the function $\beta h \mapsto \psi$ is convex, even, and nondecreasing on $[0, \infty)$.
- (b) Prove that the one-sided derivatives $m^\pm(\beta, h) = \partial^\pm \psi / \partial(\beta h)$ exist everywhere, are nondecreasing in h , satisfy $m^- \leq m^+$, and coincide except at countably many h ; the magnetization $m(\beta, h) = \partial \psi / \partial(\beta h)$ is defined where they coincide, and $|m| \leq 1$.
- (c) Prove that if ψ is differentiable in βh at (β, h) then for every van Hove sequence and every boundary condition

$$\frac{1}{|\Lambda^{(n)}|} \left\langle \sum_{i \in \Lambda^{(n)}} \sigma_i \right\rangle_{\Lambda^{(n)}}^{\eta_n} \rightarrow m(\beta, h)$$

(a sequence of convex functions converging pointwise has derivatives converging at every point of differentiability of the limit).

- (d) Define the spontaneous magnetization $m^*(\beta) = \lim_{h \downarrow 0} m(\beta, h)$. Prove that the limit exists, that $m^*(\beta) = m^+(\beta, 0) = -m^-(\beta, 0) \geq 0$, and that ψ is differentiable in h at $h = 0$ if and only if $m^*(\beta) = 0$.
- (e) Prove that $\beta \mapsto \beta f(\beta, 0) = -\psi(\beta, 0)$ is concave, that its one-sided derivatives $u^\pm(\beta)$ (the internal energy per site) exist and are nonincreasing, and that for $\beta_1 < \beta_2$, $\psi(\beta_2, 0) - \psi(\beta_1, 0) = -\int_{\beta_1}^{\beta_2} u(\beta) d\beta$ with $u = u^+$ almost everywhere.

6 Boundary conditions

Exercise 6.1 (The FKG lattice condition, the four functions theorem, Holley's inequality, and the FKG inequality). Let Λ be a finite set and order $\{\pm 1\}^\Lambda$ pointwise, with $(\sigma \vee \sigma')_i = \max(\sigma_i, \sigma'_i)$ and $(\sigma \wedge \sigma')_i = \min(\sigma_i, \sigma'_i)$. A function $f : \{\pm 1\}^\Lambda \rightarrow \mathbb{R}$ is increasing if $\sigma \leq \sigma'$ implies $f(\sigma) \leq f(\sigma')$;

an event is increasing if its indicator is. A strictly positive measure μ on $\{\pm 1\}^\Lambda$ satisfies the FKG lattice condition if

$$\mu(\sigma \vee \sigma') \mu(\sigma \wedge \sigma') \geq \mu(\sigma) \mu(\sigma') \quad \text{for all } \sigma, \sigma'.$$

Two positive measures μ_1, μ_2 satisfy the Holley condition if $\mu_1(\sigma \vee \sigma') \mu_2(\sigma \wedge \sigma') \geq \mu_1(\sigma) \mu_2(\sigma')$ for all σ, σ' . For probability measures, $\mu_2 \preceq \mu_1$ (stochastic domination) means $\mathbb{E}_{\mu_2} f \leq \mathbb{E}_{\mu_1} f$ for every increasing f .

- (a) (Four functions, one site.) Let $\alpha, \beta, \gamma, \delta : \{\pm 1\} \rightarrow [0, \infty)$ satisfy $\alpha(s)\beta(s') \leq \gamma(s \vee s')\delta(s \wedge s')$ for all s, s' . Prove $(\alpha(+) + \alpha(-))(\beta(+) + \beta(-)) \leq (\gamma(+) + \gamma(-))(\delta(+) + \delta(-))$. (If $\gamma(+)\delta(-) = 0$ argue directly; otherwise normalize $\gamma(+) = \delta(-) = 1$, replace $\gamma(-)$ and $\delta(+)$ by the smaller values $\alpha(-)\beta(-)$ and $\alpha(+)\beta(+)$, and reduce to $(1 - \alpha(-)\beta(+))(1 - \alpha(+)\beta(-)) \geq 0$.)
- (b) (Four functions theorem.) Let $\alpha, \beta, \gamma, \delta : \{\pm 1\}^\Lambda \rightarrow [0, \infty)$ satisfy $\alpha(\sigma)\beta(\sigma') \leq \gamma(\sigma \vee \sigma')\delta(\sigma \wedge \sigma')$ for all σ, σ' . Prove by induction on $|\Lambda|$ that $\alpha(\{\pm 1\}^\Lambda)\beta(\{\pm 1\}^\Lambda) \leq \gamma(\{\pm 1\}^\Lambda)\delta(\{\pm 1\}^\Lambda)$, where $\alpha(A) = \sum_{\sigma \in A} \alpha(\sigma)$: fix a site i , define $\alpha'(\tau) = \alpha(\tau, +) + \alpha(\tau, -)$ on $\{\pm 1\}^{\Lambda \setminus \{i\}}$ and similarly β', γ', δ' , and verify the hypothesis for the primed functions using (a). Deduce $\alpha(A)\beta(B) \leq \gamma(A \vee B)\delta(A \wedge B)$ for all $A, B \subset \{\pm 1\}^\Lambda$, where $A \vee B = \{\sigma \vee \sigma' : \sigma \in A, \sigma' \in B\}$.
- (c) (Holley's inequality.) If probability measures μ_1, μ_2 satisfy the Holley condition, prove $\mu_2 \preceq \mu_1$ (apply (b) to $\alpha = f\mu_2, \beta = \mu_1, \gamma = f\mu_1, \delta = \mu_2$ for $f \geq 0$ increasing).
- (d) (FKG inequality.) If a probability measure μ satisfies the lattice condition, prove $\mathbb{E}_\mu(fg) \geq \mathbb{E}_\mu f \mathbb{E}_\mu g$ for all increasing f, g , and $\mathbb{E}_\mu(fg) \leq \mathbb{E}_\mu f \mathbb{E}_\mu g$ if f is increasing and g decreasing.
- (e) Prove that if μ satisfies the lattice condition, then so does every conditional measure $\mu(\cdot \mid \sigma_i = s)$ on $\{\pm 1\}^{\Lambda \setminus \{i\}}$, and that the pair $\mu(\cdot \mid \sigma_i = +1), \mu(\cdot \mid \sigma_i = -1)$ satisfies the Holley condition.

Exercise 6.2 (Monotone couplings from the Holley condition). Let μ_1, μ_2 be probability measures on $\{\pm 1\}^\Lambda$ satisfying the Holley condition. Prove, by induction on $|\Lambda|$, that there is a probability measure P on $\{\pm 1\}^\Lambda \times \{\pm 1\}^\Lambda$ with marginals μ_1 and μ_2 and $P(\sigma^{(1)} \geq \sigma^{(2)}) = 1$: choose a site i ; from Holley's inequality $\mu_1(\sigma_i = +1) \geq \mu_2(\sigma_i = +1)$, so the two Bernoulli laws of σ_i admit a monotone coupling; and for $s \geq s'$ the conditional measures $\mu_1(\cdot \mid \sigma_i = s)$ and $\mu_2(\cdot \mid \sigma_i = s')$ again satisfy the Holley condition (compare the points (σ, s) and (σ', s')). Deduce that for any $f : \{\pm 1\}^\Lambda \rightarrow \mathbb{R}$,

$$|\mathbb{E}_{\mu_1} f - \mathbb{E}_{\mu_2} f| \leq \sum_{i \in \Lambda} \text{osc}_i(f) \frac{\mathbb{E}_{\mu_1} \sigma_i - \mathbb{E}_{\mu_2} \sigma_i}{2},$$

where $\text{osc}_i(f)$ is the maximal change of f under a flip of σ_i .

Exercise 6.3 (The Ising model satisfies the lattice condition; Griffiths inequalities via duplicate variables). Let $\Lambda \subset \mathbb{Z}^d$ be a box. The boundary conditions $+, -$ are $\eta \equiv +1, \eta \equiv -1$; $\emptyset$ is free; periodic means the Hamiltonian of the torus. More generally admit real boundary values $\eta_j \in \mathbb{R}$ and site-dependent fields $h_i \in \mathbb{R}$ in H_Λ^η .

- (a) Prove that for $a, a', b, b' \in \{\pm 1\}$, $(a \vee a')(b \vee b') + (a \wedge a')(b \wedge b') \geq ab + a'b'$, and deduce that μ_Λ^η satisfies the lattice condition for every $J \geq 0$, every $\eta \in \mathbb{R}^{\mathbb{Z}^d \setminus \Lambda}$, and all fields $h_i \in \mathbb{R}$; likewise the free and periodic measures.
- (b) Prove that if $\eta \leq \eta'$ pointwise and $h_i \leq h'_i$, then the pair $\mu_\Lambda^{\eta'}(h'), \mu_\Lambda^\eta(h)$ satisfies the Holley condition, hence $\mu_\Lambda^\eta(h) \preceq \mu_\Lambda^{\eta'}(h')$; in particular $\mu_\Lambda^- \preceq \mu_\Lambda^\eta \preceq \mu_\Lambda^+$ for every $\eta \in \Omega$, and $\mu_\Lambda^\emptyset \preceq \mu_\Lambda^+$.
- (c) (Griffiths I.) For $A \subset \Lambda$ put $\sigma_A = \prod_{i \in A} \sigma_i$. Expanding every factor $e^{K\sigma_i \sigma_j}, e^{\beta h_i \sigma_i}$, and $e^{K\sigma_i \eta_j}$ in its power series, prove $\langle \sigma_A \rangle_\Lambda^\eta \geq 0$ whenever $J \geq 0, h_i \geq 0$, and $\eta_j \geq 0$ for all i, j .
- (d) (Griffiths II, Ginibre's argument.) Under the same hypotheses and for $A, B \subset \Lambda$, write $\langle \sigma_A \sigma_B \rangle - \langle \sigma_A \rangle \langle \sigma_B \rangle = (Z_\Lambda^\eta)^{-2} \sum_{\sigma, \sigma'} \sigma_A (\sigma_B - \sigma'_B) e^{-\beta H_\Lambda^\eta(\sigma) - \beta H_\Lambda^\eta(\sigma')}$ and substitute $\sigma_i = (x_i +$

$y_i)/2$, $\sigma'_i = (x_i - y_i)/2$. Prove that $\sigma_A(\sigma_B - \sigma'_B)$, $\sigma_i\sigma_j + \sigma'_i\sigma'_j = (x_ix_j + y_iy_j)/2$, and $\sigma_i + \sigma'_i = x_i$ are polynomials in (x, y) with nonnegative coefficients, that the exponential of a polynomial with nonnegative coefficients has a power series with nonnegative coefficients, and that $\sum_{\sigma_i, \sigma'_i \in \{\pm 1\}} x_i^a y_i^b \geq 0$ for all $a, b \geq 0$. Conclude

$$\langle \sigma_A \sigma_B \rangle_\Lambda^\eta \geq \langle \sigma_A \rangle_\Lambda^\eta \langle \sigma_B \rangle_\Lambda^\eta,$$

and that $\langle \sigma_A \rangle_\Lambda^\eta$ is nondecreasing in each coupling K , in each field $\beta h_i \geq 0$, and in each boundary value $\eta_j \geq 0$.

Exercise 6.4 (Probability measures on spin products and conditional expectations). For the countable product $\Omega = \{\pm 1\}^{\mathbb{Z}^d}$ a cylinder set fixes finitely many spins. Suppose probability laws on all finite spin sets are consistent under restriction.

- Define their common finitely additive law on the algebra of cylinder sets. Use compactness to show that a decreasing sequence of cylinder sets with empty intersection is eventually empty. Extend the resulting premeasure and prove uniqueness on the product sigma-algebra by approximation with cylinder functions.
- For a probability measure μ , the space $L^2(\mu)$ is the completion of bounded measurable functions modulo zero-norm functions for $\|f\|_2^2 = \int |f|^2 d\mu$. For a sub-sigma-algebra $\mathcal{A}$, define $\mathbb{E}_\mu(f | \mathcal{A})$ as the orthogonal projection onto its closed subspace of $\mathcal{A}$ -measurable functions. Prove the integral characterization against bounded $\mathcal{A}$ -measurable functions, positivity, and the tower property; extend to L^1 by truncation.
- Prove that orthogonal projections onto a decreasing sequence of closed Hilbert subspaces converge in norm to the projection onto their intersection, by comparing the squared norms and applying the Cauchy criterion. Deduce the corresponding convergence of bounded conditional expectations; treat sigma-algebras modulo null sets.

Exercise 6.5 (Monotonicity in the volume and the infinite-volume limits $\mu^\pm$). Let $\Lambda \subset \Lambda'$ be boxes in $\mathbb{Z}^d$. A function $f : \Omega \rightarrow \mathbb{R}$ is local if it depends on finitely many coordinates.

- Prove $\mu_{\Lambda'}^+ \preceq \mu_\Lambda^+$ and $\mu_\Lambda^- \preceq \mu_{\Lambda'}^-$ (condition $\mu_{\Lambda'}^+$ on $\sigma|_{\Lambda' \setminus \Lambda}$ and use the finite-volume consistency of Part I together with the previous exercise).
- Prove that $\langle \sigma_0 \rangle_\Lambda^+$ is nondecreasing in h and, for $h \geq 0$, nondecreasing in β ; and that $\langle \sigma_0 \rangle_\Lambda^+ \geq \langle \sigma_0 \rangle_\Lambda^0 \geq 0$ for $h \geq 0$.
- Prove that every local f is a difference of two increasing local functions. Deduce that for every local f the limits $\mu^+(f) = \lim_{L \rightarrow \infty} \mu_{\Lambda_L^c}^+(f)$ and $\mu^-(f) = \lim_{L \rightarrow \infty} \mu_{\Lambda_L^c}^-(f)$ exist, are the same along every sequence of boxes increasing to $\mathbb{Z}^d$, and define (Kolmogorov extension) probability measures $\mu^\pm$ on Ω with $\mu^- \preceq \mu^+$; that $\mu^\pm$ are invariant under translations of $\mathbb{Z}^d$ and under the lattice symmetries fixing the origin; and that $\mu^+(f) = \mu^-(f \circ (-1))$ at $h = 0$, where (-1) is global spin reversal.
- Prove that $\mu^\pm$ satisfy the FKG inequality for increasing local functions, and that $\langle \sigma_A \rangle^+ \geq 0$, $\langle \sigma_A \sigma_B \rangle^+ \geq \langle \sigma_A \rangle^+ \langle \sigma_B \rangle^+$ for $h \geq 0$.

7 Gibbs states

Exercise 7.1 (The Gibbs specification, DLR equations, and the Gibbs property of $\mu^\pm$). Fix $\beta > 0$, $h \in \mathbb{R}$. For a box Λ let $\mathcal{F}_{\Lambda^c}$ be the σ -algebra on Ω generated by $(\sigma_j)_{j \notin \Lambda}$. The Gibbs specification is the family of probability kernels $\gamma_\Lambda(A | \eta) = \sum_{\tau \in \Omega_\Lambda} \mu_\Lambda^\eta(\tau) \mathbf{1}_A(\tau\eta)$, A measurable, $\eta \in \Omega$, where $\tau\eta$

agrees with τ on Λ and with η off Λ ; for bounded measurable f put $(\gamma_\Lambda f)(\eta) = \sum_\tau \mu_\Lambda^\eta(\tau) f(\tau\eta)$. A probability measure μ on Ω is a Gibbs state (for β, h) if it satisfies the DLR equations

$$\mu(A \mid \mathcal{F}_{\Lambda^c}) = \gamma_\Lambda(A \mid \cdot) \quad \mu\text{-a.s.}, \quad \text{for every box } \Lambda \text{ and measurable } A,$$

equivalently $\mu(\gamma_\Lambda f) = \mu(f)$ for all bounded measurable f and all Λ . Write $\mathcal{G}(\beta, h)$ for the set of Gibbs states.

- (a) Prove that $\gamma_\Lambda f$ is $\mathcal{F}_{\Lambda^c}$ -measurable, that $\gamma_\Lambda(gf) = g\gamma_\Lambda f$ for g $\mathcal{F}_{\Lambda^c}$ -measurable, that $\gamma_\Lambda f$ is local whenever f is local, and that $\gamma_{\Lambda'}\gamma_\Lambda = \gamma_{\Lambda'}$ for $\Lambda \subset \Lambda'$ (the finite-volume consistency of Part I). Prove the equivalence of the two formulations of the DLR equations.
- (b) Prove that μ^+ and μ^- are Gibbs states: for local f and $\Lambda \subset \Lambda_L^c$, $\mu_{\Lambda_L^c}^+(\gamma_\Lambda f) = \mu_{\Lambda_L^c}^+(f)$, and $\gamma_\Lambda f$ is local.
- (c) Prove that $\mathcal{G}(\beta, h)$ is convex and closed in the weak topology of probability measures on the compact space Ω , hence compact, and that every Gibbs state μ satisfies $\mu^- \preceq \mu \preceq \mu^+$ on increasing local functions (write $\mu(f) = \mu(\gamma_\Lambda f)$ and bound $\gamma_\Lambda f$ pointwise by $\mu_{\Lambda_L^c}^\pm(f)$ using the section Boundary conditions).
- (d) Prove that if μ is a translation-invariant Gibbs state then $\mu(\sigma_0) \in [\langle \sigma_0 \rangle^-, \langle \sigma_0 \rangle^+]$, where $\langle \sigma_0 \rangle^\pm = \mu^\pm(\sigma_0)$.

Exercise 7.2 (Extremal Gibbs states and tail triviality). The tail σ -algebra is $\mathcal{T} = \bigcap_L \mathcal{F}_{(\Lambda_L^c)^c}$. A probability measure μ is tail trivial if $\mu(A) \in \{0, 1\}$ for all $A \in \mathcal{T}$. A Gibbs state μ is extremal if $\mu = t\nu_1 + (1-t)\nu_2$ with $\nu_1, \nu_2 \in \mathcal{G}(\beta, h)$ and $t \in (0, 1)$ implies $\nu_1 = \nu_2 = \mu$.

- (a) Let $\mu \in \mathcal{G}(\beta, h)$ and let $\nu \in \mathcal{G}(\beta, h)$ be absolutely continuous with respect to μ , with density g . For a box Λ let $g_\Lambda = \mathbb{E}_\mu(g \mid \mathcal{F}_{\Lambda^c})$. Prove $\nu(A) = \mu(g\gamma_\Lambda(A)) = \mu(g_\Lambda\gamma_\Lambda(A)) = \mu(g_\Lambda \mathbf{1}_A)$ for every measurable A , and conclude $g = g_\Lambda$ μ -a.s. for every box Λ .
- (b) Deduce that for every $c \in \mathbb{Q}$ the event $\{g > c\}$ coincides μ -a.s. with the tail event $\limsup_L \{g_{\Lambda_L^c} > c\}$. Prove: if μ is tail trivial then g is μ -a.s. constant, hence $\nu = \mu$.
- (c) Prove that a tail-trivial Gibbs state is extremal (if $\mu = t\nu_1 + (1-t)\nu_2$ then $\nu_1 \ll \mu$). Conversely, if $\mu \in \mathcal{G}(\beta, h)$ and $A \in \mathcal{T}$ with $0 < \mu(A) < 1$, prove that $\mu(\cdot \mid A)$ and $\mu(\cdot \mid A^c)$ are Gibbs states (use $\gamma_\Lambda(f\mathbf{1}_A) = \mathbf{1}_A\gamma_\Lambda f$), so that μ is not extremal. Conclude that a Gibbs state is extremal if and only if it is tail trivial.
- (d) Prove that a tail-trivial probability measure μ on Ω has the mixing property

$$\lim_{L \rightarrow \infty} \sup_{B \in \mathcal{F}_{(\Lambda_L^c)^c}} |\mu(A \cap B) - \mu(A)\mu(B)| = 0$$

for every measurable A (approximate $\mathbf{1}_A$ by local functions and use that $\mathbb{E}_\mu(\mathbf{1}_A \mid \mathcal{F}_{(\Lambda_L^c)^c})$ converges in $L^1(\mu)$ to $\mathbb{E}_\mu(\mathbf{1}_A \mid \mathcal{T}) = \mu(A)$; prove the convergence first for bounded variables by the decreasing-subspace projection argument, and then use L^1 approximation).

Exercise 7.3 (Uniqueness of the Gibbs state and the magnetization criterion). Fix $\beta > 0$ and $h \in \mathbb{R}$.

- (a) Prove that the following are equivalent: (i) $\mathcal{G}(\beta, h)$ has exactly one element; (ii) $\mu^+ = \mu^-$; (iii) $\langle \sigma_0 \rangle^+ = \langle \sigma_0 \rangle^-$. For (iii) $\Rightarrow$ (ii) apply the coupling bound of the section Boundary conditions to $\mu_\Lambda^+, \mu_\Lambda^-$, then let $\Lambda \uparrow \mathbb{Z}^d$ and use the translation invariance of $\mu^\pm$.
- (b) Prove that at $h = 0$ the condition (iii) reads $\langle \sigma_0 \rangle_{\beta, 0}^+ = 0$.
- (c) Prove that $\beta \mapsto \langle \sigma_0 \rangle_{\beta, 0}^+$ is nondecreasing, so that $\beta_c = \inf\{\beta > 0 : \langle \sigma_0 \rangle_{\beta, 0}^+ > 0\} \in [0, \infty]$ satisfies: $\mathcal{G}(\beta, 0)$ is a singleton for $\beta < \beta_c$ and has at least the two elements $\mu^+ \neq \mu^-$ for $\beta > \beta_c$.

- (d) Prove that μ^+ and μ^- are extremal Gibbs states, in the following two steps. First, if $\mu^+ = t\nu_1 + (1-t)\nu_2$ with $\nu_1, \nu_2 \in \mathcal{G}$, then $\nu_1 \preceq \mu^+$ and $\nu_2 \preceq \mu^+$ by the previous exercise, and $t\nu_1(f) + (1-t)\nu_2(f) = \mu^+(f)$ for increasing f forces $\nu_1(f) = \nu_2(f) = \mu^+(f)$; second, increasing local functions determine a measure.

8 Correlation functions

Exercise 8.1 (Correlation functions, truncated correlations, and their monotonicity). For a Gibbs state μ and a finite $A \subset \mathbb{Z}^d$ the n -point function is $\langle \sigma_A \rangle = \mu(\sigma_A)$, $n = |A|$; the truncated two-point function is $\langle \sigma_x; \sigma_y \rangle = \langle \sigma_x \sigma_y \rangle - \langle \sigma_x \rangle \langle \sigma_y \rangle$.

- (a) Prove that for $h \geq 0$, $\langle \sigma_A \rangle^+$ is nondecreasing in β and in h , that $\langle \sigma_x; \sigma_y \rangle^+ \geq 0$, and that $\langle \sigma_x; \sigma_y \rangle^+ \leq 1 - (\langle \sigma_0 \rangle^+)^2$.
(b) Prove that at $h = 0$, $\langle \sigma_A \rangle_\Lambda^\emptyset = 0$ for $|A|$ odd, $\langle \sigma_A \rangle^+ \geq \langle \sigma_A \rangle_\Lambda^\emptyset \geq 0$ for $A \subset \Lambda$, and $\langle \sigma_x \sigma_y \rangle^+ \geq (\langle \sigma_0 \rangle^+)^2$.
(c) Prove that μ^+ is mixing in the sense of the section Gibbs states, so that $\langle \sigma_0; \sigma_x \rangle^+ \rightarrow 0$ as $|x| \rightarrow \infty$, and hence $\lim_{|x| \rightarrow \infty} \langle \sigma_0 \sigma_x \rangle^+ = (\langle \sigma_0 \rangle^+)^2$.
(d) Define the correlation length $\xi(\beta, h) \in [0, \infty]$ by

$$\frac{1}{\xi} = \liminf_{n \rightarrow \infty} \left(-\frac{1}{n} \log \langle \sigma_0; \sigma_{ne_1} \rangle^+ \right),$$

with $-\log 0 = +\infty$ in the limit inferior. Prove that ξ is unchanged when e_1 is replaced by any other unit vector of $\mathbb{Z}^d$, that $\xi \leq 1/c$ whenever $\langle \sigma_0; \sigma_x \rangle^+ \leq Ce^{-c|x|}$ for all x .

Exercise 8.2 (Exponential decay at high temperature from the path expansion). Let $\Lambda \subset \mathbb{Z}^d$ be a box, $h = 0$, and $t = \tanh K$. Adjoin a ghost site g joined by one edge to each boundary site $i \in \partial\Lambda$ for each of its neighbours outside Λ , so that the $+$ boundary term is $K \sum_i n_i \sigma_i = K \sum_{ig} \sigma_i \sigma_g$ with $\sigma_g = +1$ fixed. Write $\bar{E}$ for the edge set of Λ together with the ghost edges.

- (a) Prove $e^{K\sigma_i \sigma_j} = \cosh K (1 + t\sigma_i \sigma_j)$, and by expanding the product over edges and summing over $\sigma \in \Omega_\Lambda$,

$$Z_\Lambda^+ = 2^{|\Lambda|} (\cosh K)^{|\bar{E}|} \sum_{E' \subset \bar{E}: \deg_{E'}(i) \text{ even } \forall i \in \Lambda} t^{|E'|}, \quad \langle \sigma_0 \rangle_\Lambda^+ = \frac{\sum_{E': \deg_{E'}(i) \text{ odd iff } i=0} t^{|E'|}}{\sum_{E': \deg_{E'}(i) \text{ even } \forall i \in \Lambda} t^{|E'|}},$$

where degrees are counted at sites of Λ only; the same formulas with $\bar{E}$ replaced by $E(\Lambda)$ give $Z_\Lambda^\emptyset$ and $\langle \sigma_0 \sigma_x \rangle_\Lambda^\emptyset$ (odd degree exactly at 0 and x).

- (b) Prove that every E' in the numerator for $\langle \sigma_0 \rangle_\Lambda^+$ contains a self-avoiding path from 0 to g (the total number of odd-degree vertices of a finite graph is even), and every E' in the numerator of $\langle \sigma_0 \sigma_x \rangle_\Lambda^\emptyset$ contains a self-avoiding path from 0 to x . Removing a canonically chosen such path γ from E' leaves a set with all degrees even; deduce

$$\langle \sigma_0 \rangle_\Lambda^+ \leq \sum_{\gamma: 0 \rightarrow g} t^{|\gamma|}, \quad \langle \sigma_0 \sigma_x \rangle_\Lambda^\emptyset \leq \sum_{\gamma: 0 \rightarrow x} t^{|\gamma|},$$

the sums running over self-avoiding paths.

- (c) Prove that the number of paths of length n from 0 in $\mathbb{Z}^d$ is at most $(2d)^n$, and conclude that for $2d \tanh K < 1$,

$$\langle \sigma_0 \rangle_\Lambda^+ \leq \frac{(2dt)^{\text{dist}(0, \partial\Lambda)+1}}{1-2dt}, \quad \langle \sigma_0 \sigma_x \rangle_\Lambda^\emptyset \leq \frac{(2dt)^{|x|_1}}{1-2dt},$$

hence $\langle \sigma_0 \rangle_{\beta,0}^+ = 0$, the Gibbs state at $(\beta, 0)$ is unique, and $\beta_c \geq \beta_0$ with $\tanh(\beta_0 J) = 1/(2d)$.

- (d) Prove that for $2d \tanh K < 1$ the unique Gibbs state satisfies $\langle \sigma_0 \sigma_x \rangle \leq (2dt)^{|x|_1}/(1 - 2dt)$ (every weak limit point of the free measures $\mu_{\Lambda_L^c}^\emptyset$ is a Gibbs state), so that $\xi(\beta, 0) \leq 1/\log(1/(2d \tanh K))$.

Exercise 8.3 (Two-point functions on the cylinder from the spectrum of the transfer matrix). Let T_L be the transfer matrix of the section Transfer matrices of Part I on the row space $\mathcal{H}_L$, with spectral decomposition $T_L = \sum_a \lambda_a P_a$, $\lambda_{\max} = \lambda_0 > \lambda_1 \geq \lambda_2 \geq \dots > 0$ (P_a are mutually orthogonal spectral projections, refined by spin reversal when $h = 0$, with their full multiplicities), and let Σ_j be multiplication by τ_j on $\mathcal{H}_L$. Consider the torus $\mathbb{T}_{N,L}$ and the sites $(0, 0)$ and $(n, 0)$ in rows 0 and n .

- (a) Prove

$$\langle \sigma_{(0,0)} \sigma_{(n,0)} \rangle_{\mathbb{T}_{N,L}} = \frac{\sum_{a,b} \lambda_a^n \lambda_b^{N-n} \text{Tr}(P_a \Sigma_0 P_b \Sigma_0)}{\sum_a \lambda_a^N \text{Tr} P_a}, \quad 0 \leq n \leq N.$$

- (b) Prove that as $N \rightarrow \infty$ with n and L fixed (the infinite cylinder of circumference L),

$$\langle \sigma_{(0,0)} \sigma_{(n,0)} \rangle_L = \sum_a \left(\frac{\lambda_a}{\lambda_{\max}} \right)^n \text{Tr}(P_0 \Sigma_0 P_a \Sigma_0),$$

$$\text{Tr}(P_0 \Sigma_0 P_a \Sigma_0) \geq 0, \quad \sum_a \text{Tr}(P_0 \Sigma_0 P_a \Sigma_0) = 1.$$

- (c) At $h = 0$ prove that Σ_0 anticommutes with the spin reversal F , that P_0 lies in the $F = +1$ eigenspace, hence $\text{Tr}(P_0 \Sigma_0) = 0$ and only a with P_a in the $F = -1$ eigenspace contribute; conclude that $\langle \sigma_{(0,0)}; \sigma_{(n,0)} \rangle_L$ decays as e^{-n/ξ_L} with $1/\xi_L = \log(\lambda_{\max}/\lambda_{\max}^{(-)})$, where $\lambda_{\max}^{(-)}$ is the largest eigenvalue λ_a in the $F = -1$ sector with $\text{Tr}(P_0 \Sigma_0 P_a \Sigma_0) \neq 0$.

9 Phase transitions

Exercise 9.1 (Phase transitions: nonanalyticity of the pressure, nonuniqueness of Gibbs states, and their equivalence at zero field). A phase transition occurs at (β, h) in the thermodynamic sense if ψ is not real analytic at (β, h) , and in the sense of Gibbs states if $|\mathcal{G}(\beta, h)| > 1$.

- (a) Prove that $h \mapsto \langle \sigma_0 \rangle_{\beta,h}^+$ is nondecreasing and right-continuous (a decreasing limit of continuous nondecreasing functions of h is upper semicontinuous).
(b) Prove, for $0 \leq h < h'$ and every box Λ ,

$$\begin{aligned} \beta \int_h^{h'} \langle \sigma_0 \rangle_{\beta,s}^+ ds &\leq \frac{\log Z_\Lambda^+(\beta, h') - \log Z_\Lambda^+(\beta, h)}{|\Lambda|} \\ &\leq \beta \int_h^{h'} \left(\langle \sigma_0 \rangle_{\Lambda'_\ell, s}^+ + \frac{|\{i \in \Lambda : \text{dist}(i, \Lambda^c) \leq \ell\}|}{|\Lambda|} \right) ds. \end{aligned}$$

for every ℓ , using $\langle \sigma_i \rangle_\Lambda^+ \geq \langle \sigma_i \rangle^+$ and $\langle \sigma_i \rangle_\Lambda^+ \leq \langle \sigma_i \rangle_{i+\Lambda'_\ell}^+$ for the box $i + \Lambda'_\ell$ of side $2\ell + 1$ centred at i when it lies in Λ . Deduce $\psi(\beta, h') - \psi(\beta, h) = \beta \int_h^{h'} \langle \sigma_0 \rangle_{\beta,s}^+ ds$ and hence

$$m^*(\beta) = m^+(\beta, 0) = \langle \sigma_0 \rangle_{\beta,0}^+, \quad m(\beta, h) = \langle \sigma_0 \rangle_{\beta,h}^+ \text{ at every } h > 0 \text{ where } \psi \text{ is differentiable.}$$

- (c) Conclude that at $h = 0$ the following are equivalent: $\psi(\beta, \cdot)$ is differentiable at 0; $m^*(\beta) = 0$; $\mathcal{G}(\beta, 0)$ is a singleton. Conclude also that β_c of the section Gibbs states equals $\sup\{\beta : m^*(\beta) = 0\}$, that $m^*(\beta) > 0$ for $\beta > \beta_c$, and that for $d = 1$, $\beta_c = \infty$ (the section Transfer matrices of Part I).

Exercise 9.2 (The Peierls argument on $\mathbb{Z}^2$). Let $d = 2$, $h = 0$, and let $\Lambda \subset \mathbb{Z}^2$ be a box with + boundary condition. The dual lattice is $(\mathbb{Z}^2)^* = \mathbb{Z}^2 + (\frac{1}{2}, \frac{1}{2})$; each edge ij of $\mathbb{Z}^2$ crosses exactly one dual edge $(ij)^*$. For $\sigma \in \Omega_\Lambda$ (extended by +1 outside Λ) let $E^*(\sigma)$ be the set of dual edges $(ij)^*$ with $\sigma_i \neq \sigma_j$. A contour is a cycle (closed self-avoiding polygon) of the dual lattice; its length $|\gamma|$ is its number of edges; it surrounds 0 if 0 lies in the bounded component of its complement in $\mathbb{R}^2$.

- (a) Prove that $E^*(\sigma)$ has even degree at every dual vertex. Prove that if $\sigma_0 = -1$ then $E^*(\sigma)$ contains a contour surrounding 0: let C be the connected component of $\{i : \sigma_i = -1\}$ containing 0 (finite, contained in Λ), and take the boundary of the unbounded component of the complement of the union of the closed unit squares centred at the sites of C .
- (b) For a contour $\gamma \subset E^*(\sigma)$ let σ^γ be obtained by reversing every spin inside γ . Prove that $\sigma \mapsto \sigma^\gamma$ is an injection of $\{\sigma : \gamma \subset E^*(\sigma)\}$ into Ω_Λ with $H_\Lambda^+(\sigma^\gamma) = H_\Lambda^+(\sigma) - 2J|\gamma|$, and deduce $\mu_\Lambda^+(\gamma \subset E^*(\sigma)) \leq e^{-2K|\gamma|}$.
- (c) Prove that the number of contours of length ℓ surrounding 0 is at most $\frac{\ell}{2} 3^{\ell-1}$ (such a contour crosses the half-line $\{(x, 0) : x \geq 0\}$ at a vertical dual edge within distance $\ell/2$ of 0, and a self-avoiding polygon is determined by its successive turns) and that ℓ is even and $\ell \geq 4$.
- (d) Conclude, with $y = 9e^{-4K} < 1$,

$$\mu_\Lambda^+(\sigma_0 = -1) \leq \sum_{\ell \geq 4 \text{ even}} \frac{\ell}{2} 3^{\ell-1} e^{-2K\ell} = \frac{1}{3} \cdot \frac{y^2(2-y)}{(1-y)^2}, \quad \langle \sigma_0 \rangle_{\beta,0}^+ \geq 1 - \frac{2}{3} \cdot \frac{y^2(2-y)}{(1-y)^2},$$

uniformly in Λ ; hence $m^*(\beta) > 0$ for $K \geq \frac{3}{4}$ (where the bound gives $m^* \geq 0.318$), and $\beta_c(2)J \leq \frac{3}{4}$. Combined with the section Correlation functions, $\operatorname{artanh}(\frac{1}{4}) \leq \beta_c J \leq \frac{3}{4}$ for $d = 2$, so that $0 < \beta_c < \infty$ and there is a phase transition at $(\beta_c, 0)$; and $\beta_c(d) < \infty$ for every $d \geq 2$ (embed $\mathbb{Z}^2$ and use monotonicity of $\langle \sigma_0 \rangle^+$ in the couplings).

Exercise 9.3 (Critical exponents and the Rushbrooke inequality). Suppose $0 < \beta_c < \infty$ and $m^*(\beta_c) = 0$. Write $\tau = (\beta - \beta_c)/\beta_c$, and regard ψ as a function of (β, η) with $\eta = \beta h$. Where the derivatives exist put $\chi = \beta \partial^2 \psi / \partial \eta^2$ (the susceptibility $\partial m / \partial h$), $c = k_B \beta^2 \partial^2 \psi / \partial \beta^2$ (the specific heat per site; at $\eta = 0$ the derivative at fixed η coincides with the derivative at fixed $h = 0$, so this agrees with Part I). The critical exponents are defined, when the limits exist, by

$$\begin{aligned} c(\beta, 0) &\sim |\tau|^{-\alpha} \quad (\tau \rightarrow 0), & m^*(\beta) &\sim \tau^{\beta_{\text{mag}}} \quad (\tau \downarrow 0), \\ \chi(\beta, 0) &\sim |\tau|^{-\gamma} \quad (\tau \rightarrow 0), & m(\beta_c, h) &\sim h^{1/\delta} \quad (h \downarrow 0), \\ \xi(\beta, 0) &\sim |\tau|^{-\nu} \quad (\tau \rightarrow 0), & \langle \sigma_0; \sigma_x \rangle_{\beta_c, 0} &\sim |x|^{-(d-2+\eta_{\text{corr}})} \quad (|x| \rightarrow \infty), \end{aligned}$$

in this display interpret the power laws as $\log c / \log |\tau| \rightarrow -\alpha$ and the corresponding logarithmic ratios for the other observables; retain the usual ratio-one meaning of $\sim$ elsewhere. A logarithmic divergence of c is recorded as $\alpha = 0$ (log). Primed exponents α', γ' refer to $\tau \downarrow 0$.

- (a) Prove that the Hessian of ψ in (β, η) is positive semidefinite wherever ψ is C^2 , hence $c\chi \geq k_B \beta^3 (\partial m / \partial \beta)^2$ there, where $m = \partial \psi / \partial \eta$ and $\partial m / \partial \beta$ is taken at fixed η .
- (b) Assume: ψ is C^2 on $\{\beta_c < \beta < \beta_c + \epsilon, 0 < \eta < \epsilon\}$; there are constants with $c \leq A\tau^{-\alpha'}$ and $\chi \leq B\tau^{-\gamma'}$ on that region; $m(\beta, \eta) \rightarrow m^*(\beta)$ as $\eta \downarrow 0$ for each β ; and $m^*(\beta) \geq D\tau^{\beta_{\text{mag}}}$. Prove, by integrating (a) in β from β_c to β at fixed η and letting $\eta \downarrow 0$, that

$$\alpha' + 2\beta_{\text{mag}} + \gamma' \geq 2.$$

(If $\alpha' + \gamma' \geq 2$ there is nothing to prove.)

- (c) For the Ising chain at $h = 0$ prove $\chi/(\beta\xi) \rightarrow 2$ as $K \rightarrow \infty$. Distinguish this zero-temperature relation from exponents defined at a finite critical temperature.

10 Lee–Yang zeros

Exercise 10.1 (The partition function as a polynomial in the fugacity and the Asano contraction lemma). Let $\Gamma = (V, E)$ be a finite graph, $K \geq 0$, and for $\sigma \in \{\pm 1\}^V$ let $n_-(\sigma) = |\{i : \sigma_i = -1\}|$. Put

$$P_\Gamma(z_1, \dots, z_n) = \sum_{\sigma} \left(\prod_{ij \in E} e^{K\sigma_i\sigma_j} \right) \prod_{i: \sigma_i = -1} z_i, \quad V = \{1, \dots, n\},$$

a polynomial of degree at most one in each variable (multiaffine), and $P_\Gamma(z) = P_\Gamma(z, \dots, z)$, so that $Z_\Gamma(K, \beta h) = e^{\beta h|V|} P_\Gamma(e^{-2\beta h})$ with $z = e^{-2\beta h}$. Let $D = \{z \in \mathbb{C} : |z| < 1\}$.

- (a) Prove $P_\Gamma(z) = z^{|V|} P_\Gamma(1/z)$ (spin reversal), that P_Γ has real nonnegative coefficients, leading and constant coefficients $e^{K|E|}$, and that the zero set of P_Γ is invariant under $z \mapsto 1/z$ and under complex conjugation.
- (b) (Asano contraction lemma.) Let $A(z_1, z_2) = a + bz_1 + cz_2 + dz_1z_2$ with $a, b, c, d \in \mathbb{C}$ and $A(z_1, z_2) \neq 0$ for all $(z_1, z_2) \in D \times D$. Prove that $a \neq 0$; that $|a + bz_1| \geq |c + dz_1|$ for all $z_1 \in \overline{D}$ and $|a + cz_2| \geq |b + dz_2|$ for all $z_2 \in \overline{D}$; by averaging the squares of these inequalities over $|z_1| = 1$ and $|z_2| = 1$, that $|a| \geq |d|$; and conclude that $a + dz \neq 0$ for $z \in D$.
- (c) (Edge polynomials.) Prove that

$$A_K(z, z') = e^K + e^{-K}z + e^{-K}z' + e^Kzz' = \sum_{\sigma, \sigma' \in \{\pm 1\}} e^{K\sigma\sigma'} z^{[\sigma=-1]} z'^{[\sigma'=-1]}$$

has no zero in $D \times D$ for $K \geq 0$: solving $A_K = 0$ for z' , show that $\left| e^K + e^{-K}z \right|^2 - \left| e^{-K} + e^Kz \right|^2 = 2 \sinh(2K)(1 - |z|^2) \geq 0$.

Exercise 10.2 (The Lee–Yang circle theorem). Keep the notation of the previous exercise.

- (a) Introduce a variable $z_{e,i}$ for each pair (edge e , endpoint i of e), and put $Q = \prod_{e=ij \in E} A_K(z_{e,i}, z_{e,j})$, a multiaffine polynomial in the $2|E|$ variables with no zero in the polydisc $D^{2|E|}$. Define the contraction of two variables $z_{e,i}, z_{e',i}$ at the same site i into one variable z_i as the operation replacing $a + bz_{e,i} + cz_{e',i} + dz_{e,i}z_{e',i}$ (with a, b, c, d polynomials in the remaining variables) by $a + dz_i$. Prove, by (b) of the previous exercise applied with the remaining variables fixed in the polydisc, that a contraction of a multiaffine polynomial without zeros in the polydisc has no zeros in the polydisc.
- (b) Prove that contracting, for each site i , all variables $z_{e,i}$ ($e \ni i$) into z_i turns Q into $P_\Gamma(z_1, \dots, z_n)$ (a term of Q survives all contractions exactly when, at every site, either all or none of the variables $z_{e,i}$ occur, which is the consistency of the spin values σ_i across the edges at i); treat isolated sites separately. Conclude that $P_\Gamma(z_1, \dots, z_n) \neq 0$ on D^n .
- (c) (Circle theorem.) Prove that all zeros of $P_\Gamma(z)$ lie on the unit circle $|z| = 1$, for every finite graph Γ and every $K \geq 0$.
- (d) Prove that the same holds for the Ising model on a box Λ with free or periodic boundary conditions and with site-dependent couplings $K_{ij} \geq 0$, and that with $+$ boundary conditions the zeros of the polynomial $e^{-\beta h|\Lambda|} Z_\Lambda^+$ in $z = e^{-2\beta h}$ lie in $\{|z| \geq 1\}$ (a boundary field $K n_i \sigma_i$ rescales z_i by $e^{-2K n_i}$).

Exercise 10.3 (Consequences: analyticity of the pressure in the field and the density of zeros).

Let $\Lambda = \Lambda_L \subset \mathbb{Z}^d$ with free boundary conditions, and write $P_\Lambda(z) = e^{K|E(\Lambda)|} \prod_{k=1}^{|\Lambda|} (z - e^{i\theta_k})$ with $\theta_k \in (-\pi, \pi]$ (the circle theorem). Let $\rho_\Lambda = |\Lambda|^{-1} \sum_k \delta_{\theta_k}$, a probability measure on the circle, symmetric under $\theta \mapsto -\theta$.

- (a) Prove that for $z \notin \{|z| = 1\}$,

$$\frac{\log Z_\Lambda(K, \beta h)}{|\Lambda|} = \beta h + K \frac{|E(\Lambda)|}{|\Lambda|} + \int \log |z - e^{i\theta}| d\rho_\Lambda(\theta), \quad z = e^{-2\beta h} > 0,$$

and that every subsequential weak limit ρ of (ρ_{Λ_L}) satisfies $\psi(\beta, h) = \beta h + dK + \int \log |z - e^{i\theta}| d\rho(\theta)$ for all $h \neq 0$ (the integrand is continuous in θ for $|z| \neq 1$).

- (b) Prove that $u(z) = \int \log |z - e^{i\theta}| d\rho(\theta)$ is harmonic on D and on $\mathbb{C} \setminus \overline{D}$, and conclude that $h \mapsto \psi(\beta, h)$ is real analytic on $(0, \infty)$ and on $(-\infty, 0)$ for every $\beta > 0$; hence in the thermodynamic sense a phase transition in h at fixed β can occur only at $h = 0$, and $m(\beta, h)$, $\chi(\beta, h)$ are real analytic in $h \neq 0$.
- (c) Prove that if there is $\delta > 0$ with $\rho_{\Lambda_L}((-\delta, \delta)) = 0$ for all large L , then $\psi(\beta, \cdot)$ is real analytic in a neighbourhood of $h = 0$ and $m^*(\beta) = 0$.
- (d) Suppose that a subsequential limit ρ has, near $\theta = 0$, a density $g(\theta)$ with respect to $d\theta$ that is continuous at 0. Prove

$$m(\beta, h) = 1 - 2z \partial_z u(z) = 1 - 2z \int \frac{z - \cos \theta}{1 - 2z \cos \theta + z^2} d\rho(\theta), \quad z = e^{-2\beta h},$$

and, letting $z \uparrow 1$ (Poisson kernel), $m^*(\beta) = 2\pi g(0)$.

11 High/low-temperature expansions

Exercise 11.1 (High-temperature and low-temperature expansions and Kramers–Wannier duality on the torus). Let $\mathbb{T}_L = (\mathbb{Z}/L\mathbb{Z})^2$ with edge set E , $|E| = 2L^2$, $h = 0$, $t = \tanh K$, and let $K^* > 0$ be defined by $\tanh K^* = e^{-2K}$, equivalently $\sinh 2K \sinh 2K^* = 1$. The dual torus $\mathbb{T}_L^*$ has one vertex per face of $\mathbb{T}_L$ and one edge e^* crossing each edge e of $\mathbb{T}_L$.

- (a) (High temperature.) Prove $Z_{\mathbb{T}_L}(K) = 2^{L^2} (\cosh K)^{2L^2} \sum_{E' \subset E \text{ even}} t^{|E'|}$, the sum over edge sets with even degree at every vertex, and $\langle \sigma_x \sigma_y \rangle = \sum_{E' \text{ odd exactly at } x, y} t^{|E'|} / \sum_{E' \text{ even}} t^{|E'|}$.
- (b) (Low temperature.) For a configuration σ on $\mathbb{T}_L$ let $\Gamma(\sigma) \subset E^*$ be the set of dual edges crossing edges with $\sigma_i \neq \sigma_j$. Prove that $\Gamma(\sigma)$ is even, that $\sigma \mapsto \Gamma(\sigma)$ is two-to-one onto its image, and that $Z_{\mathbb{T}_L}(K) = 2e^{2KL^2} \sum_{\Gamma \in \mathcal{B}} e^{-2K|\Gamma|}$, where $\mathcal{B}$ is the image: the even subsets of E^* that are boundaries, that is, those crossing every closed loop of $\mathbb{T}_L$ an even number of times.
- (c) Prove that an even subset of E^* has a homology class in $(\mathbb{Z}/2)^2$ (its parities of crossings with a horizontal and a vertical loop), that $\mathcal{B}$ is the class $(0, 0)$, and that the even subsets of class (a, b) are exactly the images $\Gamma(\sigma)$ of configurations with twisted boundary conditions $\varepsilon = (\varepsilon_1, \varepsilon_2) = ((-1)^b, (-1)^a)$ (the coupling across the seam in direction r is $\varepsilon_r K \sigma_i \sigma_j$). Denote the corresponding partition functions $Z_{\mathbb{T}_L}^\varepsilon(K)$.
- (d) Combining (a) with (b)–(c) on the dual torus (which is again $\mathbb{T}_L$), prove the Kramers–Wannier identity

$$\frac{Z_{\mathbb{T}_L}(K)}{(\sinh 2K)^{L^2/2}} = \frac{1}{2} \sum_{\varepsilon \in \{\pm 1\}^2} \frac{Z_{\mathbb{T}_L}^\varepsilon(K^*)}{(\sinh 2K^*)^{L^2/2}}.$$

- (e) Prove $|\log Z_{\mathbb{T}_L}^\varepsilon - \log Z_{\mathbb{T}_L}| \leq 4KL$ and deduce that $\Phi(K) = \psi(K) - \frac{1}{2} \log \sinh 2K$ (with $\psi(K) = \psi(\beta, 0)$, $K = \beta J$) satisfies $\Phi(K) = \Phi(K^*)$. The map $K \mapsto K^*$ is a decreasing involution of $(0, \infty)$ with unique fixed point

$$K_{\text{sd}} = \frac{1}{2} \log(1 + \sqrt{2}) = \frac{1}{2} \operatorname{arsinh} 1 = 0.4407 \dots,$$

and if ψ has exactly one point of nonanalyticity in $K \in (0, \infty)$, it is K_{sd} .

- (f) Prove that on a bipartite graph at $h = 0$, $Z(K) = Z(-K)$ and $\langle \sigma_x \sigma_y \rangle_K = (-1)^{|x-y|_1} \langle \sigma_x \sigma_y \rangle_{-K}$ (reverse the spins of one sublattice), so that the antiferromagnet $J < 0$ on $\mathbb{Z}^2$ has the same pressure and the same critical coupling.

Exercise 11.2 (Prediction: the square-lattice Ising critical temperature and singularities). Consider the infinite square-lattice nearest-neighbour Ising model with $J > 0$, $h = 0$, and $K = J/(k_B T)$. For this final exercise assume, as proved by Onsager [1], the pressure formula

$$\psi(K) = \frac{1}{2} \log(2 \sinh 2K) + \frac{1}{2\pi} \int_0^\pi \text{arcosh}(\cosh 2K \coth 2K - \cos \theta) d\theta.$$

For the plus Gibbs state assume, as proved by Yang [2], that $m^*(K) = (1 - \sinh^{-4} 2K)^{1/8}$ for $K > K_{\text{sd}}$ and $m^*(K) = 0$ for $K \leq K_{\text{sd}}$, with K_{sd} from the duality exercise.

- (a) Set $s = \sinh 2K$ and use $\cosh 2K \coth 2K = s + s^{-1} \geq 2$ to show that the pressure is analytic away from K_{sd} . Locate the singular integrand at $\theta = 0$ and derive

$$K_c = K_{\text{sd}} = \frac{1}{2} \log(1 + \sqrt{2}), \quad k_B T_c = 2.269185 \dots J.$$

- (b) Split the integral at a fixed small θ_0 . Near $(K_c, 0)$ use $\text{arcosh}(\cosh 2K \coth 2K - \cos \theta) = \sqrt{16(K - K_c)^2 + \theta^2}$ to leading order. Control the differentiated remainder and derive

$$\frac{c}{k_B} = \frac{8K_c^2}{\pi} \log \frac{1}{|1 - T/T_c|} + O(1).$$

Calculate the logarithmic amplitude and its ratio above and below the transition.

- (c) Expand the magnetization formula to obtain $m^*(K) \sim (8\sqrt{2})^{1/8} (K - K_c)^{1/8}$ and hence $\beta_{\text{mag}} = 1/8$. Use the section Gibbs states to identify the ordered and unique-state regimes.
- (d) Under the bipartite spin reversal derive the corresponding prediction for staggered magnetization in the square-lattice antiferromagnet. State which fitted quantities test the exponent and which only set the nonuniversal energy scale J . Separate the model's strictly two-dimensional, nearest-neighbour, zero-field assumptions from any material's interlayer couplings or anisotropy corrections.

Part III

Percolation / Random-Cluster Model

12 Probability on lattices

Exercise 12.1 (Independent edges, monotone events, and lattice probability). Let $\mathbb{E}^d$ be the nearest-neighbour edges of $\mathbb{Z}^d$. A bond configuration is $\omega \in \{0, 1\}^{\mathbb{E}^d}$; an edge is open when its value is one. The measure $\mathbb{P}_p$ makes the edge values independent with probability p of being open. An event is increasing if opening more edges cannot destroy it. A cylinder event depends on finitely many edges.

- (a) Construct the monotone coupling $\omega_e(p) = \mathbf{1}_{U_e \leq p}$ from independent uniform variables. Prove that probabilities of increasing events are nondecreasing in p .

- (b) Prove the Harris inequality $\mathbb{P}_p(A \cap B) \geq \mathbb{P}_p(A)\mathbb{P}_p(B)$ for increasing cylinder events, by induction on the number of edges and conditioning on the last edge. Extend it to increasing connection events by monotone limits.
- (c) An edge is pivotal for a cylinder event when changing that edge alone changes whether the event occurs. Differentiate first with independent densities p_e and then set $p_e = p$ to prove Russo's formula $\frac{d}{dp}\mathbb{P}_p(A) = \sum_e \mathbb{P}_p(e \text{ pivotal})$ for increasing cylinder events.

Exercise 12.2 (Random-cluster weights and the Edwards–Sokal coupling). On a finite graph (V, E) write $o(\omega)$ for the number of open edges and $k(\omega)$ for the number of connected components, including isolated vertices. For $Q > 0$ define

$$\phi_{p,Q}(\omega) = Z_{\text{RC}}^{-1} p^{o(\omega)} (1-p)^{|E|-o(\omega)} Q^{k(\omega)}.$$

A boundary wiring is an equivalence relation on specified boundary vertices; count components after making those identifications. For integer $Q \geq 2$ the Potts Hamiltonian is $H = -J \sum_{ij} \mathbf{1}_{\sigma_i = \sigma_j}$, $\sigma_i \in \{1, \dots, Q\}$.

- (a) Put $p = 1 - e^{-\beta J}$ and assign joint weight $\prod_{ij} [(1-p)\mathbf{1}_{\omega_{ij}=0} + p\mathbf{1}_{\omega_{ij}=1}\mathbf{1}_{\sigma_i=\sigma_j}]$. Prove that its spin marginal is the Potts measure and its edge marginal is $\phi_{p,Q}$, with $Z_{\text{RC}} = e^{-\beta J|E|} Z_{\text{Potts}}$.
- (b) Conditional on edges, colour each cluster independently and uniformly. Derive

$$\langle \mathbf{1}_{\sigma_x = \sigma_y} \rangle - \frac{1}{Q} = \left(1 - \frac{1}{Q}\right) \phi_{p,Q}(x \leftrightarrow y).$$

For $Q = 2$ identify the Ising coupling $K = \beta J/2$ and prove $\langle \sigma_x \sigma_y \rangle = \phi_{p,2}(x \leftrightarrow y)$.

- (c) For $Q \geq 1$ prove supermodularity of k by considering the effect of opening two edges. Apply the FKG lattice condition of the section Boundary conditions to prove positive association. Calculate the conditional open-edge probability as p if its endpoints are already connected and $p/(p + Q(1-p))$ otherwise.

13 Connectivity

Exercise 13.1 (Open clusters and bounds on the percolation threshold). Let $C(x)$ be the cluster containing x , let $\theta(p) = \mathbb{P}_p(|C(0)| = \infty)$, and put $p_c = \sup\{p : \theta(p) = 0\}$. For $\Lambda_n = \{-n, \dots, n\}^d$, define its boundary as in the section Thermodynamic limit.

- (a) Prove that $\{|C(0)| = \infty\}$ is the decreasing intersection of $\{0 \leftrightarrow \partial\Lambda_n\}$. Show that an infinite cluster contains an infinite self-avoiding path by choosing successive extensions in the finitely branching tree of finite paths.
- (b) Count at most $2d(2d-1)^{n-1}$ self-avoiding n -edge paths from zero. Deduce $p_c(\mathbb{Z}^d) \geq 1/(2d-1)$ for $d \geq 2$, and prove $p_c(\mathbb{Z}) = 1$ separately.
- (c) On $\mathbb{Z}^2$, if the origin's cluster is finite, trace a closed dual circuit of closed primal edges around it. Bound the number of possible length- n circuits by $4(2n+1)^2 3^{n-1}$, by specifying a nearby starting edge and a nonbacktracking continuation. Deduce $\theta(p) > 0$ when p is sufficiently close to one and hence $p_c < 1$.

Exercise 13.2 (Planar duality and random-cluster self-duality). Let G be a connected finite plane graph and let G^* have one vertex for each face, including the unbounded face. Declare the dual edge open precisely when its crossed primal edge is closed.

- (a) Apply Euler's formula to the open subgraph to prove $k(\omega) = k(\omega^*) + o(\omega^*) + |V| - |E| - 1$.

- (b) Substitute this equality in the random-cluster weight and derive

$$\frac{p^*}{1-p^*} = \frac{Q(1-p)}{p}, \quad p_{\text{sd}} = \frac{\sqrt{Q}}{1+\sqrt{Q}}.$$

Explain the exchange of a free boundary with a single wired outer dual vertex through the graph construction.

- (c) For a rectangle prove that exactly one of a primal left-right open crossing and a dual top-bottom open crossing occurs, with the dual boundary arcs specified by the crossed boundary edges. Use planar separation. Distinguish this finite combinatorial identity from an assertion that $p_{\text{sd}} = p_c$ in an infinite lattice.

14 Critical phenomena

Exercise 14.1 (An exact percolation transition on a branching lattice). Consider independent bond percolation on the rooted tree with $b \geq 2$ children at each vertex. Write $\theta_b(p)$ for the probability that the root has infinitely many descendants in its open cluster.

- (a) Let x_n be the probability of an open path from the root to generation n . Prove $x_0 = 1$, $x_{n+1} = 1 - (1 - px_n)^b$, and $x_n \downarrow \theta_b(p)$.
- (b) Prove that $\theta_b(p)$ is the largest solution in $[0, 1]$ of $x = 1 - (1 - px)^b$. Use strict concavity to show $p_c = 1/b$, with zero survival at p_c .
- (c) Expand at the bifurcation to obtain $\theta_b(p) \sim 2b^2(p - 1/b)/(b - 1)$ as $p \downarrow 1/b$. Count open descendants by generations to prove $\mathbb{E}_p|C(0)| = (1 - bp)^{-1}$ for $p < 1/b$.

15 Correlation length

Exercise 15.1 (Exponential connectivity decay and the inverse correlation seminorm). For $0 < p < 1$ on $\mathbb{Z}^d$ set $\tau_n(x) = -\log \mathbb{P}_p(0 \leftrightarrow nx)$. Use the correlation-length convention of the section Correlation functions, with spin correlation replaced by connection probability.

- (a) Use Harris' inequality and translation invariance to prove $\tau_{n+m}(x) \leq \tau_n(x) + \tau_m(x)$. Prove the subadditive limit formula $\tau(x) = \lim_n \tau_n(x)/n = \inf_n \tau_n(x)/n$ by division with remainder.
- (b) Prove integer homogeneity, the triangle inequality, symmetry, and $\tau(x) \leq -\log p |x|_1$. Extend to a continuous seminorm on $\mathbb{R}^d$. If $\xi^{-1} = \tau(e_1)$, use coordinate-reflection symmetry to prove $\tau(x) \geq |x|_\infty/\xi$ and hence $\mathbb{P}_p(0 \leftrightarrow x) \leq e^{-|x|_\infty/\xi}$.
- (c) Count self-avoiding paths to prove $\xi < \infty$ for $p < 1/(2d - 1)$. With $\chi(p) = \mathbb{E}_p|C(0)| = \sum_x \mathbb{P}_p(0 \leftrightarrow x)$, derive $\chi(p) \leq C_d \max(1, \xi)^d$ whenever $\xi < \infty$.
- (d) On the rooted tree a fixed length- n path has connection probability p^n . Compute its decay length $-1/\log p$ and show it stays finite at $p = 1/b$ even though the mean cluster size diverges. Identify the exponential growth in the number of sites per generation.

16 Scaling and universality

Exercise 16.1 (Critical exponents and finite-size crossing windows). When the indicated asymptotics hold, define exponents by $\theta(p) \sim C(p - p_c)^{\beta_{\text{perc}}}$, $\chi(p) \sim C'(p_c - p)^{-\gamma}$, and $\xi(p) \sim C''|p - p_c|^{-\nu}$ on the specified sides. Universality of an exponent means its equality for a specified class of models after identifying the observables; it does not assert equality of thresholds or amplitudes.

- (a) Determine β_{perc} and γ on the rooted branching lattice from the section Critical phenomena. Explain why no divergent path-decay exponent ν is obtained there.
- (b) Let $\pi_n(p)$ be finite-volume crossing probabilities. Suppose, as mathematical data for this conditional assertion, $\pi_n(p_c + tn^{-1/\nu}) \rightarrow \Phi(t)$, locally uniformly in t , where Φ is continuous and strictly increasing from zero to one. Put $p_n(a) = \inf\{p : \pi_n(p) \geq a\}$. Use monotonicity to prove
$$n^{1/\nu}(p_n(b) - p_n(a)) \longrightarrow \Phi^{-1}(b) - \Phi^{-1}(a) \quad (0 < a < b < 1).$$
- (c) For a change of parameter $\tilde{p} = f(p)$ with $f'(p_c) > 0$, determine which exponents, amplitudes, thresholds, and window coefficients change. State the normalization needed when comparing crossing windows on two lattices.

17 SLE / conformal invariance

Exercise 17.1 (Loewner evolution and Brownian scaling). For a given continuous real driving function W define $g_t(z)$ in the upper half-plane, up to its collision lifetime, by $\partial_t g_t(z) = 2/(g_t(z) - W_t)$, $g_0(z) = z$. A standard Brownian motion is a continuous process $B_0 = 0$ with independent Gaussian increments of mean zero and variance $t - s$ on an interval $[s, t]$. Given such a process, the random Loewner evolution driven by $W_t = \sqrt{\kappa}B_t$ is chordal SLE_κ .

- (a) Solve the evolution for $W = 0$ and obtain $g_t(z) = \sqrt{z^2 + 4t}$, with the branch asymptotic to z . Determine the removed vertical slit.
- (b) For general continuous W prove, by rescaling the ODE, that $\tilde{g}_t(z) = ag_{t/a^2}(z/a)$ is driven by $\tilde{W}_t = aW_{t/a^2}$. Use Gaussian increment laws to prove scale invariance of SLE_κ up to the corresponding time change.
- (c) For deterministic s , compose g_{s+t} with $g_s^{-1}(\cdot + W_s)$ and subtract W_s . Show that the resulting evolution is driven by $W_{s+t} - W_s$, independent of the past with the original driving law. Prove injectivity and existence of the inverse on the surviving domain using the forward and reverse ODEs.

Exercise 17.2 (Prediction: conformally invariant percolation crossing probabilities). Let D be a Jordan domain with four counterclockwise marked boundary points a, b, c, d , and fix a conformal map to the upper half-plane sending them to $0, x, 1, \infty$, $0 < x < 1$. Use independent site percolation at density $1/2$ on a triangular lattice of mesh δ approximating D . The observable is a black crossing from arc (ab) to arc (cd) . For this model assume, as proved by Smirnov [3], that its probability converges to

$$P(x) = \frac{\int_0^x t^{-2/3}(1-t)^{-2/3} dt}{\int_0^1 t^{-2/3}(1-t)^{-2/3} dt}.$$

- (a) Prove $P(x) + P(1-x) = 1$, $P(1/2) = 1/2$, and $P(x) \sim 3x^{1/3} / \int_0^1 t^{-2/3}(1-t)^{-2/3} dt$ as $x \downarrow 0$. Calculate $P(1/4)$ numerically.
- (b) Use a conformal symmetry exchanging opposite pairs of sides to derive crossing probability $1/2$ for a square. Specify that x is the cross-ratio of the mapped real boundary points, not of arbitrary points in the original domain.
- (c) For M independent lattice samples compute the mean and variance of the empirical crossing frequency. Separate this sampling uncertainty from the finite-mesh error, for which the stated hypothesis supplies no rate. State precisely the triangular-site model covered by the hypothesis; a corresponding square-bond assertion requires an additional universality result.

Part IV

Six-Vertex Model

18 Row-to-row transfer matrices

Exercise 18.1 (Ice configurations and the row transfer operator). Let $S = \{+, -\}$ label arrows pointing up or right, and down or left, respectively, and put $V = \mathbb{C}^S$ with counting inner product. At each square-lattice vertex require two incoming and two outgoing arrows. Regard the left and bottom arrows as inputs and the right and top arrows as outputs. The vertex operator on $V \otimes V$ has kernel

$$R_{++}^{++} = R_{--}^{--} = a, \quad R_{+-}^{+-} = R_{-+}^{-+} = b, \quad R_{+-}^{--} = R_{-+}^{++} = c,$$

with all other values zero. For $a, b, c > 0$ the weight of a configuration is the product of its vertex weights; normalize these weights by their sum to obtain its Gibbs measure. On a cylinder of circumference L set $\mathcal{H}_L = V^{\otimes L}$ and

$$\mathcal{T}_0 = R_{0L} \cdots R_{01}, \quad T = \text{Tr}_0 \mathcal{T}_0.$$

Here 0 is the horizontal arrow space and Tr_0 is the partial trace of the section Transfer matrices.

- Expand the partial trace as a sum over horizontal arrow configurations and prove that $Z_{L,N} = \text{Tr}(T^N)$ on the L by N torus.
- Let $(Zf)(s) = sf(s)$, with $+$ = 1, $-$ = -1, and $(Xf)(s) = f(-s)$. Prove that T commutes with $S_{\text{tot}}^z = \frac{1}{2} \sum_j Z_j$, with $X^{\otimes L}$, and with cyclic translation. Decompose $\mathcal{H}_L$ into the sectors $\mathcal{H}_{L,M}$ with exactly M down arrows.
- Prove that the two polarized sectors have transfer eigenvalue $a^L + b^L$. Show that multiplying all vertex weights by $\rho > 0$ adds $\log \rho$ to the pressure per vertex.

19 Spectral parameter

Exercise 19.1 (The trigonometric vertex operator and anisotropy). For $\eta > 0$ and a complex spectral parameter u use the preceding kernel with

$$a(u) = \sinh(u + \eta), \quad b(u) = \sinh u, \quad c(u) = \sinh \eta.$$

The anisotropy is $\Delta = (a^2 + b^2 - c^2)/(2ab)$ wherever $ab \neq 0$. Let $P(v \otimes w) = w \otimes v$.

- Prove $\Delta = \cosh \eta$, $R(0) = \sinh \eta P$, and

$$R(u)R(-u) = (\sinh^2 \eta - \sinh^2 u) \text{id}.$$

Determine the exceptional parameters where $R(u)$ is singular.

- Continue to $(u, \eta) = (iv, i\gamma)$ and divide all weights by i . Prove that the resulting weights are positive for $0 < \gamma < \pi$ and $0 < v < \pi - \gamma$, with $\Delta = \cos \gamma$.
- Show that under $u = \epsilon x$, $\eta = \epsilon$, $\epsilon^{-1}R(u) \rightarrow x \text{id} + P$. Identify the three limiting weights.

20 Commuting families

Exercise 20.1 (Monodromy, the RTT relation, and conserved operators). An RLL relation for operators $L_{0j}(u)$ on an auxiliary copy of V and a site space W_j is

$$R_{00'}(u-v)L_{0j}(u)L_{0'j}(v) = L_{0'j}(v)L_{0j}(u)R_{00'}(u-v).$$

Let operators at disjoint sites and auxiliary factors commute, and put $\mathcal{T}_0(u) = L_{0L}(u) \cdots L_{01}(u)$ and $t(u) = \text{Tr}_0 \mathcal{T}_0(u)$. These equations are the defining relations for the operator families in this exercise.

- Multiply the local relations, keeping the order of factors, to prove the same relation with L replaced by $\mathcal{T}$ (the RTT relation). Use invertibility of $R(u-v)$ and cyclicity in the auxiliary factors to prove $[t(u), t(v)] = 0$, first away from its exceptional parameters and then by analytic continuation.
- For the distinguished arrow space define $A(u), B(u), C(u), D(u)$ by the four input-output kernels of $\mathcal{T}_0(u)$ on the auxiliary arrows, so $t(u) = A(u) + D(u)$. Extract $[B(u), B(v)] = 0$ from RTT.
- Wherever $t(u_0)$ is invertible prove that $t(u_0)^{-1}t'(u_0)$ commutes with every $t(v)$ and that the coefficients of the local power series for $\log(t(u_0)^{-1}t(u))$ commute with one another.

21 Yang–Baxter equation

Exercise 21.1 (The Yang–Baxter identity and the XXZ Hamiltonian). The Yang–Baxter equation is the operator identity

$$R_{12}(u-v)R_{13}(u)R_{23}(v) = R_{23}(v)R_{13}(u)R_{12}(u-v).$$

On the arrow space use $S^z = Z/2$, $S^x = X/2$, $S^y = Y/2$, where $(Yf)(s) = -isf(-s)$.

- Prove the equation for the trigonometric kernel of the section Spectral parameter. The sectors of three equal arrows are scalar; on the sectors with one or two down arrows reduce the identity to the addition formulas for $\sinh$. Deduce the RLL relation for $L_{0j}(u) = R_{0j}(u - \theta_j)$ with arbitrary fixed inhomogeneities θ_j .
- For $\theta_j = 0$ show that $t(0) = (\sinh \eta)^L U$, where U is cyclic translation in the direction determined by the ordered monodromy. Differentiate one factor at a time to prove

$$t(0)^{-1}t'(0) = \frac{1}{\sinh \eta} \sum_{j=1}^L P_{j,j+1} R'_{j,j+1}(0).$$

- Compute the local derivative and deduce that

$$H_{\Delta,h} = J \sum_{j=1}^L \left(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta (S_j^z S_{j+1}^z - \frac{1}{4}) \right) - h \sum_j \left(S_j^z - \frac{1}{2} \right)$$

commutes with every $t(u)$. State the additive constant and multiplicative factor relating $H_{\Delta,0}$ to $t(0)^{-1}t'(0)$. Take $L \geq 3$ and periodic indices throughout.

22 Bethe ansatz

Exercise 22.1 (Magnon scattering and periodic Bethe equations). In $\mathcal{H}_{L,M}$ a configuration is the set of down-arrow sites $1 \leq x_1 < \dots < x_M \leq L$. For nonzero complex $z_1, \dots, z_M$ consider the function on configurations

$$\psi(x_1, \dots, x_M) = \sum_{\pi \in \mathfrak{S}_M} A_\pi \prod_{a=1}^M z_{\pi(a)}^{x_a}.$$

This is the coordinate Bethe ansatz for the specified lattice configuration space. Assume all denominators below are nonzero and the resulting function is not identically zero.

- (a) Away from adjacent down arrows compute the action of $H_{\Delta,h}$ and obtain $E = \sum_a [h + \frac{J}{2}(z_a + z_a^{-1} - 2\Delta)]$.
- (b) Impose the equation at adjacent sites to derive

$$\frac{A_{\pi \circ (a, a+1)}}{A_\pi} = S(z_{\pi(a)}, z_{\pi(a+1)}), \quad S(z, w) = -\frac{1 + zw - 2\Delta w}{1 + zw - 2\Delta z}.$$

Check $S(z, w)S(w, z) = 1$ and compatibility of successive swaps. Verify directly that the same pairwise conditions handle a string of three or more adjacent down arrows.

- (c) Impose periodicity and derive $z_a^L = \prod_{b \neq a} S(z_b, z_a)$. Prove that a nonzero function satisfying these conditions is an eigenfunction of $H_{\Delta,h}$.

Exercise 22.2 (Prediction: the one-magnon band above a saturated chain). Consider $H_{\Delta,h}$ with $J > 0$, $\Delta \geq 0$, lattice spacing a_0 , and even $L \geq 4$. The polarized state has energy zero. The measured energy variable is $\hbar\omega$ and the wave number is k/a_0 , with k defined modulo 2π .

- (a) Use the $M = 1$ Bethe equation to derive

$$k = \frac{2\pi n}{L}, \quad \hbar\omega(k) = h + J(\cos k - \Delta).$$

Calculate the band width $2J$ and the gap $h - J(1 + \Delta)$.

- (b) Write the exchange on each bond plus half the field on each endpoint as a positive operator when $h \geq J(1 + \Delta)$. Conclude that the polarized state is a ground state above this field. For $h < J(1 + \Delta)$ use the $k = \pi$ magnon to show that it is not a ground state.
- (c) Let $S_k^- = L^{-1/2} \sum_j e^{ikj} (S_j^x - iS_j^y)$. Compute its spectral measure in the polarized state: a single unit-weight line at $\hbar\omega(k)$. Specify the zero-temperature, nearest-neighbour, and fully polarized assumptions under which this is a prediction for momentum-resolved spin spectroscopy.

Part V

XXZ / Vertex Models

23 Quantum groups

Exercise 23.1 (The open XXZ chain and Hopf algebra symmetry). Let $V = \mathbb{C}^{\{+, -\}}$ be the physical arrow space of the section Row-to-row transfer matrices and let $q > 0$, $q \neq 1$, $d = q + q^{-1}$. On

$V \otimes V$ define a kernel e that vanishes on equal arrows and has mixed-arrow kernel

$$\begin{pmatrix} q & -1 \\ -1 & q^{-1} \end{pmatrix}$$

in the order $(+-, -+)$. On $V^{\otimes N}$ put $H^{\text{open}} = -J \sum_{i=1}^{N-1} e_i/2$, $J > 0$. An algebra action on a tensor product is specified by a coproduct: a bialgebra is a unital associative algebra A with algebra maps $\Delta : A \rightarrow A \otimes A$ and $\varepsilon : A \rightarrow \mathbb{C}$ satisfying coassociativity and $(\varepsilon \otimes \text{id})\Delta = \text{id} = (\text{id} \otimes \varepsilon)\Delta$. A Hopf algebra also has an antipode S satisfying $m(S \otimes \text{id})\Delta = m(\text{id} \otimes S)\Delta = \eta\varepsilon$.

- (a) Show that e is positive, $e^2 = de$, and

$$H^{\text{open}} = J \sum_{i=1}^{N-1} \left(S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \frac{d}{2} (S_i^z S_{i+1}^z - \frac{1}{4}) \right) - \frac{J(q - q^{-1})}{4} (S_1^z - S_N^z).$$

Use the spin conventions of the section Yang–Baxter equation.

- (b) For a Hopf algebra prove that $x \cdot (v \otimes w) = \Delta(x)(v \otimes w)$ defines an action and that iterating Δ is independent of parentheses. Prove uniqueness of S as the inverse of id under convolution $f * g = m(f \otimes g)\Delta$.
- (c) On a finite-dimensional module define $(xf)(v) = f(S(x)v)$ on its dual. Prove that evaluation and the coevaluation of the section Transfer matrices are module maps, using the antipode and snake identities.

Exercise 23.2 (Monoidal categories and quasitriangular symmetry). A category consists of objects, morphism sets, associative composition, and identity morphisms. A monoidal category has a tensor bifunctor, a unit, an associator $\alpha_{U,V,W} : (U \otimes V) \otimes W \rightarrow U \otimes (V \otimes W)$, and left and right unit isomorphisms. Their coherence equations are the pentagon

$$\alpha_{U,V,W \otimes X} \alpha_{U \otimes V, W, X} = (\text{id}_U \otimes \alpha_{V,W,X}) \alpha_{U,V \otimes W, X} (\alpha_{U,V,W} \otimes \text{id}_X)$$

and the triangle equating the two unit-removal maps on $(U \otimes \mathbf{1}) \otimes V$. A braiding consists of natural isomorphisms $c_{U,V} : U \otimes V \rightarrow V \otimes U$ satisfying, with associators inserted, the two hexagon identities

$$c_{U \otimes V, W} = (c_{U,W} \otimes \text{id}_V)(\text{id}_U \otimes c_{V,W}), \quad c_{U,V \otimes W} = (\text{id}_V \otimes c_{U,W})(c_{U,V} \otimes \text{id}_W).$$

- (a) Verify these structures for vector spaces and for modules over a bialgebra, using its coproduct for the latter.
- (b) A quasitriangular Hopf algebra has an invertible $\mathcal{R} \in A \otimes A$ with $\mathcal{R}\Delta(x) = \Delta^{\text{op}}(x)\mathcal{R}$, $(\Delta \otimes \text{id})\mathcal{R} = \mathcal{R}_{13}\mathcal{R}_{23}$, and $(\text{id} \otimes \Delta)\mathcal{R} = \mathcal{R}_{13}\mathcal{R}_{12}$. Prove that $c_{U,V} = \tau\mathcal{R}$ gives a braiding on its modules and derive the Yang–Baxter identity.
- (c) Prove the braid relation for c directly from its hexagons and naturality. Check that a cocommutative Hopf algebra has $\mathcal{R} = 1$ and the usual tensor symmetry.

24 $U_q(\mathfrak{sl}_2)$

Exercise 24.1 (The quantum enveloping algebra and its fundamental spin action). For $q \in \mathbb{C}^\times$, $q^2 \neq 1$, define $U_q(\mathfrak{sl}_2)$ by generators E, F, K, K^{-1} and relations

$$KEK^{-1} = q^2 E, \quad KFK^{-1} = q^{-2} F, \quad [E, F] = \frac{K - K^{-1}}{q - q^{-1}}, \quad KK^{-1} = K^{-1}K = 1.$$

Set

$$\Delta(E) = E \otimes K + 1 \otimes E, \quad \Delta(F) = F \otimes 1 + K^{-1} \otimes F, \quad \Delta(K) = K \otimes K,$$

$\varepsilon(E) = \varepsilon(F) = 0$, $\varepsilon(K) = 1$, $S(E) = -EK^{-1}$, $S(F) = -KF$, $S(K) = K^{-1}$. Write $[n]_q = (q^n - q^{-n})/(q - q^{-1})$.

- (a) Verify that these maps respect all the defining relations and give a Hopf algebra. Prove $S^2(x) = KxK^{-1}$.
- (b) On the physical arrow space let K multiply the two arrow states by q, q^{-1} , let E flip $-$ to $+$ and annihilate $+$, and let F flip $+$ to $-$ and annihilate $-$. Verify the relations and prove that e commutes with the coproduct action on two sites.
- (c) By induction using the commutator relation prove

$$[E, F^r] = [r]_q F^{r-1} \frac{q^{1-r} K - q^{r-1} K^{-1}}{q - q^{-1}}.$$

Define a type-1 module to be a module with a direct decomposition into K -eigenspaces of weights in $q^{\mathbb{Z}}$. Show that E and F move these spaces by weights $+2$ and -2 .

Exercise 24.2 (Quantum symmetric powers and highest-weight modules). Assume q is not a root of unity. In the physical tensor space $V^{\otimes n}$ define $V_n = \bigcap_{i=1}^{n-1} \ker e_i$ and $V_0 = \mathbb{C}$.

- (a) In the sector with r down arrows show that the equations $e_i \psi = 0$ relate values at configurations differing by an adjacent exchange by a factor of q or q^{-1} . Check consistency by counting inversions of the down-arrow set. Deduce that this sector of V_n is a line for $0 \leq r \leq n$.
- (b) Prove that V_n is a submodule. If L is its highest-weight line, prove that $F^r : L \rightarrow V_n[q^{n-2r}]$ is an isomorphism for $0 \leq r \leq n$, and that $EF^r|_L = [r]_q[n-r+1]_q F^{r-1}|_L$. Deduce irreducibility and $\dim V_n = n+1$.
- (c) Let a finite-dimensional simple type-1 module have highest nonzero weight space L . Apply the same commutator identity at the first vanishing lowering map to prove that its highest weight is q^n for some $n \geq 0$ and that the module is isomorphic to V_n .
- (d) Compute the intrinsic weighted trace $\text{tr}_q(f) = \text{Tr}(Kf)$ on V_n and obtain $\text{tr}_q(\text{id}_{V_n}) = [n+1]_q$. Prove cyclicity for module maps and multiplicativity under tensor products.

25 Hecke / Temperley–Lieb algebras

Exercise 25.1 (Braid, Hecke, and Temperley–Lieb relations). The braid group B_N has generators σ_i with $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$ and commuting generators for $|i-j| \geq 2$. The Hecke algebra $H_N(q)$ imposes additionally $(g_i - q)(g_i + q^{-1}) = 0$. The Temperley–Lieb algebra $TL_N(d)$ has generators e_i with

$$e_i^2 = de_i, \quad e_i e_{i \pm 1} e_i = e_i, \quad [e_i, e_j] = 0 \quad (|i-j| \geq 2).$$

- (a) Verify these relations for the two-arrow kernel e of the section Quantum groups, by checking the three-site arrow configurations.
- (b) In $TL_N(q + q^{-1})$ put $g_i = q - e_i$. Prove the braid and Hecke relations and derive $g_i^{-1} = g_i - (q - q^{-1})$. Deduce the homomorphism $H_N(q) \rightarrow TL_N(d)$.
- (c) Define $TL_N(d)$ also by planar noncrossing pairings of N input and N output endpoints, with multiplication by stacking and with each closed loop replaced by d . Reduce an arbitrary pairing using an innermost arc and prove that the generators satisfy the presentation above:

straightening adjacent cups and caps gives the three relations, and the relations reduce every word to a scalar times a word

$$(e_{i_1} e_{i_1-1} \cdots e_{j_1}) \cdots (e_{i_r} e_{i_r-1} \cdots e_{j_r}), \quad i_1 < \cdots < i_r, \quad j_1 < \cdots < j_r, \quad j_a \leq i_a.$$

Here $1 \leq j_a \leq i_a < N$, and the empty word is allowed. Match these words bijectively with the planar pairings. Count pairings by the arc at the first boundary point to derive the Catalan recurrence and $\dim TL_N(d) = \binom{2N}{N}/(N+1)$.

Exercise 25.2 (Baxterization of the Temperley–Lieb interaction). Let $q = e^\eta$ and $d = 2 \cosh \eta$. In $TL_N(d)$ put $X_i(u) = \sinh(\eta - u) \text{id} + \sinh u e_i$.

- (a) Expand both sides and use the Temperley–Lieb relations to prove

$$X_i(u)X_{i+1}(u+v)X_i(v) = X_{i+1}(v)X_i(u+v)X_{i+1}(u).$$

Reduce the remaining scalar condition to hyperbolic addition.

- (b) Prove $X_i(0) = \sinh \eta \text{id}$ and $X_i(u)X_i(-u) = (\sinh^2 \eta - \sinh^2 u) \text{id}$. Calculate the logarithmic derivative at zero and identify its nonconstant part with $e_i / \sinh \eta$.

26 Representation theory

Exercise 26.1 (Intrinsic isotypic decomposition and Clebsch–Gordan fusion). Let q be not a root of unity. The Casimir element is $C = FE + (qK + q^{-1}K^{-1})/(q - q^{-1})^2$. A module is semisimple if it is a direct sum of simple modules.

- (a) Check that C is central and acts on V_n by $c_n = (q^{n+1} + q^{-n-1})/(q - q^{-1})^2$. Show that the c_n are distinct for $n \geq 0$.
- (b) For a finite-dimensional type-1 module decompose by generalized Casimir eigenspaces. In a summand whose simple quotients are V_n , show that its weight- q^n space generates a sum of copies of V_n : the lowering maps are injective until level n , and level $n+1$ is absent. Compare dimensions with a composition series to prove complete reducibility.
- (c) Prove that the evaluation map gives the canonical decomposition

$$W \simeq \bigoplus_{n \geq 0} V_n \otimes \text{Hom}_{U_q}(V_n, W).$$

Compute multiplicities by subtracting consecutive weight dimensions, and derive $V_m \otimes V_n \simeq \bigoplus_{r=0}^{\min(m,n)} V_{m+n-2r}$.

- (d) Apply this to $V^{\otimes N}$ and obtain the multiplicity $\binom{N}{r} - \binom{N}{r-1}$ of V_{N-2r} . Show that the open-chain Hamiltonian acts only on the multiplicity spaces, so that every energy in such a space has multiplicity at least $N - 2r + 1$.

Exercise 26.2 (The classical spin limit). Let $\mathfrak{sl}_2$ be the trace-zero endomorphisms of a two-dimensional vector space with bracket $[A, B] = AB - BA$. Its enveloping algebra is the tensor algebra modulo $x \otimes y - y \otimes x - [x, y]$; it acts on tensor powers by the coproduct $x \mapsto x \otimes 1 + 1 \otimes x$.

- (a) Set $q = e^\eta$ and write $K = e^{\eta H}$ on the weight spaces of V_n . Take $\eta \rightarrow 0$ in the ladder relations and obtain $[H, E] = 2E$, $[H, F] = -2F$, $[E, F] = H$. Identify V_n with $\text{Sym}^n(V)$ by its tensor action.
- (b) Put $S^z = H/2$, $S^x = (E+F)/2$, $S^y = (E-F)/(2i)$, and $s = n/2$. Compute the central operator $(S^x)^2 + (S^y)^2 + (S^z)^2 = s(s+1) \text{id}$ and the character $\text{Tr } e^{tS^z} = \sinh((2s+1)t/2) / \sinh(t/2)$. Use the induced inner product on $\text{Sym}^n(V)$ to verify self-adjointness of the spin operators.

27 Rational/trigonometric hierarchy

Exercise 27.1 (Trigonometric transfer operators and the rational limit). Use the trigonometric $R(u)$ and $H_{\Delta,h}$ of the sections Spectral parameter and Yang–Baxter equation.

- (a) Under $u = \epsilon x$, $\eta = \epsilon$ prove $\epsilon^{-1}R(u) \rightarrow x \text{id} + P$ and $\Delta \rightarrow 1$. Pass to the limit in the Yang–Baxter and RTT relations.
- (b) Show that the classical limit of the open-chain kernel is $e = \text{id} - P$. Derive $H^{\text{open}} = \frac{J}{2} \sum_i (P_{i,i+1} - \text{id})$ and its global rotation symmetry.
- (c) On a periodic chain derive the one-magnon energy $h + J(\cos k - 1)$ at $\Delta = 1$ and compare with the trigonometric band $h + J(\cos k - \cosh \eta)$. Keep the different open and periodic boundary terms explicit.

Exercise 27.2 (Prediction: quantum-group multiplets of an open XXZ segment). For $q > 0$, $q \neq 1$, take the Hermitian open Hamiltonian $H^{\text{open}} = -J \sum_i e_i/2$ with the boundary field fixed in the section Quantum groups.

- (a) For two sites use $e^2 = de$ and the intrinsic decomposition $V \otimes V = V_2 \oplus V_0$ to derive a triplet at energy zero and a singlet at energy $-Jd/2 = -J\Delta$.
- (b) For N sites prove that any simple energy in the multiplicity space of V_{N-2r} has exactly $N - 2r + 1$ states unless its energy coincides with that of another multiplet. Show that it has one state in each magnetization sector $S_{\text{tot}}^z = (N - 2r)/2 - m$ for $0 \leq m \leq N - 2r$.
- (c) For an operator in the V_2 component of $\text{End}(\mathcal{H}_N)$, with action $x \cdot O = \sum x_{(1)} O S(x_{(2)})$, apply Clebsch–Gordan fusion to derive the selection rule $V_n \rightarrow V_{n-2} \oplus V_n \oplus V_{n+2}$, omitting negative labels. Prove that a scalar sector V_0 couples only to V_2 through such an operator.
- (d) State the two-site absorption energy $J\Delta$, the multiplet degeneracies, and the selection rule as separate spectroscopic predictions. Identify the boundary field, anisotropy, and transformation law of the probe required to test them.

Part VI

Eight-Vertex / RSOS Models

28 Theta and elliptic functions

Exercise 28.1 (Restricted solid-on-solid configurations). Fix an integer $k \geq 1$. The height set of the restricted solid-on-solid (RSOS) model is $\{1, \dots, k+1\}$; neighbouring lattice sites must have heights differing by one. On a finite graph take the uniform measure on admissible configurations, with zero weight for inadmissible ones. Equivalently the energy is zero on the allowed configuration space. Prove invariance under the height reflection $a \mapsto k+2-a$. On a path of N edges with initial height a and terminal height b , express the partition function as $(A^N)_{ab}$, where A is the adjacency kernel of the path graph A_{k+1} .

Exercise 28.2 (Jacobi theta functions: series, quasi-periodicity, half-period shifts, and zeros). Let

$\tau \in \mathbb{C}$ with $\text{Im } \tau > 0$ and $p = e^{i\pi\tau}$ (the nome), so $|p| < 1$; fix $p^{1/4} = e^{i\pi\tau/4}$. Define, for $z \in \mathbb{C}$,

$$\begin{aligned}\vartheta_1(z|\tau) &= 2 \sum_{n \geq 0} (-1)^n p^{(n+1/2)^2} \sin((2n+1)z) = -i \sum_{n \in \mathbb{Z}} (-1)^n p^{(n+1/2)^2} e^{i(2n+1)z}, \\ \vartheta_2(z|\tau) &= 2 \sum_{n \geq 0} p^{(n+1/2)^2} \cos((2n+1)z), \quad \vartheta_3(z|\tau) = 1 + 2 \sum_{n \geq 1} p^{n^2} \cos(2nz), \\ \vartheta_4(z|\tau) &= 1 + 2 \sum_{n \geq 1} (-1)^n p^{n^2} \cos(2nz) = \sum_{n \in \mathbb{Z}} (-1)^n p^{n^2} e^{2inz}.\end{aligned}$$

The dependence on τ is suppressed when it is fixed.

- (a) Prove that the series converge absolutely and uniformly on compact sets, so the ϑ_j are entire; ϑ_1 is odd and $\vartheta_2, \vartheta_3, \vartheta_4$ are even.
- (b) Prove the quasi-periodicity relations

$$\begin{aligned}\vartheta_1(z + \pi) &= -\vartheta_1(z), & \vartheta_1(z + \pi\tau) &= -p^{-1} e^{-2iz} \vartheta_1(z), \\ \vartheta_2(z + \pi) &= -\vartheta_2(z), & \vartheta_2(z + \pi\tau) &= p^{-1} e^{-2iz} \vartheta_2(z), \\ \vartheta_3(z + \pi) &= \vartheta_3(z), & \vartheta_3(z + \pi\tau) &= p^{-1} e^{-2iz} \vartheta_3(z), \\ \vartheta_4(z + \pi) &= \vartheta_4(z), & \vartheta_4(z + \pi\tau) &= -p^{-1} e^{-2iz} \vartheta_4(z),\end{aligned}$$

and the half-period shifts $\vartheta_1(z + \frac{\pi}{2}) = \vartheta_2(z)$, $\vartheta_2(z + \frac{\pi}{2}) = -\vartheta_1(z)$, $\vartheta_3(z + \frac{\pi}{2}) = \vartheta_4(z)$, $\vartheta_4(z + \frac{\pi}{2}) = \vartheta_3(z)$, $\vartheta_1(z + \frac{\pi\tau}{2}) = ip^{-1/4} e^{-iz} \vartheta_4(z)$, $\vartheta_4(z + \frac{\pi\tau}{2}) = ip^{-1/4} e^{-iz} \vartheta_1(z)$.

- (c) Zero count. Let f be entire, not identically zero, with $f(z + \pi) = \epsilon f(z)$, $\epsilon \in \{\pm 1\}$, and $f(z + \pi\tau) = C e^{-2inz} f(z)$ for some $C \in \mathbb{C}^\times$ and $n \in \mathbb{Z}_{\geq 0}$. Prove by integrating f'/f around a period parallelogram avoiding zeros that f has exactly n zeros modulo the lattice $\pi\mathbb{Z} + \pi\tau\mathbb{Z}$, counted with multiplicity. Deduce that ϑ_1 vanishes exactly on the lattice, with simple zeros, and that the zeros of $\vartheta_2, \vartheta_3, \vartheta_4$ are the translates of the lattice by $\frac{\pi}{2}$, $\frac{\pi}{2} + \frac{\pi\tau}{2}$, $\frac{\pi\tau}{2}$ respectively.
- (d) Dimension count. For $\epsilon \in \{\pm 1\}$, $n \geq 1$, $C \in \mathbb{C}^\times$ let $\Theta_n(\epsilon, C)$ be the space of entire f with $f(z + \pi) = \epsilon f(z)$ and $f(z + \pi\tau) = C e^{-2inz} f(z)$. Expanding $f(z) = \sum_m c_m e^{i(2m+\delta)z}$ with $\delta \in \{0, 1\}$ determined by ϵ , show that the second relation is a recursion $c_{m+n} = \gamma_m c_m$ with explicit $\gamma_m \neq 0$, whose solutions converge, and conclude $\dim \Theta_n(\epsilon, C) = n$. Verify that a product of n theta functions $\vartheta_{j_1}(z + w_1) \cdots \vartheta_{j_n}(z + w_n)$ lies in $\Theta_n(\epsilon, C)$ for suitable ϵ, C depending only on the j_i and $\sum w_i$.

Exercise 28.3 (Addition identities of degree two). Write $\vartheta_j = \vartheta_j(\cdot|\tau)$. Prove each of the following identities by the same method: both sides lie in a space $\Theta_2(\epsilon, C)$ of dimension two (as functions of x , with the other variables fixed), the right side vanishes at a point where the left side also vanishes, so that the two sides are proportional, and the constant is found by evaluating at $x = 0$ or $x = \frac{\pi}{2}$.

- (a) $\vartheta_1(x+y)\vartheta_1(x-y)\vartheta_4(0)^2 = \vartheta_1(x)^2\vartheta_4(y)^2 - \vartheta_4(x)^2\vartheta_1(y)^2$ and $\vartheta_4(x+y)\vartheta_4(x-y)\vartheta_4(0)^2 = \vartheta_4(x)^2\vartheta_4(y)^2 - \vartheta_1(x)^2\vartheta_1(y)^2$.
- (b) $\vartheta_1(x+y)\vartheta_4(x-y)\vartheta_2(0)\vartheta_3(0) = \vartheta_1(x)\vartheta_4(x)\vartheta_2(y)\vartheta_3(y) + \vartheta_2(x)\vartheta_3(x)\vartheta_1(y)\vartheta_4(y)$.
- (c) $\vartheta_4(0)^2\vartheta_3(x)^2 = \vartheta_3(0)^2\vartheta_4(x)^2 - \vartheta_2(0)^2\vartheta_1(x)^2$ and $\vartheta_4(0)^2\vartheta_2(x)^2 = \vartheta_2(0)^2\vartheta_4(x)^2 - \vartheta_3(0)^2\vartheta_1(x)^2$; deduce $\vartheta_3(0)^4 = \vartheta_2(0)^4 + \vartheta_4(0)^4$ by putting $x = \frac{\pi}{2}$.
- (d) For a fixed shift η put $f(x) = \vartheta_4(x)\vartheta_1(x+\eta)$, $g(x) = \vartheta_1(x)\vartheta_4(x+\eta)$, $h(x) = \vartheta_4(x)\vartheta_4(x+\eta)$, and $j(x) = \vartheta_1(x)\vartheta_1(x+\eta)$. Prove the four identities

$$\begin{aligned}f(x)f(y) - g(x)g(y) &= \vartheta_4(0)\vartheta_1(\eta)\vartheta_4(x-y)\vartheta_1(x+y+\eta), \\ h(x)h(y) - j(x)j(y) &= \vartheta_4(0)\vartheta_4(\eta)\vartheta_4(x-y)\vartheta_4(x+y+\eta), \\ h(x)j(y) - j(x)h(y) &= \vartheta_4(0)\vartheta_4(\eta)\vartheta_1(y-x)\vartheta_1(x+y+\eta), \\ f(x)g(y) - g(x)f(y) &= \vartheta_4(0)\vartheta_1(\eta)\vartheta_1(y-x)\vartheta_4(x+y+\eta).\end{aligned}$$

For the first use the common zero $x = -y - \eta$; for the second use $x = y + \pi\tau/2$ and the half-period shifts; for the last two use $x = y$.

- (e) The three-term identity. Put $[x, y] = \vartheta_1(x+y)\vartheta_1(x-y)$. Deduce from (a) that $[x, y]\vartheta_4(0)^2 = P_x Q_y - Q_x P_y$ with $P_x = \vartheta_1(x)^2$, $Q_x = \vartheta_4(x)^2$, and from the Plücker relation for 2×2 determinants that

$$[c, d][a, b] + [d, b][a, c] + [b, c][a, d] = 0 \quad (a, b, c, d \in \mathbb{C}).$$

- (f) Wronskian. Show that $\vartheta_1'\vartheta_4 - \vartheta_1\vartheta_4'$ lies in the same Θ_2 as $\vartheta_2\vartheta_3$ and is even, and conclude $(\vartheta_1/\vartheta_4)' = \kappa \vartheta_2\vartheta_3/\vartheta_4^2$ with $\kappa = \vartheta_1'(0)\vartheta_4(0)/(\vartheta_2(0)\vartheta_3(0))$.

Exercise 28.4 (Jacobi's triple product and the product formulas).

- (a) For $|p| < 1$ and $w \in \mathbb{C}^\times$ put $F(w) = \prod_{n \geq 1} (1 + p^{2n-1}w)(1 + p^{2n-1}w^{-1})$. Prove that F is holomorphic on $\mathbb{C}^\times$ with $F(p^2w) = (pw)^{-1}F(w)$, expand $F(w) = \sum_{m \in \mathbb{Z}} c_m w^m$, derive $c_m = p^{2m-1}c_{m-1}$, hence $c_m = p^{m^2}c_0$, so that $\prod_{n \geq 1} (1 - p^{2n})F(w) = \phi(p) \sum_m p^{m^2} w^m$ for some $\phi(p)$ depending only on p .
- (b) Determine $\phi(p) = 1$. Put $w = i$: the series becomes $\sum_j (-1)^j p^{4j^2}$ and $F(i) = \prod_n (1 + p^{4n-2})$. Apply (a) with p replaced by p^4 and $w = -1$ to the same series, and use $\prod_n (1 - p^{2n}) = \prod_n (1 - p^{8n})(1 - p^{8n-4})(1 - p^{4n-2})$ to obtain $\phi(p) = \phi(p^4)$; iterate and let $p \rightarrow 0$.
- (c) Deduce the product formulas

$$\begin{aligned} \vartheta_3(z) &= \prod_{n \geq 1} (1 - p^{2n})(1 + 2p^{2n-1} \cos 2z + p^{4n-2}), \\ \vartheta_4(z) &= \prod_{n \geq 1} (1 - p^{2n})(1 - 2p^{2n-1} \cos 2z + p^{4n-2}), \\ \vartheta_2(z) &= 2p^{1/4} \cos z \prod_{n \geq 1} (1 - p^{2n})(1 + 2p^{2n} \cos 2z + p^{4n}), \\ \vartheta_1(z) &= 2p^{1/4} \sin z \prod_{n \geq 1} (1 - p^{2n})(1 - 2p^{2n} \cos 2z + p^{4n}), \end{aligned}$$

and the derivative formula $\vartheta_1'(0) = \vartheta_2(0)\vartheta_3(0)\vartheta_4(0)$ from $\prod_n (1 + p^{2n})(1 + p^{2n-1})(1 - p^{2n-1}) = 1$. Conclude that the constant κ of the previous exercise equals $\vartheta_4(0)^2$.

- (d) Degeneration. Show that as $p \rightarrow 0$ with z fixed, $\vartheta_1(z) = 2p^{1/4} \sin z (1 + O(p^2))$, $\vartheta_2(z) = 2p^{1/4} \cos z (1 + O(p^2))$, $\vartheta_3(z), \vartheta_4(z) = 1 + O(p)$, uniformly on compact sets.

Exercise 28.5 (Modular transformation, Jacobi elliptic functions, and the modulus).

- (a) Prove the Fourier identity $\sum_{n \in \mathbb{Z}} e^{-\pi n^2 t + 2\pi i n z} = t^{-1/2} \sum_{n \in \mathbb{Z}} e^{-\pi(n-z)^2/t}$ for $\operatorname{Re} t > 0$, $z \in \mathbb{C}$ by integrating a Gaussian against $e^{-2\pi i n x}$ and periodizing the result (the branch of $t^{-1/2}$ is positive for $t > 0$). Derive

$$\vartheta_3(z|\tau) = (-i\tau)^{-1/2} e^{-iz^2/(\pi\tau)} \vartheta_3\left(\frac{z}{\tau} \middle| -\frac{1}{\tau}\right),$$

and, using the half-period shifts, $\vartheta_4(z|\tau) = (-i\tau)^{-1/2} e^{-iz^2/(\pi\tau)} \vartheta_2(z/\tau | -1/\tau)$, $\vartheta_2(z|\tau) = (-i\tau)^{-1/2} e^{-iz^2/(\pi\tau)} \vartheta_4(z/\tau | -1/\tau)$, $\vartheta_1(z|\tau) = i(-i\tau)^{-1/2} e^{-iz^2/(\pi\tau)} \vartheta_1(z/\tau | -1/\tau)$.

- (b) Define the modulus $k = \vartheta_2(0)^2/\vartheta_3(0)^2$, the complementary modulus $k' = \vartheta_4(0)^2/\vartheta_3(0)^2$, and $K = \frac{\pi}{2} \vartheta_3(0)^2$, $K' = -i\tau K$. Show $k^2 + k'^2 = 1$ and that $\tau \mapsto -1/\tau$ exchanges k with k' and K with K' .

(c) Define, with $u = \vartheta_3(0)^2 z$,

$$\operatorname{sn} u = \frac{\vartheta_3(0) \vartheta_1(z)}{\vartheta_2(0) \vartheta_4(z)}, \quad \operatorname{cn} u = \frac{\vartheta_4(0) \vartheta_2(z)}{\vartheta_2(0) \vartheta_4(z)}, \quad \operatorname{dn} u = \frac{\vartheta_4(0) \vartheta_3(z)}{\vartheta_3(0) \vartheta_4(z)}.$$

Prove from the identities of degree two that $\operatorname{sn}^2 + \operatorname{cn}^2 = 1$, $\operatorname{dn}^2 + k^2 \operatorname{sn}^2 = 1$, $\frac{d}{du} \operatorname{sn} = \operatorname{cn} \operatorname{dn}$, that sn is meromorphic with periods $4K$ and $2iK'$, simple poles at $iK' + 2K\mathbb{Z} + 2iK'\mathbb{Z}$, and that sn is determined by the initial value problem $(\operatorname{sn}')^2 = (1 - \operatorname{sn}^2)(1 - k^2 \operatorname{sn}^2)$, $\operatorname{sn} 0 = 0$, $\operatorname{sn}' 0 = 1$, so that it inverts the elliptic integral $u = \int_0^x dt / \sqrt{(1-t^2)(1-k^2 t^2)}$ near 0. For $0 < k < 1$ (i.e. $\tau \in i\mathbb{R}_{>0}$) show that sn increases from 0 to 1 on $[0, K]$ (use $\operatorname{sn} K = 1$ from the half-period shift) and deduce $K = \int_0^{\pi/2} (1 - k^2 \sin^2 \theta)^{-1/2} d\theta$ by the substitution $\operatorname{sn} u = \sin \theta$.

(d) Show that as $p \rightarrow 0$: $k \rightarrow 0$, $K \rightarrow \pi/2$, $K' \rightarrow \infty$, $\operatorname{sn} u \rightarrow \sin u$, $\operatorname{cn} u \rightarrow \cos u$, $\operatorname{dn} u \rightarrow 1$, uniformly on compact sets; and that $\tau \mapsto -1/\tau$ followed by $p \rightarrow 0$ gives the hyperbolic degeneration $\operatorname{sn}(u; k) \rightarrow \tanh u$ as $k \rightarrow 1$.

29 Elliptic R-matrices

Exercise 29.1 (Eight-vertex configurations, weights, invariants, and the elliptic parametrization). On the square lattice place an arrow on every edge and keep the vertices at which the number of incoming arrows is even: the six vertices of the section Row-to-row transfer matrices in Part IV, with weights a (both arrows continue straight, parallel), b (straight, antiparallel), c (turning), together with the two vertices with four incoming or four outgoing arrows (source and sink), both of weight d . With $V = \mathbb{C}^2$, e_+ (arrow up or right), e_- , the vertex operator is

$$R = \begin{pmatrix} a & 0 & 0 & d \\ 0 & b & c & 0 \\ 0 & c & b & 0 \\ d & 0 & 0 & a \end{pmatrix} \in \operatorname{End}(V \otimes V)$$

in the basis $e_+ \otimes e_+, e_+ \otimes e_-, e_- \otimes e_+, e_- \otimes e_-$, and the partition function on the $L \times N$ torus is $Z = \operatorname{Tr}(T^N)$ with T the row-to-row transfer matrix built from R exactly as in Part IV. Put

$$\Delta = \frac{a^2 + b^2 - c^2 - d^2}{2(ab + cd)}, \quad \Gamma = \frac{ab - cd}{ab + cd}.$$

- (a) Prove that R commutes with $\sigma^z \otimes \sigma^z$ and with $\sigma^x \otimes \sigma^x$ (parity of the number of down arrows is conserved, and arrow reversal is a symmetry), and that R commutes with $\sigma^z \otimes 1 + 1 \otimes \sigma^z$ if and only if $d = 0$.
- (b) Elliptic parametrization. Fix $\tau, \eta \in \mathbb{C}$ and define, with $\vartheta_j = \vartheta_j(\cdot|\tau)$,

$$\begin{aligned} a(u) &= \vartheta_4(\eta) \vartheta_4(u) \vartheta_1(u + \eta), & b(u) &= \vartheta_4(\eta) \vartheta_1(u) \vartheta_4(u + \eta), \\ c(u) &= \vartheta_1(\eta) \vartheta_4(u) \vartheta_4(u + \eta), & d(u) &= \vartheta_1(\eta) \vartheta_1(u) \vartheta_1(u + \eta), \end{aligned}$$

and $R(u)$ accordingly. Show that these are Baxter's weights [4] $\vartheta_4(2\eta')\vartheta_4(u' - \eta')\vartheta_1(u' + \eta')$, etc., under $u' = u + \eta/2$, $\eta' = \eta/2$, up to the common factor. Prove $\Gamma = \frac{\vartheta_4(\eta)^2 - \vartheta_1(\eta)^2}{\vartheta_4(\eta)^2 + \vartheta_1(\eta)^2}$ directly, and

$$\Delta = \frac{\vartheta_2(\eta)\vartheta_3(\eta)\vartheta_4(0)^2}{\vartheta_2(0)\vartheta_3(0)(\vartheta_4(\eta)^2 + \vartheta_1(\eta)^2)}$$

by combining the identities (a) and (b) of the exercise on addition identities: write $a^2 + b^2 - c^2 - d^2$ as $\vartheta_1(u + \eta)^2[\vartheta_4(\eta)^2\vartheta_4(u)^2 - \vartheta_1(\eta)^2\vartheta_1(u)^2] + \vartheta_4(u + \eta)^2[\vartheta_4(\eta)^2\vartheta_1(u)^2 - \vartheta_1(\eta)^2\vartheta_4(u)^2]$, factor each bracket, and use (b) at $y = \pm\eta$. Both invariants are independent of u .

- (c) Trigonometric limit. Show that as $p \rightarrow 0$, $(a, b, c, d)/(2p^{1/4}) \rightarrow (\sin(u + \eta), \sin u, \sin \eta, 0)$, which is the six-vertex parametrization of Part IV with (u, η) replaced by $(iu, i\eta)$ up to a common factor, and $\Gamma \rightarrow 1$, $\Delta \rightarrow \cos \eta$.
- (d) Positivity. For $\tau \in i\mathbb{R}_{>0}$ (so $0 < p < 1$) show from the product formulas that $\vartheta_4(x) > 0$ for real x and $\vartheta_1(x) > 0$ for $0 < x < \pi$; deduce that for real η, u with $0 < u$ and $u + \eta < \pi$, $0 < \eta < \pi$, all four weights are positive, and compute the signs of Δ and Γ there.

Exercise 29.2 (The eight-vertex Yang–Baxter equation). Let $R(u)$ be the elliptic eight-vertex R -matrix.

- (a) Write the equation $R_{12}(u - v)R_{13}(u)R_{23}(v) = R_{23}(v)R_{13}(u)R_{12}(u - v)$ on $V^{\otimes 3}$ with three sets of weights (a, b, c, d) , (a', b', c', d') , (a'', b'', c'', d'') for the arguments $u - v$, u , v . Using the parity conservation and arrow reversal of the previous exercise, show that it is equivalent to the six polynomial identities

$$\begin{aligned} c(a'a'' - b'b'') &= a(c''d' - c'd''), & c(a'b'' - a''b') &= b(d'd'' - c'c''), \\ c'(aa'' - bb'') &= a'(cc'' - dd''), & c''(ab' - a'b) &= b''(cc' - dd'), \\ d''(aa' - bb') &= a''(cd' - c'd), & d'(ab'' - a''b) &= b'(cd'' - c'd). \end{aligned}$$

- (b) Using the four shifted identities of the exercise on addition identities, prove for all x, y :

$$\begin{aligned} a(x)a(y) - b(x)b(y) &= \vartheta_4(\eta)^2\vartheta_4(0)\vartheta_1(\eta)\vartheta_4(x - y)\vartheta_1(x + y + \eta), \\ c(x)c(y) - d(x)d(y) &= \vartheta_1(\eta)^2\vartheta_4(0)\vartheta_4(\eta)\vartheta_4(x - y)\vartheta_4(x + y + \eta), \\ c(x)d(y) - c(y)d(x) &= \vartheta_1(\eta)^2\vartheta_4(0)\vartheta_4(\eta)\vartheta_1(y - x)\vartheta_1(x + y + \eta), \\ a(x)b(y) - a(y)b(x) &= \vartheta_4(\eta)^2\vartheta_4(0)\vartheta_1(\eta)\vartheta_1(y - x)\vartheta_4(x + y + \eta). \end{aligned}$$

- (c) Substitute $(x, y) = (u - v, v)$, $(u - v, u)$, (u, v) as appropriate into (b) and verify each of the six identities of (a); for instance the third becomes $c(u)\vartheta_4(\eta)^2\vartheta_4(0)\vartheta_1(\eta)\vartheta_4(u - 2v)\vartheta_1(u + \eta) = a(u)\vartheta_1(\eta)^2\vartheta_4(0)\vartheta_4(\eta)\vartheta_4(u - 2v)\vartheta_4(u + \eta)$, which holds by the definitions of $a(u)$ and $c(u)$. Conclude that the eight-vertex Yang–Baxter equation holds identically in u, v, η, τ .
- (d) Deduce the three-parameter form $R_{12}(u - v)R_{13}(u - w)R_{23}(v - w) = R_{23}(v - w)R_{13}(u - w)R_{12}(u - v)$ and, in the limit $p \rightarrow 0$, recover the six-vertex Yang–Baxter equation of Part IV.

Exercise 29.3 (Regularity, unitarity, crossing symmetry, and commuting eight-vertex transfer matrices).

- (a) Prove $R(0) = a(0)P$ with $a(0) = \vartheta_4(\eta)\vartheta_4(0)\vartheta_1(\eta)$ and P the flip (regularity).
- (b) Prove $R_{12}(u)R_{12}(-u) = \rho(u)1$ with $\rho(u) = a(u)a(-u) + d(u)d(-u) = b(u)b(-u) + c(u)c(-u)$ (unitarity; use the identities of the previous exercise at $(x, y) = (u, -u)$ to show the two expressions agree and the cross terms

$$a(u)d(-u) + d(u)a(-u), \quad b(u)c(-u) + c(u)b(-u)$$

vanish).

- (c) Prove the crossing relations $a(-u - \eta) = -b(u)$, $b(-u - \eta) = -a(u)$, $c(-u - \eta) = c(u)$, $d(-u - \eta) = d(u)$, and deduce

$$R_{12}(-u - \eta) = -(\sigma^y \otimes 1)R_{12}(u)^{t_1}(\sigma^y \otimes 1)$$

where t_1 is the partial transpose in the first factor (crossing symmetry).

- (d) Define $L_j(u) = R_{0j}(u)$, $\mathcal{T}(u) = L_N(u) \cdots L_1(u)$, $t(u) = \text{Tr}_0 \mathcal{T}(u)$ on $\mathcal{H}_N$. Prove the RLL relation and $[t(u), t(v)] = 0$ exactly as in the section Commuting families in Part IV, now without conservation of S^z but with $[t(u), \sigma_1^z \cdots \sigma_N^z] = 0$ and $[t(u), \sigma_1^x \cdots \sigma_N^x] = 0$; show that the row-to-row transfer matrix of the eight-vertex model on the torus is $t(u)$ for the appropriate normalization, so that on an N -site row and M rows, $Z = \text{Tr } t(u)^M$.

Exercise 29.4 (The XYZ chain from the logarithmic derivative and the hierarchy of degenerations).

- (a) Repeating the derivation of the section Rational/trigonometric hierarchy in Part V with $a(0)$ in place of $\sinh \eta$, prove

$$t(0)^{-1} t'(0) = \frac{1}{a(0)} \sum_{j=1}^N P_{j,j+1} R'_{j,j+1}(0),$$

$$PR'(0) = \frac{a' + c'}{2} + \frac{a' - c'}{2} \sigma^z \otimes \sigma^z$$

$$+ \frac{b' + d'}{2} \sigma^x \otimes \sigma^x + \frac{b' - d'}{2} \sigma^y \otimes \sigma^y.$$

with $a' = a'(0)$ etc.

- (b) Compute $a'(0) = \vartheta_4(\eta) \vartheta_4(0) \vartheta_1'(\eta)$, $b'(0) = \vartheta_4(\eta)^2 \vartheta_1'(0)$, $c'(0) = \vartheta_1(\eta) \vartheta_4(0) \vartheta_4'(\eta)$, $d'(0) = \vartheta_1(\eta)^2 \vartheta_4'(0)$, and, using the Wronskian identity $(\vartheta_1/\vartheta_4)' = \vartheta_4(0)^2 \vartheta_2 \vartheta_3 / \vartheta_4^2$ and $\vartheta_1'(0) = \vartheta_2(0) \vartheta_3(0) \vartheta_4(0)$, prove that

$$H_{XYZ} = \sum_{j=1}^N (J_x \sigma_j^x \sigma_{j+1}^x + J_y \sigma_j^y \sigma_{j+1}^y + J_z \sigma_j^z \sigma_{j+1}^z), \quad J_x : J_y : J_z = 1 : \Gamma : \Delta,$$

is, up to the additive constant $N(a' + c')/(2a(0))$ and the factor $2a(0)/(b' + d')$, the logarithmic derivative of $t(u)$ at $u = 0$; hence $[H_{XYZ}, t(u)] = 0$.

- (c) Hierarchy. Show: $p \rightarrow 0$ gives $\Gamma = 1$, $J_x = J_y$, the XXZ chain with $\Delta = \cos \eta$ (after $\eta \rightarrow i\eta$, the $\Delta = \cosh \eta$ form of Part V); the further limit $\eta \rightarrow 0$ with u/η fixed gives the XXX chain and the Yang R -matrix; and the limits $k \rightarrow 1$ of the modulus (via $\tau \mapsto -1/\tau$) give the hyperbolic (massive XXZ) forms. State the resulting inclusions rational $\subset$ trigonometric $\subset$ elliptic of solutions of the Yang–Baxter equation.

30 Fusion

Exercise 30.1 (The fusion procedure: degeneration of R at $u = -\eta$ and the fused R -matrix on $V_2 \otimes V_1$). Let $R(u)$ be either the six-vertex R -matrix of Part IV or the elliptic eight-vertex R -matrix, both with $R(0) \propto P$. Write $P^\pm = \frac{1}{2}(1 \pm P)$ for the projections onto the symmetric and antisymmetric parts of $V \otimes V$.

- (a) Prove $R(-\eta) = 2b(-\eta)P^-$ in both cases (six-vertex: $a(-\eta) = 0$, $c(-\eta) = -b(-\eta)$; eight-vertex: $a(-\eta) = d(-\eta) = 0$, $c(-\eta) = -b(-\eta)$).
- (b) From the Yang–Baxter equation at $u - v = -\eta$ deduce $P_{12}^- R_{13}(u) R_{23}(u + \eta) = R_{23}(u + \eta) R_{13}(u) P_{12}^-$ and hence that $R_{13}(u) R_{23}(u + \eta)$ maps $(\text{im } P_{12}^+) \otimes V$ into itself and $R_{23}(u + \eta) R_{13}(u)$ maps $(\text{im } P_{12}^-) \otimes V$ into itself.
- (c) Define the fused R -matrix $R^{(2,1)}(u) = R_{13}(u) R_{23}(u + \eta)|_{(\text{im } P_{12}^+) \otimes V} \in \text{End}(\text{Sym}^2 V \otimes V)$ and prove the mixed Yang–Baxter equation

$$R_{A3}^{(2,1)}(u - v) R_{A4}^{(2,1)}(u) R_{34}(v) = R_{34}(v) R_{A4}^{(2,1)}(u) R_{A3}^{(2,1)}(u - v)$$

on $\text{Sym}^2 V \otimes V \otimes V$ (A the fused pair of factors 1, 2), by expanding each fused matrix, commuting factors on disjoint spaces, and applying the Yang–Baxter equation twice.

- (d) Quantum determinant. Prove that $R_{23}(u + \eta)R_{13}(u)$ acts on $(\text{im } P_{12}^-) \otimes V$ as the scalar $\delta(u) = b(u)a(u + \eta) - d(u)d(u + \eta) = a(u)b(u + \eta) - c(u)c(u + \eta)$ (equal expressions by the identities of the eight-vertex section; for the six-vertex case $\delta(u) = \sinh u \sinh(u + 2\eta)$).

Exercise 30.2 (The fusion hierarchy of transfer matrices and the first functional relation). Let $\mathcal{T}(u) = L_N(u) \cdots L_1(u)$ with $L_j(u) = R_{0j}(u)$ and $t(u) = \text{Tr}_0 \mathcal{T}(u)$. Define the fused monodromy $\mathcal{T}^{(2)}(u) = \mathcal{T}_0(u)\mathcal{T}_{0'}(u + \eta)|_{\text{im } P_{00'}^+ \otimes \mathcal{H}_N}$ and $t^{(2)}(u) = \text{Tr}_{\text{Sym}^2 V} \mathcal{T}^{(2)}(u)$.

- (a) Prove that $\mathcal{T}_0(u)\mathcal{T}_{0'}(u + \eta) = \prod_{j=N}^1 (L_{j,0}(u)L_{j,0'}(u + \eta))$ (factors with distinct j commute), that $P_{00'}^- \mathcal{T}_0(u)\mathcal{T}_{0'}(u + \eta) = \mathcal{T}_{0'}(u + \eta)\mathcal{T}_0(u)P_{00'}^-$, and that $\mathcal{T}_{0'}(u + \eta)\mathcal{T}_0(u)$ acts on $\text{im } P_{00'}^- \otimes \mathcal{H}_N$ as the scalar $\delta(u)^N$.
- (b) Prove the functional relation

$$t(u)t(u + \eta) = t^{(2)}(u) + \delta(u)^N,$$

by decomposing the trace over $V_0 \otimes V_{0'}$ according to $\text{im } P^+ \oplus \text{im } P^-$ and using that the operator $\mathcal{T}_0(u)\mathcal{T}_{0'}(u + \eta)$ is block triangular for this decomposition.

- (c) Deduce $[t^{(2)}(u), t(v)] = 0$ and $[t^{(2)}(u), t^{(2)}(v)] = 0$. If a common eigenspace of the $t(u)$ has eigenvalue function $\Lambda(u)$, compute the eigenvalue of $t^{(2)}(u)$ as $\Lambda(u)\Lambda(u + \eta) - \delta(u)^N$.

31 Quantum dimensions

Exercise 31.1 (Restricted fusion and quantum dimensions of RSOS heights). Put $\gamma = \pi/(k + 2)$, $q = e^{i\gamma}$, and $d_r = \sin((r + 1)\gamma)/\sin \gamma$ for $0 \leq r \leq k$. These are the quantum dimensions of the height labels $r + 1$; they are the specialization of the weighted traces $\text{tr}_q(\text{id}_{V_r})$ of the section $U_q(\mathfrak{sl}_2)$. Define $U_0(x) = 1$, $U_1(x) = 2x$, $U_{r+1}(x) = 2xU_r(x) - U_{r-1}(x)$ and $N_r = U_r(A/2)$.

- (a) Show that $\psi(a) = \sin(a\gamma)$ satisfies $A\psi = 2 \cos \gamma \psi$, that $d_0 = 1$, and $d_1 d_r = d_{r-1} + d_{r+1}$ with $d_{-1} = d_{k+1} = 0$.
- (b) Define $S_{ar} = \sqrt{2/(k + 2)} \sin(a(r + 1)\gamma)$, $1 \leq a \leq k + 1$, $0 \leq r \leq k$. Sum geometric series to prove $\sum_a S_{ar} S_{as} = \delta_{rs}$ and the inversion $\sum_r S_{ar} S_{br} = \delta_{ab}$. Deduce the kernel resolution

$$(A^N)_{ab} = \sum_{r=0}^k S_{ar} S_{br} (2 \cos((r + 1)\gamma))^N.$$

- (c) Use the Chebyshev recurrence and $U_{k+1}(A/2) = 0$ to prove

$$N_r N_s = \sum_{\substack{t=|r-s| \\ t \equiv r+s \pmod{2}}}^{\min(r+s, 2k-r-s)} N_t.$$

Show by induction that all N_r have nonnegative integer entries and that evaluation on ψ gives $d_r d_s = \sum_t N_{rs}^t d_t$ with the displayed fusion coefficients.

Exercise 31.2 (Critical RSOS face weights and the Temperley–Lieb identity). For an admissible height path $(a_0, \dots, a_N)$ define

$$(e_i f)(a) = \mathbf{1}_{a_{i-1}=a_{i+1}} \sum_{b: |b-a_{i-1}|=1} \frac{\sqrt{\psi(a_i)\psi(b)}}{\psi(a_{i-1})} f(a_0, \dots, a_{i-1}, b, a_{i+1}, \dots, a_N),$$

with ψ from the preceding exercise and $1 \leq i < N$.

- (a) Prove $e_i^* = e_i$, $e_i^2 = d_1 e_i$, $e_i e_{i+1} e_i = e_i$, and distant commutativity. Identify e_i/d_1 as a rank-one orthogonal projection on each returning-path fibre.
- (b) Put $X_i(u) = \sin(\gamma - u) \text{id} + \sin u e_i$. Prove positivity of its admissible kernel for $0 < u < \gamma$ and derive the face Yang–Baxter equation from the Baxterization exercise of the section Hecke / Temperley–Lieb algebras by analytic continuation.

32 Elliptic quantum groups

Exercise 32.1 (A dynamical elliptic vertex operator). Write $\theta = \vartheta_1(\cdot|\tau)$ and $s(a, z) = \theta(a - z)\theta'(0)/(\theta(a)\theta(z))$. For $\gamma \neq 0$ define a meromorphic arrow kernel $\mathcal{R}(u, \lambda)$ equal to one on $++$ and $--$, with mixed-arrow block

$$\frac{1}{s(\gamma, u)} \begin{pmatrix} s(\gamma, \lambda) & s(\lambda, u) \\ s(-\lambda, u) & s(\gamma, -\lambda) \end{pmatrix}.$$

Set $\epsilon = \gamma/2$, and interpret $f(\lambda + \epsilon Z_j)$ through the two spectral projections of the physical arrow involution Z_j .

- (a) Use the theta addition identity to verify

$$\begin{aligned} & \mathcal{R}_{12}(u - v, \lambda + \epsilon Z_3) \mathcal{R}_{13}(u, \lambda - \epsilon Z_2) \mathcal{R}_{23}(v, \lambda + \epsilon Z_1) \\ &= \mathcal{R}_{23}(v, \lambda - \epsilon Z_1) \mathcal{R}_{13}(u, \lambda + \epsilon Z_2) \mathcal{R}_{12}(u - v, \lambda - \epsilon Z_3). \end{aligned}$$

Work away from poles and extend meromorphically.

- (b) Define a representation of this elliptic exchange relation to be a weight space W and an operator $L(u, \lambda)$ satisfying the same equation with the last factor replaced by W and $\mathcal{R}_{13}, \mathcal{R}_{23}$ replaced by L_{13}, L_{23} . Verify that $W = V$, $L(u, \lambda) = \mathcal{R}(u - a, \lambda)$ is such a representation for each a . Compare conventions with Felder [5].

Exercise 32.2 (Prediction: residual entropy of a constrained RSOS strip). Use the uniform RSOS chain of the section Theta and elliptic functions, with N bonds, fixed initial height a , and freely summed terminal height. Let $Z_N(a) = \sum_b (A^N)_{ab}$, and define the residual entropy per bond as $s_0 = k_B \lim_{N \rightarrow \infty} N^{-1} \log Z_N(a)$.

- (a) Since ψ is strictly positive, bound $Z_N(a)$ above and below by positive constants times $(2 \cos \gamma)^N \psi(a)$, using $\sum_b (A^N)_{ab} \psi(b) = (2 \cos \gamma)^N \psi(a)$. Derive

$$s_0 = k_B \log(2 \cos(\pi/(k+2))).$$

Explain why summing the endpoint avoids the parity zeros of a fixed-endpoint partition function.

- (b) Obtain $s_0 = 0$ for $k = 1$, $s_0 = \frac{1}{2} k_B \log 2$ for $k = 2$, and $s_0 = k_B \log((1 + \sqrt{5})/2)$ for $k = 3$. Derive the physical free energy per bond $-T s_0$ for the exactly degenerate allowed configurations.
- (c) Use the spectral kernel to calculate finite- N corrections. Specify that an entropy measurement tests this counting law only when forbidden neighbouring heights cost energy much larger than $k_B T$ and energy splittings among allowed configurations are much smaller than $k_B T$.

Part VII

Loop Models / Anyonic Chains

33 Temperley–Lieb

Exercise 33.1 (The Temperley–Lieb diagram category and the diagram basis of $TL_n(d)$). Fix $d \in \mathbb{R}$. A *planar pairing* from n to m is an isotopy class of $(n+m)/2$ pairwise disjoint smooth arcs in the strip $[0, 1] \times [0, 1]$ whose endpoints are exactly the points $(i/(n+1), 0)$, $1 \leq i \leq n$, and $(j/(m+1), 1)$, $1 \leq j \leq m$; it exists only when $n+m$ is even. Define the category $\mathcal{TL}(d)$: objects are the integers $n \geq 0$; $\text{Hom}(n, m)$ is the complex vector space with basis the planar pairings from n to m ; the composite $g \circ f$ of pairings $f: n \rightarrow m$ and $g: m \rightarrow p$ is obtained by stacking g on top of f , rescaling, deleting each closed loop so formed, and multiplying by d for each deleted loop; composition is extended bilinearly. The tensor product $n \otimes m = n + m$ places diagrams side by side. The involution $f \mapsto f^*$ reflects a pairing in the line $\{t = 1/2\}$ and is extended antilinearly. Write $\cap \in \text{Hom}(2, 0)$ and $\cup \in \text{Hom}(0, 2)$ for the single arc, 1_n for the identity pairing, and $e_i = 1_{i-1} \otimes (\cup \circ \cap) \otimes 1_{n-i-1} \in \text{End}(n)$ for $1 \leq i < n$.

Prove that composition is associative and unital, that $\otimes$ is a strict monoidal structure with $(g \circ f) \otimes (g' \circ f') = (g \otimes g') \circ (f \otimes f')$, and that $*$ is an antilinear contravariant monoidal involution. Prove the snake identities $(\cap \otimes 1_1) \circ (1_1 \otimes \cup) = 1_1 = (1_1 \otimes \cap) \circ (\cup \otimes 1_1)$ and that $e_i^2 = de_i$, $e_i e_{i \pm 1} e_i = e_i$, $e_i e_j = e_j e_i$ for $|i - j| \geq 2$. Deduce a homomorphism $\phi_n: TL_n(d) \rightarrow \text{End}(n)$ from the algebra $TL_n(d)$ of the section Hecke / Temperley–Lieb algebras in Part V. Prove that every planar pairing $n \rightarrow n$ other than 1_n is a product of the e_i , by induction on the number of arcs joining two bottom points: choose an innermost such arc, joining adjacent points $i, i+1$, and factor the diagram through e_i . Prove that the number of planar pairings from n to n is the Catalan number $C_n = \binom{2n}{n}/(n+1)$ by exhibiting a bijection with planar pairings of $2n$ points on a line (rotate the bottom row to the right of the top row) and deriving the recursion $C_n = \sum_k C_{k-1} C_{n-k}$ from the arc through the first point. Using the dimension C_n of $TL_n(d)$ established in Part V, conclude that ϕ_n is an isomorphism and that the planar pairings form a basis of $TL_n(d)$, the *diagram basis*.

Exercise 33.2 (Jones–Wenzl idempotents). Let $q \in \mathbb{C}^\times$ with $d = q + q^{-1}$ and $[n]_q = (q^n - q^{-n})/(q - q^{-1})$, so $[2]_q = d$ and $[n+1]_q = d[n]_q - [n-1]_q$. Assume $[k]_q \neq 0$ for $1 \leq k \leq n$. Define $f_0 = 1_0$, $f_1 = 1_1$ and, for $1 \leq m < n$,

$$f_{m+1} = f_m \otimes 1_1 - \frac{[m]_q}{[m+1]_q} (f_m \otimes 1_1) e_m (f_m \otimes 1_1) \in TL_{m+1}(d).$$

Prove by simultaneous induction on $m \leq n$ the following statements. (i) f_m is an idempotent with $e_i f_m = f_m e_i = 0$ for $1 \leq i < m$, and $f_m - 1_m$ lies in the two-sided ideal of $TL_m(d)$ generated by $e_1, \dots, e_{m-1}$. (ii) f_m is the unique element with these two properties. (iii) The partial closure identity

$$(1_{m-1} \otimes \cap) (f_m \otimes 1_1) (1_{m-1} \otimes \cup) = \frac{[m+1]_q}{[m]_q} f_{m-1}$$

holds in $TL_{m-1}(d)$. (iv) $f_m(f_{m-1} \otimes 1_1) = f_m = (f_{m-1} \otimes 1_1)f_m$ and $f_m TL_m(d) f_m = \mathbb{C} f_m$; in particular $\text{End}(f_m) = \mathbb{C}$ in the additive idempotent completion of $\mathcal{TL}(d)$, whose objects are finite direct sums of pairs (n, p) with $p \in \text{End}(n)$ idempotent and whose morphisms $(n, p) \rightarrow (n', p')$ are the elements $p' \text{Hom}(n, n') p$.

For the induction step of (iii) expand f_{m+1} by its definition, use $e_m(f_m \otimes 1_1)e_m = (1_{m-1} \otimes \cup) [(1_{m-1} \otimes \cap)(f_m \otimes 1_1)(1_{m-1} \otimes \cup)] (1_{m-1} \otimes \cap)$ together with the induction hypothesis, and simplify with $[m+2]_q = d[m+1]_q - [m]_q$.

Put $p = \frac{[m]_q}{[m+1]_q} (f_m \otimes 1_1)e_m(f_m \otimes 1_1)$ for $m < n$, so that $f_m \otimes 1_1 = f_{m+1} + p$. Prove that p is an idempotent orthogonal to f_{m+1} and that the morphisms $\iota = \frac{[m]_q}{[m+1]_q} (f_m \otimes 1_1)(1_{m-1} \otimes \cup): (m-1, f_{m-1}) \rightarrow (m+1, p)$ and $\pi = (1_{m-1} \otimes \cap)(f_m \otimes 1_1): (m+1, p) \rightarrow (m-1, f_{m-1})$ satisfy $\pi\iota = f_{m-1}$, $\iota\pi = p$. Conclude the isomorphism $(m, f_m) \otimes (1, f_1) \simeq (m-1, f_{m-1}) \oplus (m+1, f_{m+1})$ in the additive idempotent completion, for $1 \leq m < n$.

Exercise 33.3 (The Markov trace, negligible morphisms, and the semisimplification at roots of unity). Define the *Markov trace* $\text{tr}_n: TL_n(d) \rightarrow \mathbb{C}$ on a planar pairing $x: n \rightarrow n$ by joining the i -th top point to the i -th bottom point by n nested arcs to the right of the diagram and setting $\text{tr}_n(x) = d^{\ell(x)}$, $\ell(x)$ the number of closed loops so obtained. Prove that $\text{tr}_n(xy) = \text{tr}_n(yx)$, $\text{tr}_{n+1}(x \otimes 1_1) = d \text{tr}_n(x)$, $\text{tr}_{n+1}(xe_n) = \text{tr}_n(x)$ for $x \in TL_n(d)$, $\text{tr}_{n+m}(x \otimes y) = \text{tr}_n(x) \text{tr}_m(y)$, and $\text{tr}_n(x^*) = \text{tr}_n(\overline{x})$ for real d . Prove from the partial closure identity of the preceding exercise that $\text{tr}_m(f_m) = [m+1]_q$ whenever f_m is defined.

Call $x \in \text{Hom}(n, m)$ *negligible* if $\text{tr}_n(yx) = 0$ for all $y \in \text{Hom}(m, n)$. Prove that negligible morphisms form a two-sided ideal $\mathcal{N}$ closed under $\otimes$ with arbitrary morphisms and under $*$ (for the tensor property, write $\text{tr}_{n+p}(z(x \otimes 1_p))$ as $\text{tr}_n(z'x)$ with z' a partial closure of z), so that the quotient $\mathcal{TL}(d)/\mathcal{N}$ is again a $\mathbb{C}$ -linear monoidal category with involution.

Now let $k \geq 1$ and $q = e^{i\pi/(k+2)}$, $d = 2 \cos(\pi/(k+2))$, so that $[m]_q = \sin(m\pi/(k+2))/\sin(\pi/(k+2)) > 0$ for $1 \leq m \leq k+1$ and $[k+2]_q = 0$. Prove that f_{k+1} is negligible: for $y \in TL_{k+1}(d)$ write $f_{k+1}yf_{k+1} = \lambda(y)f_{k+1}$ and compute $\text{tr}(f_{k+1}y) = \lambda(y)[k+2]_q$. Prove that in the additive idempotent completion of $\mathcal{TL}(d)/\mathcal{N}$: (a) $\text{Hom}((a, f_a), (b, f_b)) = 0$ for $a \neq b$ in $\{0, \dots, k\}$ (every pairing $a \rightarrow b$ with $a \neq b$ has an arc joining two bottom or two top points); (b) $\text{End}((a, f_a)) = \mathbb{C}$; (c) every object $(n, 1_n)$ is isomorphic to a direct sum of objects (a, f_a) , $0 \leq a \leq k$, by induction on n using $(m, f_m) \otimes (1, f_1) \simeq (m-1, f_{m-1}) \oplus (m+1, f_{m+1})$ and $(k+1, f_{k+1}) \simeq 0$. Write $\mathcal{C}_k$ for this category and a for the object (a, f_a) ; call the objects $0, \dots, k$ simple and $\mathcal{C}_k$ semisimple. Prove the fusion rule

$$a \otimes b \simeq \bigoplus_{\substack{c \equiv a+b \pmod{2} \\ |a-b| \leq c \leq \min(a+b, 2k-a-b)}} c$$

by induction on b , using $a \otimes (b \otimes 1) \simeq (a \otimes b) \otimes 1$ and the rule for $\otimes 1$. Compare with the fusion rules of the A_{k+1} RSOS heights of the section Quantum dimensions in Part VI, and verify $\text{tr}(f_a) = [a+1]_q$ against the quantum dimensions $d_a = [a+1]_q$ computed there.

Exercise 33.4 (The dense loop model and its six-vertex representation). Let Λ be the square lattice rotated by $\pi/4$, drawn on the strip $\{1, \dots, L\} \times \{1, \dots, N\}$ so that each vertex has two edges entering from below (south-west, south-east) and two leaving above (north-west, north-east). A *dense loop configuration* chooses at every vertex one of the two non-crossing pairings of its four half-edges: the pairing $\{\text{SW}, \text{NW}\}, \{\text{SE}, \text{NE}\}$ with weight $\rho_1 > 0$ or the pairing $\{\text{SW}, \text{SE}\}, \{\text{NW}, \text{NE}\}$ with weight $\rho_2 > 0$. The union of the chosen arcs is a disjoint union of loops and of open arcs ending on the boundary. With free boundary in the horizontal direction and the strands joined across the top and bottom as in the Markov closure, define

$$Z_{L,N}(n; \rho_1, \rho_2) = \sum_{\text{configurations}} \rho_1^{\#\{\rho_1\text{-vertices}\}} \rho_2^{\#\{\rho_2\text{-vertices}\}} n^{\#\{\text{loops}\}}.$$

Prove that $Z_{L,N} = \text{tr}_L(T^N)$ with $d = n$ and

$$T = \prod_{i \text{ even}} (\rho_1 + \rho_2 e_i) \prod_{i \text{ odd}} (\rho_1 + \rho_2 e_i) \in TL_L(n)$$

(the two products over the two sublattices of vertices of a double row), identifying the vertex pairings with 1 and e_i .

Now let $n = 2 \cos \gamma$ with $0 \leq \gamma < \pi$ and orient every loop in one of its two directions. Give each oriented loop the weight $e^{i\gamma\omega/(2\pi)}$ where $\omega \in \{\pm 2\pi\}$ is its total turning angle, and prove that summing over the two orientations of a contractible loop yields n . Draw the local pairings with quarter-circle arcs. Prove that each passage through such an arc turns by $\pm\pi/2$, so that the loop weight factorizes as a product over vertices of $e^{\pm i\gamma/4}$ per turn. Orienting each edge by its loop, prove that every vertex has two entering and two leaving arrows (the ice rule of Part IV), and that the sum over pairings compatible with a given arrow configuration, weighted by ρ_i and the turn factors, produces the six-vertex weights

$$a = \rho_1, \quad b = \rho_2, \quad c_{\pm} = \rho_1 e^{\pm i\gamma/2} + \rho_2 e^{\mp i\gamma/2},$$

where a (b) is the weight of the two vertices at which both lines pass straight with parallel (antiparallel) orientation and $c_{\pm}$ are the weights of the two vertices with sources or sinks along a line. Prove that in every configuration on a domain with prescribed boundary arrows the difference between the numbers of c_+ and c_- vertices is determined by the boundary, so that Z , after division by the boundary factor $(c_+/c_-)^{(N_+ - N_-)/2}$, depends on $c_{\pm}$ only through $c^2 := c_+ c_-$, and compute

$$\Delta = \frac{a^2 + b^2 - c^2}{2ab} = -\cos \gamma = -\frac{n}{2}.$$

State precisely which loops receive the weight n under this correspondence (contractible loops) and which receive the weight 2 (loops whose turning angle vanishes, that is, loops winding around the periodic direction), and conclude the equality of the dense loop partition function on the plane with the six-vertex partition function of Part IV at $\Delta = -n/2$, for $0 < n \leq 2$.

34 Fusion graphs

Exercise 34.1 (Fusion rings, fusion matrices, and fusion graphs). A *fusion ring* is a ring K that is free as a $\mathbb{Z}$ -module on a finite set $B \ni 1$ containing the unit, such that the structure constants $ab = \sum_{c \in B} N_{ab}^c c$ are nonnegative integers and such that there is an involution $a \mapsto \bar{a}$ of B , extending to an anti-automorphism of K , with $N_{ab}^1 = \delta_{b, \bar{a}}$. Elements of B are the simple elements. Prove the reciprocity identities $N_{ab}^c = N_{\bar{a}\bar{c}}^{\bar{b}} = N_{cb}^{\bar{a}} = N_{\bar{b}\bar{a}}^{\bar{c}}$ (write N_{ab}^c as the coefficient of 1 in $(ab)\bar{c}$ and reassociate). Define the *fusion matrices* $N_a \in \text{Mat}_B(\mathbb{Z}_{\geq 0})$ by $(N_a)_{cb} = N_{ab}^c$, so that N_a is the matrix of left multiplication by a in the basis B ; prove $N_a N_b = \sum_c N_{ab}^c N_c$ and $N_{\bar{a}} = N_a^T$. Define the *fusion graph* Γ_x of $x \in B$: vertices B and N_{xa}^b edges from a to b ; it is undirected when $\bar{x} = x$. Call x a *generator* if every $b \in B$ occurs in some power x^m , and prove that x is a generator if and only if Γ_x is strongly connected.

Prove that K is commutative if and only if all N_a commute. For the fusion rule of the section Temperley–Lieb at level k (simples $0, \dots, k$, $\bar{a} = a$), prove that it defines a commutative fusion ring, that 1 is a generator, and that Γ_1 is the path graph with vertices $0, \dots, k$ and edges $\{a, a+1\}$, that is, the Dynkin diagram A_{k+1} of the RSOS heights $a+1 \in \{1, \dots, k+1\}$ of Part VI. Prove that the ring $\mathbb{Z}[x]/(U_{k+1}(x/2))$, with U_m the Chebyshev polynomial of the second kind $U_m(\cos \theta) = \sin((m+1)\theta)/\sin \theta$, is isomorphic to this fusion ring via $x \mapsto 1$, with $a \mapsto U_a(x/2)$.

Exercise 34.2 (Anyonic-chain Hilbert spaces and the Temperley–Lieb path representation). Let K be a commutative fusion ring, $x = \bar{x} \in B$ a generator with $N_{xa}^b \in \{0, 1\}$ for all a, b , and $\Gamma = \Gamma_x$. For $a, b \in B$ and $N \geq 0$ define the *anyonic-chain Hilbert space*

$$\mathcal{H}_N(a, b) = \text{span}_{\mathbb{C}}\{(a_0, \dots, a_N) : a_0 = a, a_N = b, N_{xa_{i-1}}^{a_i} = 1 \text{ for } 1 \leq i \leq N\},$$

the space of functions on admissible paths of length N in Γ from a to b , with the paths orthonormal. Prove $\dim \mathcal{H}_N(a, b) = (N_x^N)_{ba}$ and $\sum_b \dim \mathcal{H}_N(a, b) = \|N_x^N \delta_a\|_1$.

Let $\psi : B \rightarrow (0, \infty)$ satisfy $N_x \psi = \lambda \psi$, that is, $\sum_b N_{xa}^b \psi_b = \lambda \psi_a$ for all a . For $1 \leq i \leq N-1$ define e_i on $\mathcal{H}_N(a, b)$ by

$$(e_i \Psi)(a_0, \dots, a_N) = \delta_{a_{i-1}, a_{i+1}} \sum_{c: N_{xa_{i-1}}^c = 1} \frac{\sqrt{\psi_{a_i} \psi_c}}{\psi_{a_{i-1}}} \Psi(a_0, \dots, a_{i-1}, c, a_{i+1}, \dots, a_N).$$

Prove that each e_i is self-adjoint and that $e_i^2 = \lambda e_i$, $e_i e_{i \pm 1} e_i = e_i$, $e_i e_j = e_j e_i$ for $|i - j| \geq 2$, so that $\mathcal{H}_N(a, b)$ is a $*$ -representation of $TL_N(\lambda)$, the *path representation*. For the level- k fusion ring with $x = 1$ and $\psi_a = \sin(\pi(a+1)/(k+2))$, the sine eigenvector of Part VI with $\lambda = 2 \cos(\pi/(k+2))$, verify that e_i/λ is the orthogonal projector onto the subspace of paths in which the two steps $a_{i-1} \rightarrow a_i \rightarrow a_{i+1}$ return to a_{i-1} with the intermediate label weighted by $\sqrt{\psi_{a_i}}$; that is, e_i/λ is a rank-one projector on each fibre $\{(a_{i-1}, \cdot, a_{i+1}) : a_{i-1} = a_{i+1}\}$ and zero on the other fibres. Call $P_i^{(1)} = e_i/\lambda$ the projector onto the trivial channel at bond i : it is the orthogonal projector onto the states in which the two x -labels at positions $i, i+1$ fuse to 1. Prove $\text{Tr } P_i^{(1)} = \sum_c (N_x^{i-1})_{ca} (N_x^{N-i-1})_{bc} = (N_x^{N-2})_{ba}$ by counting the fibres on which $P_i^{(1)} \neq 0$.

Exercise 34.3 (The golden chain as a Temperley–Lieb Hamiltonian). The *Fibonacci fusion ring* has $B = \{1, \tau\}$, $\bar{\tau} = \tau$, and $\tau^2 = 1 + \tau$. Prove it is a fusion ring, that τ is a generator, that Γ_τ has the single edge $\{1, \tau\}$ and a loop at τ , that $\psi = (1, \varphi)$ with $\varphi = (1 + \sqrt{5})/2$ is a positive eigenvector of N_τ with eigenvalue φ , and that $\dim \mathcal{H}_N(1, \tau) = F_N$, $\dim \mathcal{H}_N(1, 1) = F_{N-1}$ for $N \geq 1$, with $F_0 = 0$, $F_1 = F_2 = 1$, $F_{N+1} = F_N + F_{N-1}$. Prove that the map $1 \mapsto 0$, $\tau \mapsto 2$ identifies the Fibonacci ring with the subring of the level-3 ring of the section Temperley–Lieb spanned by the even simples $0, 2$, and that $\varphi = [3]_q$ at $q = e^{i\pi/5}$.

The *golden chain* is $H = -\sum_{i=1}^{N-1} P_i^{(1)}$ on $\mathcal{H}_N(a, b)$ (open chain) or on the space of closed admissible paths $(a_0, \dots, a_N = a_0)$ with i taken modulo N (periodic chain). Prove that $H = -\varphi^{-1} \sum_i e_i$ with e_i the path representation of $TL_N(\varphi)$, and write out the 2×2 block of $P_i^{(1)}$ on the fibre $a_{i-1} = a_{i+1} = \tau$, $a_i \in \{1, \tau\}$, and its value on the fibre $a_{i-1} = a_{i+1} = 1$:

$$P_i^{(1)}|_{(\tau, \cdot, \tau)} = \begin{pmatrix} \varphi^{-2} & \varphi^{-3/2} \\ \varphi^{-3/2} & \varphi^{-1} \end{pmatrix}, \quad P_i^{(1)}|_{(1, \tau, 1)} = 1,$$

and $P_i^{(1)} = 0$ on paths with $a_{i-1} \neq a_{i+1}$. Verify $\varphi^{-1} + \varphi^{-2} = 1$ and the projector property directly. Prove that path reversal intertwines the Hamiltonians on $\mathcal{H}_N(a, b)$ and $\mathcal{H}_N(b, a)$, and that on the periodic chain H commutes with the translation $(a_i) \mapsto (a_{i+1})$. Compute the spectrum of H for $N = 3$ on $\mathcal{H}_3(1, \tau)$ and on $\mathcal{H}_3(\tau, \tau)$.

35 Perron–Frobenius

Exercise 35.1 (The Perron–Frobenius theorem for irreducible nonnegative matrices). Let B be a finite vertex set of size n and $A : \mathbb{R}^B \rightarrow \mathbb{R}^B$ a kernel with nonnegative values. Call A *irreducible* if

for all i, j there is $m \geq 1$ with $(A^m)_{ij} > 0$, and *primitive* if there is $m \geq 1$ with all entries of A^m positive. Write $x \geq 0$ and $x > 0$ for entrywise inequalities on vectors, $|y|$ for the vector of absolute values, and $r(A) = \max\{|\lambda| : \lambda \in \text{spec } A\}$. Assume A irreducible.

- (a) Prove that $P = (I + A)^{n-1}$ has all entries positive, and that A has no zero row and no zero column.
- (b) For $x \geq 0$, $x \neq 0$, put $f(x) = \min\{(Ax)_i/x_i : x_i > 0\}$. Prove $f(tx) = f(x)$ for $t > 0$, $f(Px) \geq f(x)$, and that f is continuous on $\{x > 0\}$. Prove that $r := \sup\{f(x) : x \geq 0, x \neq 0\}$ is attained: the supremum over the simplex $\{x \geq 0, \sum_i x_i = 1\}$ is bounded by the supremum over its image under P , a compact subset of $\{x > 0\}$.
- (c) Let x attain r . Prove $Ax \geq rx$; if $Ax \neq rx$ then $P(Ax - rx) > 0$, hence $A(Px) > rPx$ and $f(Px) > r$, a contradiction. Conclude $Ax = rx$, $x = (1 + r)^{-(n-1)}Px > 0$, and $r > 0$.
- (d) Prove that every eigenvalue λ of A satisfies $|\lambda| \leq r$: if $Ay = \lambda y$ then $A|y| \geq |\lambda||y|$, so $f(|y|) \geq |\lambda|$. Prove also that $Ay \geq \rho y$ with $y \geq 0$, $y \neq 0$ implies $\rho \leq r$, and $Ay \leq \rho y$ with $y > 0$ implies $r \leq \rho$ (apply the first part to $A^\top$, which is irreducible, and pair with its positive eigenvector). Hence $r = r(A)$.
- (e) Prove that the eigenspace of r is one-dimensional: if $Ay = ry$ with y real, choose $t \in \mathbb{R}$ such that $x - ty \geq 0$ has a zero entry; then $A(x - ty) = r(x - ty)$, and $x - ty \neq 0$ would give $P(x - ty) = (1 + r)^{n-1}(x - ty) > 0$. Treat complex y through its real and imaginary parts.
- (f) Prove that r is a simple root of the characteristic polynomial: let $z > 0$ satisfy $A^\top z = rz$; if $(A - r)w = x$ for some w , then $z^\top x = z^\top (A - r)w = 0$, contradicting $z^\top x > 0$.
- (g) Prove that every eigenvector $y \geq 0$ of A is a positive multiple of x : pair $Ay = \lambda y$ with z .
- (h) Assume A primitive with $A^m > 0$. Prove that $|\lambda| < r$ for every eigenvalue $\lambda \neq r$: if $Ay = \lambda y$ with $|\lambda| = r$, then $A|y| \geq r|y|$ forces $A|y| = r|y|$ by (c) applied to $|y|$, so $|y| = cx > 0$; then $|A^m y| = A^m |y|$ with $A^m > 0$ forces all entries of y to have the same argument, so $y \in \mathbb{C}x$ and $\lambda = r$. Deduce $\lim_{m \rightarrow \infty} r^{-m} A^m = xz^\top / (z^\top x)$, with an $O(\rho^m)$ error for every $\rho > |\lambda_1|/r$, $\rho < 1$, where λ_1 is a second-largest eigenvalue in modulus; account for possible Jordan blocks.
- (i) Prove that an irreducible matrix with a positive diagonal entry is primitive, and that for $n > 1$ the cyclic permutation kernel of an n -cycle is irreducible, not primitive, and has all n -th roots of unity as eigenvalues.

Exercise 35.2 (Frobenius–Perron dimensions of a fusion ring and the growth of anyonic chains). Let K be a fusion ring with simples B , N_a the matrix of left multiplication by a , and R_b the matrix of right multiplication by b , $(R_b)_{ca} = N_{ab}^c$.

- (a) Prove $N_a R_b = R_b N_a$ for all a, b (associativity), and that $M = \sum_{b \in B} R_b$ has all entries positive: for $c, e \in B$ show $\bar{c}e \neq 0$ by multiplying on the right by $\bar{e}$, and use $N_{cb}^e = N_{\bar{c}\bar{e}}^b$.
- (b) Let $v > 0$ be the Perron–Frobenius eigenvector of M . Prove that $N_a v$ is a nonnegative nonzero eigenvector of M with eigenvalue $r(M)$, hence $N_a v = \lambda_a v$ with $\lambda_a > 0$, and that $a \mapsto \lambda_a$ extends to a ring homomorphism $\text{FPdim}: K \rightarrow \mathbb{R}$, the *Frobenius–Perron dimension*, with $\text{FPdim}(1) = 1$ and $\text{FPdim}(a) \text{FPdim}(b) = \sum_c N_{ab}^c \text{FPdim}(c)$.
- (c) Prove that a nonnegative matrix N with a positive eigenvector of eigenvalue λ has $r(N) = \lambda$ (bound the entries of N^m). Conclude $\text{FPdim}(a) = r(N_a) = r(N_{\bar{a}}) = \text{FPdim}(\bar{a})$ and $\text{FPdim}(a) \geq 1$ (use $N_{a\bar{a}}^1 = 1$).
- (d) Let $x \in B$ be a generator. Prove that $w = (\text{FPdim}(a))_{a \in B}$ satisfies $N_x w = \text{FPdim}(x)w$ (use $N_{xb}^c = N_{\bar{x}\bar{c}}^b$ and multiplicativity), hence is the Perron–Frobenius eigenvector of the irreducible matrix N_x normalized by $w_1 = 1$. Write $d_a = \text{FPdim}(a)$ and compare it with the quantum dimension of the section Quantum dimensions; for the level- k fusion ring identify $d_a = [a + 1]_q$, $q = e^{i\pi/(k+2)}$, with the quantum dimensions of Part VI, using the sine eigenvector computed there and (d).

- (e) Prove that the path representation of the section Fusion graphs exists for every commutative fusion ring and every self-dual generator x with $N_{xa}^b \in \{0, 1\}$, with $\psi = w$ and $\lambda = d_x$. Prove $\sum_b \dim \mathcal{H}_N(a, b) d_b = d_x^N d_a$ exactly, and, when N_x is primitive, $\dim \mathcal{H}_N(a, b) \sim \frac{d_a d_b}{\sum_c d_c^2} d_x^N$ as $N \rightarrow \infty$. Verify for the Fibonacci ring that this reproduces $F_N \sim \varphi^N / \sqrt{5}$.

Exercise 35.3 (Nondegenerate leading eigenvalues of transfer matrices and the correlation length).

- (a) Let T be the row-to-row transfer matrix of the six-vertex model of Part IV on $(\mathbb{C}^2)^{\otimes L}$, periodic, with weights $a, b, c > 0$, and let T_M be its restriction to the sector with M down arrows, $0 \leq M \leq L$. In the arrow basis, prove that T_M has nonnegative entries and positive diagonal entries. Prove that $T(\sigma', \sigma) > 0$ when σ' is obtained from σ by moving a single down arrow to an adjacent site: the horizontal arrows are determined by the ice rule and alternate at the two flipped sites. Conclude that T_M is irreducible, hence primitive, so that its largest eigenvalue $\Lambda_0(M)$ is positive, simple, and strictly dominant, with positive eigenvector.
- (b) With $Z = \sum_M \text{Tr } T_M^N$ prove $\lim_{N \rightarrow \infty} N^{-1} \log Z = \log \max_M \Lambda_0(M)$. For a sector-preserving observable $\mathcal{O}$, subtract the leading spectral projection in its connected correlation. Prove an $O(\rho^r)$ bound for every $\rho > |\Lambda_1(M)|/\Lambda_0(M)$, $\rho < 1$. When the observable has nonzero overlap with the subleading spectral subspace, identify the exponential decay rate as $\log(\Lambda_0(M)/|\Lambda_1(M)|)$; allow polynomial factors from Jordan blocks and faster decay when that overlap vanishes.
- (c) Prove that the transfer matrix $T = \prod_{i \text{ even}} (\rho_1 + \rho_2 e_i) \prod_{i \text{ odd}} (\rho_1 + \rho_2 e_i)$ of the dense loop model, acting in the path representation $\mathcal{H}_L(a, b)$ of the level- k fusion ring with $\rho_1, \rho_2 > 0$, has nonnegative entries and positive diagonal in the path basis, and is irreducible on each nonzero fixed-endpoint path space for this path graph and $L \geq 2$. Conclude the same spectral statements for the loop model at $n = 2 \cos(\pi/(k+2))$.

36 Modular tensor categories

Exercise 36.1 (Rigid semisimple categories, ribbon traces, and modularity). A $\mathbb{C}$ -linear category is semisimple here if its objects are finite direct sums of simple objects, with endomorphism algebra $\mathbb{C}$ for a simple object and no morphisms between nonisomorphic simples. A monoidal category is rigid if every object has left and right duals with evaluation and coevaluation satisfying the snake identities. A pivotal structure is a monoidal natural isomorphism $j_X : X \rightarrow X^{**}$; it is spherical if the left and right evaluation-coevaluation traces agree. For example its right trace is

$$\text{tr}_X(f) = \text{ev}_{X^*}(j_X f \otimes \text{id}_{X^*}) \text{coev}_X.$$

A ribbon structure consists of a braiding and a compatible twist θ , with $\theta_{X \otimes Y} = c_{Y, X} c_{X, Y} (\theta_X \otimes \theta_Y)$ and $\theta_{X^*} = (\theta_X)^*$, where the last star means the categorical dual morphism. For simple representatives a let $d_a = \text{tr}_a(\text{id})$ and choose $\mathcal{D}$ with $\mathcal{D}^2 = \sum_a d_a^2 \neq 0$. Such a category, with finitely many simple classes, simple unit, finite-dimensional morphism spaces, and invertible kernel

$$S_{ab} = \mathcal{D}^{-1} \text{tr}_{a \otimes b}(c_{b, a} c_{a, b}),$$

is a modular tensor category.

- (a) Prove additivity and multiplicativity of the spherical dimension by composing evaluation and coevaluation maps. Recover the loop trace and dimensions of the section Temperley–Lieb for its rigid diagram category.

- (b) Let $A = (\mathbb{Z}/2)^2$. On finite-dimensional A -graded spaces use the usual tensor product and duals, and on homogeneous factors of degrees (a, b) and (c, d) set $c(v \otimes w) = (-1)^{ad}w \otimes v$ and $\theta(v) = (-1)^{ab}v$. Verify the hexagons, balancing, and dual compatibility. Compute all simple dimensions as one.
- (c) Show that $\mathcal{D} = 2$ and $S_{(a,b),(c,d)} = \frac{1}{2}(-1)^{ad+bc}$. Prove $S^2 = \text{id}$ by summing the four group characters.

Exercise 36.2 (Fusion diagonalization by a modular kernel in the pointed model). Use the four-charge category just constructed and let $(N_x f)(a) = f(a - x)$ on the charge-labelled function space.

- (a) Show that its fusion is group addition, all quantum dimensions are one, and the columns of S are its character functions. Prove the kernel identity $N_x S_b = (S_{xb}/S_{0b})S_b$.
- (b) Deduce the Verlinde identity in this model,

$$N_{xy}^z = \sum_b \frac{S_{xb} S_{yb} \overline{S_{zb}}}{S_{0b}},$$

and evaluate it as $\mathbf{1}_{x+y=z}$.

37 Braid group representations

Exercise 37.1 (Braiding on fusion spaces and the loop representation). Use the braid group of the section Hecke / Temperley–Lieb algebras. For an object x in a braided rigid category, its fusion space with total charge a is $\text{Hom}(a, x^{\otimes N})$.

- (a) Construct the braid action by postcomposition with the adjacent braidings, inserting associators, and verify the relations using the hexagons.
- (b) In the Temperley–Lieb diagram category choose $A \in \mathbb{C}^\times$ with $d = -A^2 - A^{-2}$. Prove that $b_i = A \text{id} + A^{-1}e_i$ satisfies the braid relations and that $b_i^{-1} = A^{-1} \text{id} + A e_i$. If $|A| = 1$, $d \in \mathbb{R}$, and $e_i^* = e_i$ in the path representation, prove that b_i is unitary.
- (c) In the four-charge category calculate the double exchange of $e = (1, 0)$ and $m = (0, 1)$ and obtain -1 . Calculate the self-twists of e , m , and $e + m$.

38 Knot invariants

Exercise 38.1 (The bracket state sum and the Jones normalization). For an unoriented link diagram D with crossings, a smoothing chooses one of the two planar pairings at each crossing. Give the two choices weights A and A^{-1} according to the crossing convention $b = A \text{id} + A^{-1}e$. If a smoothing has ℓ circles define

$$\langle D \rangle = \sum_{\text{smoothings}} A^{n_A - n_{A^{-1}}} d^{\ell-1}, \quad d = -A^2 - A^{-2}.$$

For an oriented diagram let $w(D)$ be its signed crossing number.

- (a) Prove invariance of the bracket under the second and third Reidemeister moves using the inverse and braid identities. Compute the two curl factors as $-A^3$ and $-A^{-3}$.
- (b) Deduce that $V_D(A) = (-A^3)^{-w(D)} \langle D \rangle$ is invariant under all three moves, with the positive-crossing convention chosen to give the curl factor $-A^3$. Verify that an unknot has value one and a two-component unlink has value d . Define the usual variable by $t = A^{-4}$.

- (c) Assume $d \neq 0$. For the Hopf link obtained by closing two positive two-strand crossings, calculate $\text{tr}_2(b_1^2)/d$ and its writhe correction. Identify the two contributions from the spectral projections of e_1 .

Exercise 38.2 (Prediction: anyonic entropy and a mutual braiding phase). For the Fibonacci path chain of the section Fusion graphs let all admissible paths from 1 to τ have equal energy zero. For an independent braiding experiment use the four-charge category of the section Modular tensor categories, with one probe of charge e and one enclosed charge m .

- (a) Use the fusion recurrence and Perron–Frobenius result to derive $\dim \mathcal{H}_N(1, \tau) = F_N$ and residual entropy per anyon $k_B \log \varphi$, $\varphi = (1 + \sqrt{5})/2$. Determine the leading finite- N correction from $F_N = (\varphi^N - (-\varphi^{-1})^N)/\sqrt{5}$.
- (b) Let a balanced two-path interferometer have reference relative phase ϕ and coherent amplitudes of equal magnitude. Derive $P_0(\phi) = (1 + \cos \phi)/2$ without an enclosed charge and $P_m(\phi) = (1 - \cos \phi)/2$ when the probe winds once around m . Obtain the phase change from double braiding, equivalently from $S_{em}S_{00}/(S_{e0}S_{0m})$.
- (c) Specify independently the assumptions of degenerate fusion states for the entropy prediction and preserved charge, adiabatic transport, and equal coherent path amplitudes for the interference prediction. Calculate the change under two windings and distinguish it from a change in dynamical phase.

Part VIII

String-Net / State-Sum Models

39 Fusion categories

Exercise 39.1 (Closed-string configurations and a commuting-projector Hamiltonian). Let Γ be a connected trivalent graph giving a cell decomposition of a closed oriented surface, with vertices V , edges E , and disc faces F . A string configuration is a subset of E with an even number of occupied edges incident at every vertex. On $\mathcal{H} = \mathbb{C}^{\{0,1\}^E}$ define

$$(Z_e \psi)(n) = (-1)^{n_e} \psi(n), \quad (X_e \psi)(n) = \psi(n + \mathbf{1}_e), \quad Q_v = \frac{1 + \prod_{e \ni v} Z_e}{2}, \quad B_f = \frac{1 + \prod_{e \in \partial f} X_e}{2},$$

where additions and incidences are modulo two. Assume each face boundary is an embedded cycle. For $J_v, J_f > 0$ put $H = \sum_v J_v(1 - Q_v) + \sum_f J_f(1 - B_f)$.

- (a) Prove that the Q_v impose the string constraint and that B_f averages a configuration with the configuration obtained by adding the boundary of f . Prove that all these orthogonal projections commute.
- (b) Prove that the zero-energy states are the functions supported on string configurations and constant on orbits of face-boundary addition. Construct one such state per orbit by uniform summation. Prove invariance under every automorphism of the cell decomposition that preserves the couplings.

Exercise 39.2 (Fusion categories and group-labelled strings). A fusion category over $\mathbb{C}$ is a $\mathbb{C}$ -linear semisimple rigid monoidal category with finite-dimensional morphism spaces, finitely many simple isomorphism classes, and simple tensor unit; semisimplicity and rigidity have the meanings of

the section Modular tensor categories. A spherical structure is a pivotal structure whose left and right traces agree. For a finite group G let Vect_G consist of finite-dimensional G -graded spaces, grading-preserving maps, and $(U \otimes V)_g = \bigoplus_{ab=g} U_a \otimes V_b$.

- (a) Prove that Vect_G is a fusion category with simple objects the specified graded lines $\mathbb{C}_g$, dual $\mathbb{C}_g^* = \mathbb{C}_{g^{-1}}$, and fusion $\mathbb{C}_g \otimes \mathbb{C}_h = \mathbb{C}_{gh}$. Use the vector-space evaluation and coevaluation to construct a spherical structure with all simple dimensions equal to one.
- (b) For $G = \mathbb{Z}/2$ prove that the admissible trivalent labellings $a + b + c = 0$ are exactly the string configurations of the preceding exercise. Interpret adding a face boundary as inserting the nontrivial group-labelled loop and applying the fusion rule.

40 6j-symbols

Exercise 40.1 (Associators as maps between fusion spaces). For simple objects a, b, c, d in a fusion category define $V_{ab}^c = \text{Hom}(c, a \otimes b)$. Composition and the associator induce the recoupling isomorphism

$$F_d^{abc} : \bigoplus_e V_{ab}^e \otimes V_{ec}^d \longrightarrow \bigoplus_f V_{bc}^f \otimes V_{af}^d.$$

These intrinsic recoupling maps are the 6j data used here.

- (a) Construct this map by identifying the left and right sides with $\text{Hom}(d, (a \otimes b) \otimes c)$ and $\text{Hom}(d, a \otimes (b \otimes c))$, respectively. Prove both identifications using the simple summands and composition pairings.
- (b) In Vect_G , prove that the only nonzero channel has $e = ab$, $f = bc$, $d = abc$ and that F is the identity on the canonical graded lines. Reverse the orientation of a string and verify that its label changes to its group inverse.
- (c) Replace the associator on the canonical lines by multiplication by $\omega(a, b, c) \in U(1)$. Compute the inverse recoupling map and show that it is the adjoint. Determine the normalization of ω forced by keeping the unit constraints equal to the identity.

41 Pentagon identities

Exercise 41.1 (The pentagon, three-cocycles, and changes of associator). A normalized group three-cocycle is a function $\omega : G^3 \rightarrow U(1)$, equal to one if an argument is the identity, satisfying

$$\omega(b, c, d)\omega(a, bc, d)\omega(a, b, c) = \omega(ab, c, d)\omega(a, b, cd).$$

For a normalized function $\nu : G^2 \rightarrow U(1)$ put $(\delta\nu)(a, b, c) = \nu(b, c)\nu(a, bc)/(\nu(ab, c)\nu(a, b))$.

- (a) Compare the two composites from $((a \otimes b) \otimes c) \otimes d$ to $a \otimes (b \otimes (c \otimes d))$. Prove that the associator of the preceding exercise satisfies the pentagon exactly when ω is a three-cocycle.
- (b) Prove that replacing ω by $\omega \delta\nu$ gives a monoidally equivalent category by giving the tensor constraint of the identity functor. Denote this category by Vect_G^ω . Construct its left and right duals by multiplying the evaluation and coevaluation on the canonical graded lines by unit scalars, and use the cocycle equation to check both snake identities.
- (c) On $G = \mathbb{Z}/2$ verify that $\omega(a, b, c) = (-1)^{abc}$ is a normalized three-cocycle and cannot equal $\delta\nu$ for a normalized ν , by evaluating at $(1, 1, 1)$.

42 Turaev–Viro

Exercise 42.1 (The pointed Turaev–Viro sum and local retriangulations). Let M be a closed oriented triangulated three-manifold, with an ordering of its vertices. Label an oriented edge ij by $g_{ij} \in G$, with $g_{ji} = g_{ij}^{-1}$, and impose $g_{ij}g_{jk} = g_{ik}$ on each ordered triangle. In the untwisted category Vect_G define

$$Z_G(M) = |G|^{-n_0} \sum_{\text{admissible edge labellings}} 1,$$

where n_0 is the number of vertices. This is the pointed, untwisted Turaev–Viro state sum in this exercise.

- (a) For a 2–3 move, compare two and three tetrahedra filling the same triangular bipyramid. Prove that the triangle constraints give a bijection between internal labellings for each admissible boundary labelling.
- (b) For a 1–4 move, inserting one vertex in a tetrahedron, prove that an admissible boundary labelling has exactly $|G|$ extensions, parametrized by the label of one new edge. Deduce invariance of Z_G under both moves and their inverses.
- (c) Insert the oriented recoupling weight $\omega(g_{ij}, g_{jk}, g_{kl})^{\epsilon_{ijk\ell}}$ on a tetrahedron $i < j < k < \ell$. Verify that equality of the products in the 2–3 move is precisely the pentagon identity. Keep the subsequent state-space calculation untwisted.

43 TQFT

Exercise 43.1 (Finite field groupoids and gluing of state spaces). A groupoid is a category in which every morphism is invertible. For a finite groupoid $\mathcal{G}$ define $|\mathcal{G}| = \sum_{[x]} |\text{Aut}(x)|^{-1}$. For a triangulated surface Σ , let $\mathcal{A}_\Sigma$ be the groupoid of the admissible G -edge labellings of the preceding section; a morphism is a vertex labelling (h_i) acting by $g_{ij} \mapsto h_i g_{ij} h_j^{-1}$. Set $\mathcal{V}(\Sigma) = \text{Fun}(\pi_0 \mathcal{A}_\Sigma, \mathbb{C})$ with inner product $\sum_{[a]} \overline{f(a)} g(a) / |\text{Aut}(a)|$. An oriented bordism $M : \Sigma_0 \rightarrow \Sigma_1$ is a compact oriented three-manifold with boundary identified with $\overline{\Sigma_0} \sqcup \Sigma_1$. A three-dimensional oriented TQFT assigns spaces to closed oriented surfaces and linear maps to oriented bordisms, compatibly with composition and disjoint union.

- (a) Prove the orbit-counting identity $|X/G| = |X|/|G|$ for a finite group action. Recover $Z_G(M)$ as the cardinality of the field groupoid on M .
- (b) For a bordism $M : \Sigma_0 \rightarrow \Sigma_1$ define its map by pullback along restriction to Σ_0 followed by summation over the homotopy fibres of restriction to Σ_1 , weighted by $|\text{Aut}|^{-1}$. A homotopy fibre over b consists of a field and an isomorphism from its restriction to b . Show by finite double counting that composition corresponds to gluing, the cylinder gives the identity, and disjoint union gives the tensor product. Work with fixed triangulated bordisms and the local moves proved in the section Turaev–Viro.

Exercise 43.2 (Prediction: string-net ground-state degeneracy and excitation energy). Return to the $\mathbb{Z}/2$ string Hamiltonian of the section Fusion categories on a genus- g surface, with constant $J_v, J_f > 0$.

- (a) Prove that the even edge subsets form a vector space over $\mathbb{F}_2$ of dimension $|E| - |V| + 1$ by eliminating edges of a spanning tree. Prove that face boundaries span a space of dimension $|F| - 1$: in a sum with zero boundary, coefficients of adjacent faces agree. Using $|V| - |E| + |F| =$

$2 - 2g$, derive

$$\dim \ker H = 2^{2g}.$$

- (b) The global products of the vertex involutions and of the face involutions are both one. Deduce that each type of violation occurs in pairs, and construct a string operator creating a pair of either type. On a sufficiently large cell decomposition derive the excitation gap $2 \min(J_v, J_f)$.
- (c) Identify the degeneracy with $\dim \mathcal{V}(\Sigma)$ for $G = \mathbb{Z}/2$ by exchanging the primal and dual cell decompositions. State the predicted values on the sphere and torus and the spectroscopic pair-creation thresholds $2J_v, 2J_f$. Specify that these are exact predictions for the stated commuting-projector Hamiltonian; a microscopic material requires an identification of its effective couplings and degrees of freedom. Compare the construction with Levin and Wen [6].

Part IX

Lattice Gauge Theory

44 Gauge fields on edges

Exercise 44.1 (Finite-group lattice gauge fields and gauge-invariant weights). Let G be a finite group and X a finite oriented cell complex whose faces have specified boundary edge words. A gauge field is an assignment $g_e \in G$ with $g_{\bar{e}} = g_e^{-1}$. The gauge transformations are the vertex actions of the section TQFT, $g_e \mapsto h_{s(e)} g_e h_{t(e)}^{-1}$. The face holonomy U_f is the ordered product around its boundary, based at a vertex. For a real class function w define

$$Z_X(\beta_g) = |G|^{-|E|} \sum_g \exp \left(\beta_g \sum_f w(U_f) \right).$$

- (a) Prove that changing the base vertex conjugates U_f and that the Gibbs measure is gauge invariant. State the additional condition on w for invariance under reversal of a face orientation.
- (b) For $G = \{\pm 1\}$, $w(u) = u$, identify the Hamiltonian $H = -J_g \sum_f U_f$ with $\beta_g = \beta J_g$. On a connected graph fix a root and a spanning tree. Prove that each field has a unique gauge transform trivial on the tree if the transformation at the root is required to be the identity. Determine the remaining root action.

45 Wilson loops

Exercise 45.1 (Wilson observables and the exact planar area law). For a finite-dimensional representation $\rho : G \rightarrow \text{GL}(V)$ and an oriented closed edge path C define the normalized Wilson loop $W_\rho(C) = (\dim V)^{-1} \text{Tr } \rho(U_C)$. Use the defining one-dimensional representation for $G = \{\pm 1\}$.

- (a) Prove gauge invariance and independence of the starting vertex. For the $\mathbb{Z}/2$ model show that $W(C) = \prod_{f \subset D} U_f$ when $C = \partial D$ is a simple loop bounding a union of faces.
- (b) On a planar disc cellulation with free edge boundary conditions show that the map from edge fields to face holonomies is onto and that all fibres have size $2^{|E|-|F|}$. One proof successively removes boundary faces; another uses the tree elimination of the section Gauge fields on edges. Deduce

$$Z_X = (\cosh \beta_g)^{|F|}, \quad \langle W(C) \rangle = (\tanh \beta_g)^{|D|}.$$

- (c) Prove that any gauge-noninvariant single edge has zero expectation, even when β_g is large, by changing the gauge at an endpoint. Compare this with the gauge-invariant loop.

46 Flat connections

Exercise 46.1 (Flat fields and the path groupoid). A field is flat if $U_f = 1$ for every face. The path groupoid $\Pi(X)$ has vertices as objects and edge words as morphisms, modulo immediate backtracking and face-boundary relations. Regard G as the one-object groupoid BG_{alg} .

- (a) Prove that flat fields are functors $\Pi(X) \rightarrow BG_{\text{alg}}$ and that gauge transformations are natural isomorphisms. Check that subdivision of an edge gives an equivalent groupoid of flat fields.
- (b) On a connected complex with root x prove that gauge classes are conjugacy classes of homomorphisms $\text{Aut}_{\Pi(X)}(x) \rightarrow G$. Determine the automorphism group of the field corresponding to ρ as the centralizer of $\text{im } \rho$.
- (c) For a torus use its polygon with boundary word $aba^{-1}b^{-1}$ to obtain commuting pairs $(A, B) \in G^2$ modulo simultaneous conjugation. For abelian G count $|G|^{2g}$ flat classes on an oriented genus- g surface.

47 Dijkgraaf–Witten

Exercise 47.1 (Cocycle actions for finite gauge fields). For the normalized three-cocycle ω of the section Pentagon identities and a flat field on an ordered triangulated closed oriented three-manifold define

$$Z_{G,\omega}(M) = |G|^{-n_0} \sum_{g \text{ flat}} \prod_{i < j < k < \ell} \omega(g_{ij}, g_{jk}, g_{k\ell})^{\epsilon_{ijk\ell}},$$

where the product is over tetrahedra and ϵ compares the vertex ordering with the orientation of M .

- (a) Recover the pointed state sum of the section Turaev–Viro. Using the pentagon, check invariance under the 2–3 move. For the 1–4 move show that each of the $|G|$ extensions has the same total weight as the original tetrahedron.
- (b) Prove that replacing ω by $\omega \delta \nu$ leaves the closed sum unchanged: each oriented triangle factor cancels with the factor from the tetrahedron on the other side. For a manifold with boundary identify the uncanceled boundary product explicitly.
- (c) For $\omega = 1$ prove $Z_{G,1}(M) = |\text{Hom}(\pi_1(M), G)|/|G|$ when M is connected, where $\pi_1(M) = \text{Aut}_{\Pi(M)}(x)$ for a triangulation. Calculate $Z_{G,1}(S^3)$ and $Z_{G,1}(S^2 \times S^1)$.

48 Classifying spaces

Exercise 48.1 (The nerve of a group and its classifying space). The nerve of a category has as its n -simplices the strings of n composable arrows; face maps compose or omit end arrows and degeneracy maps insert identities. Its geometric realization glues a standard topological simplex to each such string by these maps. Define BG to be the realization of the nerve of BG_{alg} . A homotopy between maps is a continuous map on the product with $[0, 1]$ restricting to the given maps at its ends.

- (a) Show that a flat field on an ordered simplicial complex specifies a simplicial map to the nerve: an ordered simplex $i_0 < \dots < i_n$ maps to $(g_{i_0 i_1}, \dots, g_{i_{n-1} i_n})$. Construct the homotopy induced

by a vertex gauge transformation.

- (b) Let $\mathcal{E}G$ have object set G and exactly one arrow between each pair of objects. Prove its nerve is contractible by inserting a fixed initial object. Prove that the free G -action on its realization has quotient BG . Deduce $\pi_1(BG) \simeq G$ and $\pi_n(BG) = 0$ for $n > 1$, where π_n denotes based maps from the n -sphere modulo based homotopy.
- (c) For a connected finite cell complex build a map to BG from a homomorphism of fundamental groups, first on its edges, then across its faces and higher cells. Use the vanishing higher homotopy groups to prove $[X, BG] \simeq \text{Hom}(\pi_1(X), G)/G$.

49 Higher gauge theory

Exercise 49.1 (Abelian two-form fields and gauge transformations between transformations). For a finite cell complex X let $C_n(X; \mathbb{Z})$ be its free group of oriented cells, with incidence boundary ∂ , and put $C^n(X; A) = \text{Hom}(C_n(X; \mathbb{Z}), A)$ for an abelian group A . The coboundary is $\delta c = c\partial$; define $H^n(X; A) = \ker \delta / \text{im } \delta$. A flat two-form field is $b \in C^2(X; A)$ with $\delta b = 0$. A gauge transformation is $b \mapsto b + \delta a$, $a \in C^1(X; A)$, and $a \mapsto a + \delta \lambda$, $\lambda \in C^0(X; A)$, is a transformation between gauge transformations.

- (a) Verify $\partial^2 = 0$ for a simplicial boundary by pairing the two orders of deleting vertices. Deduce $\delta^2 = 0$ and that flat two-form fields modulo gauge transformations are classified by $H^2(X; A)$.
- (b) For a two-cycle S and a character $\chi : A \rightarrow U(1)$ define $W_\chi(S) = \chi(b(S))$. Prove gauge invariance and invariance under replacing S by a homologous cycle when b is flat.
- (c) Give the three-torus its product cell structure with one vertex, three edges, three faces, and one three-cell. Compute its cellular boundaries and prove $H^2(T^3; A) = A^3$.

50 Homotopy types

Exercise 50.1 (Cocycle spaces as homotopy types of higher gauge fields). For an abelian group A and $n \geq 1$ define a simplicial abelian group by $K(A, n)_r = Z^n(\Delta^r; A)$, the normalized simplicial n -cocycles on the standard r -simplex, with restriction maps.

- (a) Prove that a simplicial map $X \rightarrow K(A, n)$ is the same as an n -cocycle on X : restrict a cocycle to every simplex and check the compatibility conditions.
- (b) Use the oriented prism on each simplex of $X \times [0, 1]$ to show that homotopic maps give cohomologous cocycles. Conversely extend a coboundary difference across the prism to construct a homotopy. Obtain $[X, K(A, n)] = H^n(X; A)$ in the simplicial homotopy category, allowing subdivision.
- (c) Apply this to spheres and derive $\pi_j(K(A, n)) = A$ for $j = n$ and zero otherwise. Interpret the edge-field and two-form classifications as $[X, K(A, 1)]$ and $[X, K(A, 2)]$ for abelian A .

Exercise 50.2 (Prediction: Wilson area decay and quantized higher flux sectors). For the planar square $\mathbb{Z}/2$ gauge model with lattice spacing a_0 and $\beta_g = \beta J_g > 0$, let C enclose physical area $\mathcal{A} = a_0^2 |D|$. For the flat $\mathbb{Z}/2$ two-form model on a spatial three-torus use the uniform superposition of every gauge orbit as its zero-energy state, with constraints imposed by commuting averages and flatness projections.

- (a) Derive

$$-\log \langle W(C) \rangle = \sigma_{\text{area}} \mathcal{A}, \quad \sigma_{\text{area}} = -a_0^{-2} \log \tanh \beta_g.$$

Obtain $\sigma_{\text{area}} \sim 2a_0^{-2}e^{-2\beta_g}$ at large β_g and calculate the ratio of expectations for loops whose enclosed areas differ by one plaquette.

- (b) Derive eight ground states on the three-torus, labelled by the three independent signs of the Wilson surfaces on the coordinate two-tori. Construct the commuting surface measurements distinguishing them using the pairing of the section Higher gauge theory.
- (c) State which prediction requires free planar boundary conditions and finite β_g , and which requires exact flatness and periodic spatial boundaries. Identify loop and surface expectation measurements in a lattice simulator that test the two formulas.

Part X

Topological Lattice Phases

51 Gapped Hamiltonians

Exercise 51.1 (A bipartite hopping chain and its bulk spectral gap). Consider spinless fermions with two distinguished sites A, B per cell of a one-dimensional lattice. Its one-particle Hamiltonian has hopping $v > 0$ from A_j to B_j and $w > 0$ from B_j to A_{j+1} , with Hermitian conjugate hoppings and no on-site terms. On the Brillouin circle $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ the Fourier transform gives

$$H(k) = \begin{pmatrix} 0 & \overline{q(k)} \\ q(k) & 0 \end{pmatrix}, \quad q(k) = v + we^{ik}.$$

The two entries refer to the physical sublattices. The involution $\Gamma = +1$ on A , -1 on B satisfies $\Gamma H \Gamma = -H$ (chiral symmetry). A bulk gap at Fermi energy zero is a number $\epsilon > 0$ such that $\text{Spec } H(k) \cap (-\epsilon, \epsilon) = \emptyset$ for every k .

- (a) Derive the Fourier formula directly on periodic chains and prove $H(k)^2 = |q(k)|^2 \text{id}$. Find the two band energies and their spectral projections.
- (b) Prove that a gap exists exactly when $v \neq w$ and that its half-width is $|v - w|$. At half filling calculate the minimum particle-hole excitation energy $2|v - w|$.
- (c) Define the filled-band ground state by occupying the negative spectral subspace at each allowed momentum. Show that its one-particle correlation operator is the negative spectral projection, without choosing eigenvectors.

52 Deformation classes of phases

Exercise 52.1 (Symmetry-preserving deformations and winding number). Two continuous gapped chiral one-particle Hamiltonian families on $\mathbb{T}$ are deformation equivalent if they are joined by a continuous path with a uniform gap and the same chiral grading. Stabilization allows adding decoupled copies of the constant family $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. For a smooth nonvanishing scalar q put

$$\nu(q) = \frac{1}{2\pi i} \int_{\mathbb{T}} q^{-1} dq.$$

- (a) Lift the phase of $q/|q|$ continuously on $[0, 2\pi]$ and prove that $\nu(q) \in \mathbb{Z}$. Prove its invariance under gapped deformations and additivity under multiplication.

- (b) Prove that two scalar loops are homotopic through nonvanishing loops exactly when their windings agree, by lifting their quotient's phase. Calculate $\nu(v + we^{ik}) = 0$ for $v > w$ and 1 for $w > v$.
- (c) For a finite-dimensional graded space with an invertible off-diagonal map $D(k) : V_A \rightarrow V_B$, use $U = D(D^*D)^{-1/2}$ to flatten the Hamiltonian through a gapped chiral path. For two unitary families U, U_0 with the same domain and codomain define the relative integer as the winding of $\det(U_0^{-1}U)$, using the determinant of the section Transfer matrices. Prove stabilization invariance.

53 Generalized cohomology

Exercise 53.1 (Reduced cohomology, suspension, and stable unitary families). For a based finite cell complex X , its reduced suspension is $\Sigma X = (X \times [0, 1]) / (X \times \{0, 1\} \cup \{*\} \times [0, 1])$. A reduced generalized cohomology theory consists of contravariant homotopy-invariant graded abelian groups $\tilde{E}^n(X)$, natural suspension isomorphisms $\tilde{E}^{n+1}(\Sigma X) \simeq \tilde{E}^n(X)$, and exact sequences for cell cofibrations $A \hookrightarrow X$ and their quotients X/A , with additivity on finite wedges.

- (a) From the cellular cochains of the section Higher gauge theory construct the relative complex $C^*(X, A; \mathbb{Z})$ and its long exact sequence. Compute the cellular cochains of a suspension to obtain the suspension isomorphism for ordinary reduced cohomology.
- (b) A complex vector bundle is a family of vector spaces locally isomorphic to a product $U \times \mathbb{C}^n$, with linear fibre identifications. Define $K^0(X)$ as the group completion of bundles under direct sum and $\tilde{K}^0(X)$ as the kernel of restriction to the base point. Show that a continuous constant-rank orthogonal projection family on X defines a bundle, by projecting from a nearby fibre and proving local invertibility.
- (c) Define $K^{-1}(X)$ by stable homotopy classes of continuous unitary endomorphism families, with direct sum as addition. Show that adjoining identity families does not change the class, and construct an explicit block-rotation homotopy from $U \oplus U^{-1}$ to the identity. For $X = S^1$ prove that determinant winding is an additive homotopy invariant of these classes. Relate it to the flattened hopping family.

54 Stable homotopy theory

Exercise 54.1 (Suspension spectra and ordinary cohomology). A sequential spectrum is a sequence of based spaces E_j with structure maps $\Sigma E_j \rightarrow E_{j+1}$. Its stable homotopy groups are $\pi_n(E) = \varinjlim_j \pi_{n+j}(E_j)$. The suspension spectrum of X has levels $\Sigma^j X$ and the canonical structure maps. A loop space ΩY consists of based loops in Y ; the adjunction $[\Sigma X, Y] = [X, \Omega Y]$ uses based homotopy classes.

- (a) Prove the suspension-loop adjunction from the two quotient definitions. Write the transition maps defining $\pi_n(E)$ and show that a compatible levelwise homotopy induces the same maps on these groups.
- (b) Use the cocycle models $K(A, j)$ of the section Homotopy types. Construct the maps $\Sigma K(A, j) \rightarrow K(A, j+1)$ from the suspension of cocycles. Show that the associated represented groups on finite cell complexes are ordinary cohomology:

$$\tilde{H}^n(X; A) = \varinjlim_j [\Sigma^j X, K(A, n+j)].$$

Derive the suspension isomorphism and exactness from the relative cochain calculation of the preceding section.

- (c) For $X = S^1$, identify the scalar flattened Hamiltonian $q/|q|$ with its class in $H^1(S^1; \mathbb{Z}) = \mathbb{Z}$ by phase lifting. Show that its image under each suspension isomorphism is the same integer on the suspended fundamental class.

Exercise 54.2 (Prediction: bulk gap and boundary modes of the hopping chain). For an open chain of N cells use the hoppings of the section Gapped Hamiltonians, with no bond from B_N to A_1 . Fix $0 < v < w$, $r = v/w$, cell spacing a_0 , chiral symmetry, and half filling.

- (a) On the semi-infinite chain construct the normalized zero-energy state supported on sublattice A , $\psi(A_j) = \sqrt{1-r^2}(-r)^{j-1}$, $\psi(B_j) = 0$. Derive its amplitude localization length $\ell = a_0/\log(w/v)$.
- (b) On the finite chain normalize the truncated state and compute its residual norm

$$\|H\psi\| = vr^{N-1} \sqrt{\frac{1-r^2}{1-r^{2N}}}.$$

Apply the spectral theorem to prove the existence of a pair of energies $\pm E_N$ satisfying the same upper bound on $|E_N|$. Prove that zero itself is absent for $v > 0$ by the invertibility of the triangular hopping map.

- (c) Connect $v < w$ to $v = 0$ through gapped bulk Hamiltonians and identify the two exactly unpaired boundary sites at that endpoint. Compare with the gapped deformation of $v > w$ to $w = 0$ and its absence of boundary zero modes. Relate the difference to the winding class.
- (d) State the bulk absorption threshold $2(w-v)$ and the boundary localization length as distinct measurable predictions. Identify the fixed termination and chiral symmetry as hypotheses; show directly that an on-site potential at A_1 can shift the boundary energy when that symmetry is relaxed.

Part XI

Quiver / Spin-Chain Models

55 Bethe ansatz

Exercise 55.1 (The inhomogeneous rational spin chain and a one-root Bethe state). Let $V = \mathbb{C}^{\{+, -\}}$ be the arrow space of the section Row-to-row transfer matrices. For $\epsilon \neq 0$ use $R(u) = u \text{id} + \epsilon P$ and $\mathcal{T}_0(u) = R_{0N}(u - \theta_N) \cdots R_{01}(u - \theta_1)$. With the auxiliary kernels A, B, C, D of the section Commuting families and nonzero $\kappa_{\pm}$, put $t_{\kappa}(u) = \kappa_+ A(u) + \kappa_- D(u)$. Let Ω be the all-up configuration.

- (a) Prove $[t_{\kappa}(u), t_{\kappa}(v)] = 0$ by RTT and the commutation of R with the product of the two diagonal twists. Compute $A(u)\Omega = a(u)\Omega$, $D(u)\Omega = d(u)\Omega$, $C(u)\Omega = 0$, where $a(u) = \prod_j (u - \theta_j + \epsilon)$ and $d(u) = \prod_j (u - \theta_j)$.
- (b) Extract the two exchange relations involving A, B and D, B from RTT. For a nonzero state $B(\lambda)\Omega$ derive

$$\Lambda(u) = \kappa_+ a(u) \frac{u - \lambda - \epsilon}{u - \lambda} + \kappa_- d(u) \frac{u - \lambda + \epsilon}{u - \lambda}, \quad \kappa_+ a(\lambda) = \kappa_- d(\lambda).$$

Show that the Bethe equation cancels the apparent pole at $u = \lambda$ and the unwanted vector $B(u)\Omega$.

56 Quiver varieties

Exercise 56.1 (A framed one-vertex quiver and the cotangent Grassmannian). A quiver is a finite directed graph. Consider the quiver with one vertex, no arrows, a framing space W , and a vertex space U , both finite-dimensional. Its doubled framed data are $I : W \rightarrow U$, $J : U \rightarrow W$, with action $g \cdot (I, J) = (gI, Jg^{-1})$ of $\mathrm{GL}(U)$. Use the trace pairing to equip this space with the symplectic form $\omega((I_1, J_1), (I_2, J_2)) = \mathrm{Tr}(I_1 J_2 - I_2 J_1)$. The moment map is $\mu(I, J) = IJ \in \mathrm{End}(U)$; stability means that I is surjective. The quiver variety is

$$\mathfrak{M}(r, W) = \{(I, J) : IJ = 0, I \text{ surjective, } \dim U = r\} / \mathrm{GL}(U).$$

- (a) For $\xi \in \mathrm{End}(U)$ verify $\iota_{\xi\#}\omega = d \mathrm{Tr}(\xi\mu)$, fixing this sign convention for moment maps.
- (b) Put $S = \ker I$. Prove that J has image in S and descends to a map $W/S \rightarrow S$. Identify the orbit space with the cotangent bundle of the Grassmannian of codimension- r subspaces of W , using the trace duality between $\mathrm{Hom}(S, W/S)$ and $\mathrm{Hom}(W/S, S)$.
- (c) For $r = 1$, $\dim W = 2$, identify $\mathfrak{M}$ with $T^*\mathbb{P}(W)$ by recording the kernel line. Prove that scaling the cotangent fibre retracts this space onto $\mathbb{P}(W)$. Compute the varieties for $r = 0$ and $r = 2$.

57 Cohomology and K -theory

Exercise 57.1 (The cohomology of a projective line and its line bundles). Use the cellular cohomology and bundle group completion of the sections Higher gauge theory and Generalized cohomology. The first Chern class of a complex line bundle L is the pullback of the generator of $H^2(\mathbb{P}(\mathbb{C}^\infty); \mathbb{Z})$ under its classifying map, normalized so that the dual tautological line on $\mathbb{P}(\mathbb{C}^2)$ has integral one.

- (a) Build $\mathbb{P}(\mathbb{C}^n)$ by its cells of dimensions $0, 2, \dots, 2n - 2$. Intersect transverse projective hyperplanes to show that the powers of their degree-two class give $H^*(\mathbb{P}(\mathbb{C}^n); \mathbb{Z}) = \mathbb{Z}[x]/(x^n)$. Deduce $H^*(T^*\mathbb{P}(W); \mathbb{Z}) = \mathbb{Z}[x]/(x^2)$ for $\dim W = 2$.
- (b) Let L be the tautological line on $\mathbb{P}(W)$. Use $0 \rightarrow L \rightarrow W \rightarrow W/L \rightarrow 0$ and its determinant to prove $[L] + [L]^{-1} = 2$ when W is two-dimensional and constant. Deduce $([L] - 1)^2 = 0$ in $K^0(\mathbb{P}(W))$ and calculate the rank and first Chern class of $[L] - 1$.

Exercise 57.2 (Equivariant cohomology of the two-site quiver variety). Specify the physical framing decomposition $W = W_1 \oplus W_2$ into lines and let a torus scale the two lines independently. Let an additional circle scale cotangent fibres. Equivariant cohomology is $H_T^*(X; \mathbb{C}) = H^*(ET \times_T X; \mathbb{C})$, where ET is a contractible free T -space; for circles use the union of odd unit spheres in $\mathbb{C}^\infty$. Write a_1, a_2, ϵ for the degree-two character classes, $B = \mathbb{C}[a_1, a_2, \epsilon]$, and $t = c_1(L)$.

- (a) Use finite projective-space approximations to BT and the two-cell calculation of the preceding exercise to prove

$$H_T^*(T^*\mathbb{P}(W); \mathbb{C}) = B[t]/((t - a_1)(t - a_2)).$$

Derive the relation from the tautological line and quotient line whose sum is the pulled-back framing bundle.

- (b) Restrict to the two fixed points $p_i = \mathbb{P}(W_i)$ and prove $t|_{p_i} = a_i$. Show that restriction is injective and that a pair $(f_1, f_2) \in B^2$ is in its image exactly when $f_1 - f_2$ is divisible by $a_1 - a_2$.

58 Stable envelopes

Exercise 58.1 (Two chambers and the stable-envelope maps of a projective line). For the two fixed points in the preceding exercise choose opposite orders $p_1 < p_2$ and $p_2 < p_1$, called chambers $+$ and $-$. Fix the polarization and signs by the following normalization: the stable-envelope class for p_1 in chamber $+$ vanishes at p_2 and equals $a_1 - a_2$ at p_1 ; the class for p_2 has restrictions ϵ and $a_2 - a_1 + \epsilon$. In chamber $-$ the classes have restrictions $(a_1 - a_2 + \epsilon, \epsilon)$ and $(0, a_2 - a_1)$. These support, normalization, and degree conditions define the stable envelopes in this two-fixed-point model.

- (a) Use the divisibility criterion to prove existence and uniqueness, and obtain

$$\text{Stab}_+(p_1) = t - a_2, \quad \text{Stab}_+(p_2) = t - a_1 + \epsilon, \quad \text{Stab}_-(p_1) = t - a_2 + \epsilon, \quad \text{Stab}_-(p_2) = t - a_1.$$

- (b) Put $u = a_1 - a_2$ and invert $u, u - \epsilon, u + \epsilon$. Compute the chamber-change operator $\text{Stab}_-^{-1} \text{Stab}_+$ on the fixed-point function space:

$$\frac{1}{u + \epsilon} \begin{pmatrix} u & \epsilon \\ \epsilon & u \end{pmatrix}.$$

The entries here are indexed by the two geometrically fixed points.

- (c) Add the two one-point quiver varieties $r = 0, 2$, with chamber change equal to one. Identify the resulting operator with $(u \text{id} + \epsilon P)/(u + \epsilon)$ on two arrow spaces. Verify its Yang–Baxter identity using the rational limit of the section Spectral parameter. Compare this concrete construction with [7].

59 Geometric representation theory

Exercise 59.1 (Permutation symmetry and the two-site representation). Let $\mathcal{W} = \bigoplus_{r=0}^2 H_T^*(\mathfrak{M}(r, W))$ with the character differences inverted, and identify its fixed-point function space with the four spin configurations of $V^{\otimes 2}$ by the occupied subset of $\{W_1, W_2\}$. On each arrow space use the spin operators of the section Yang–Baxter equation.

- (a) Show that the raising and lowering operators on the occupied subsets, defined by removing or adding one label with coefficient one, satisfy $[S_{\text{tot}}^z, S_{\text{tot}}^\pm] = \pm S_{\text{tot}}^\pm$ and $[S_{\text{tot}}^+, S_{\text{tot}}^-] = 2S_{\text{tot}}^z$.
- (b) Prove that chamber change commutes with this action. Use the intrinsic projections $(\text{id} \pm P)/2$ to split the two-site space into symmetric and alternating subspaces. Compute their dimensions and the chamber-change eigenvalues 1 and $(u - \epsilon)/(u + \epsilon)$.
- (c) For $\theta_1 = \theta_2 = 0$, $\kappa_+ = \kappa_- = 1$, solve the one-root Bethe equation and show that $B(-\epsilon/2)\Omega$ belongs to the alternating line. Identify the complementary spin-one multiplet by the total spin action.

Exercise 59.2 (Prediction: the exchange splitting of a two-spin bond). Let two physical spins of magnitude $\frac{1}{2}$ interact through $H = J(\mathbf{S}_1 \cdot \mathbf{S}_2 - \frac{1}{4})$ with $J > 0$ and $\hbar = 1$ in the spin commutation relations. Use the quiver chamber-change operator $\mathcal{R}(u) = (u \text{id} + \epsilon P)/(u + \epsilon)$.

- (a) Prove $H = \frac{J\epsilon}{2} \mathcal{R}(0)^{-1} \mathcal{R}'(0) = \frac{J}{2}(P - \text{id})$. Derive a singlet energy $-J$ and a triplet energy 0.
- (b) In a magnetic field with Zeeman term $-\hbar S_{\text{tot}}^z$ calculate all four energies and the first ground-state crossing at $h = J$.
- (c) Derive the zero-field partition function $Z = e^{\beta J} + 3$ and susceptibility to h , $\chi = 2\beta/(e^{\beta J} + 3)$. State the singlet-triplet spectroscopic gap J and specify that isotropic exchange, two spin- $\frac{1}{2}$ degrees of freedom, and negligible coupling to other bonds are the model assumptions.

Part XII

Gaudin Model

60 Commuting Hamiltonians

Exercise 60.1 (The rational Gaudin spin system and its invariant interaction). Let V_{s_i} carry spin $s_i \in \frac{1}{2}\mathbb{Z}_{\geq 0}$, with $[S_i^a, S_i^b] = i\varepsilon_{abc}S_i^c$, $\sum_a (S_i^a)^2 = s_i(s_i + 1)\text{id}$, and operators at different sites commuting. These are the $q \rightarrow 1$ representations of the section Representation theory. For pairwise distinct real parameters $z_1, \dots, z_N$ define

$$\Omega_{ij} = \mathbf{S}_i \cdot \mathbf{S}_j, \quad G_i = \sum_{j \neq i} \frac{\Omega_{ij}}{z_i - z_j} \quad \text{on } \mathcal{H} = \bigotimes_i V_{s_i}.$$

Intrinsically, Ω_{ij} is the contraction of the two spin actions with the invariant Euclidean pairing on the physical rotation Lie algebra. A Gaudin Hamiltonian is $H = \sum_i c_i G_i$ for specified real c_i .

- (a) Prove $[\Omega_{ij}, \Omega_{kl}] = 0$ for disjoint pairs and $[\Omega_{ij}, \Omega_{ik} + \Omega_{jk}] = 0$ for distinct i, j, k , using invariance of the pairing. Combine these identities with partial fractions to prove $[G_i, G_j] = 0$.
- (b) Prove $\sum_i G_i = 0$ and

$$\sum_i z_i G_i = \frac{1}{2} \left(\mathbf{S}_{\text{tot}}^2 - \sum_i s_i(s_i + 1)\text{id} \right).$$

Show that every G_i is self-adjoint and rotation invariant.

- (c) Use joint spectral projections to express the partition function of H as a sum over joint energies with their multiplicities.

61 Bethe equations

Exercise 61.1 (Gaudin Bethe states and cancellation of unwanted terms). Let Ω be a tensor product of highest-weight states and put $S^-(w) = \sum_i S_i^- / (w - z_i)$, $S_i^\pm = S_i^x \pm iS_i^y$. For distinct roots $w_1, \dots, w_M$ avoiding all z_i define $\Psi = \prod_a S^-(w_a)\Omega$.

- (a) Compute the action for $M = 0, 1$ directly from the spin commutation relations. Then commute G_i successively through the product to prove that, if $\Psi \neq 0$ and

$$\sum_i \frac{s_i}{w_a - z_i} - \sum_{b \neq a} \frac{1}{w_a - w_b} = 0 \quad (1 \leq a \leq M),$$

it has joint eigenvalues

$$g_i = \sum_{j \neq i} \frac{s_i s_j}{z_i - z_j} - \sum_a \frac{s_i}{z_i - w_a}.$$

Keep the terms in which one $S^-(w_a)$ is replaced by S_i^- ; their coefficients are the displayed Bethe equations.

- (b) Compute $S_{\text{tot}}^+ \Psi$ by the same method and show that it vanishes under these equations. Its total spin is $s_{\text{tot}} = \sum_i s_i - M$ when the vector is nonzero. Check both identities of the section Commuting Hamiltonians against the resulting eigenvalues.
- (c) For two spins $s_1 = s_2 = \frac{1}{2}$ solve the $M = 1$ equation, show $w = (z_1 + z_2)/2$, and identify the alternating spin state without choosing a decomposition into eigenvectors.

62 Opers

Exercise 62.1 (Projective differential operators and the Miura transformation). On a Riemann surface let K be the holomorphic cotangent line. A projective second-order differential operator acts from $K^{-1/2}$ to $K^{3/2}$, with principal symbol one and vanishing first-derivative term in a local coordinate: $D = \partial_z^2 - t(z)$. On the sphere use its fixed spin line $K^{1/2} = \mathcal{O}(-1)$, where $\mathcal{O}(-1)$ denotes the tautological line of the section Cohomology and K -theory. Such an operator, or its projective solution local system, is the scalar form of a PGL_2 -oper. Meromorphic poles are permitted at specified marked points. For the Gaudin parameters define the Miura function

$$\chi(z) = \sum_i \frac{s_i}{z - z_i} - \sum_a \frac{1}{z - w_a}, \quad D = (\partial_z - \chi)(\partial_z + \chi) = \partial_z^2 - (\chi^2 - \chi').$$

- (a) Use $T_L \mathbb{P}(W) = \text{Hom}(L, W/L)$ for $\dim W = 2$ and a fixed identification $\bigwedge^2 W = \mathbb{C}$ to prove that the cotangent line is the square of the tautological line. Under $z = z(\zeta)$ transform a solution as a half-density and derive the transformation rule $t_\zeta = (z')^2 t_z - \frac{1}{2}\{z, \zeta\}$, where $\{z, \zeta\} = z'''/z' - \frac{3}{2}(z''/z')^2$. Verify compatibility on overlapping coordinates.
- (b) Expand $\chi^2 - \chi'$ at each w_a . Show that its double pole cancels automatically and its simple pole vanishes exactly when the Bethe equation holds. At a marked point prove

$$t(z) = \frac{s_i(s_i + 1)}{(z - z_i)^2} + \frac{2g_i}{z - z_i} + O(1).$$

Recover the joint Hamiltonian eigenvalues from these coefficients.

- (c) Set $y(z) = \prod_a (z - w_a) \prod_i (z - z_i)^{-s_i}$ on a simply connected domain avoiding the marked points. Verify $Dy = 0$ and construct a second solution $y \int y^{-2} dz$. Use the Bethe equations to show that $y^{-2} dz$ has zero residues at the w_a ; it is regular at each z_i . Deduce projectively trivial monodromy for the constructed oper when $2s_i$ are nonnegative integers.

63 Langlands duality

Exercise 63.1 (Rank-one root data and the dual group of the spin symmetry). For a complex algebraic torus T , its character and cocharacter lattices are $X^*(T) = \text{Hom}(T, \mathbb{C}^\times)$ and $X_*(T) = \text{Hom}(\mathbb{C}^\times, T)$, paired by composition. A root datum consists of these paired lattices and paired finite sets of roots and coroots with $\langle \alpha, \alpha^\vee \rangle = 2$, such that the reflections $x \mapsto x - \langle x, \alpha^\vee \rangle \alpha$ and their dual reflections preserve the respective root sets. Its dual interchanges the two lattices and the two root sets.

- (a) Use the specified maximal torus $T = \{\text{diag}(t, t^{-1})\} \subset SL_2$. Compute $X^*(T) = \mathbb{Z}$, root 2, $X_*(T) = \mathbb{Z}$, and coroot 1. After interchange identify the result with the torus and root data of $PGL_2 = GL_2/\mathbb{C}^\times$.
- (b) The indicial exponents of the oper at z_i solve $r(r - 1) = s_i(s_i + 1)$. Compute them as $-s_i$ and $s_i + 1$ and their difference as $2s_i + 1$. Explain this integer through the highest weight $2s_i$ and the rank-one half-sum of positive roots, with the positive root fixed above.
- (c) Show that changing signs of a lift of the oper's two-dimensional solution transport has no effect on its projectivization. Relate this to the quotient from SL_2 to the dual group PGL_2 . Compare the constructed Bethe-to-oper map with [8].

64 Geometric Langlands

Exercise 64.1 (Line bundles, Hecke modifications, and the abelian eigencondition on the sphere). For a smooth complex curve C , a rank-one local system is a functor from its fundamental groupoid to one-dimensional complex vector spaces. A divisor is a finite integer sum of points. For a divisor D define $\mathcal{O}(D)$ by its meromorphic sections f satisfying $\text{ord}_x(f) + D(x) \geq 0$ wherever the section is required to be holomorphic. The divisor Picard group is divisors modulo the divisors of nonzero meromorphic functions; use its line bundles $\mathcal{O}(D)$ as the Picard space in this exercise. A Hecke modification at x is $\mathcal{O}(D) \mapsto \mathcal{O}(D+x)$. For a local system E on C , the abelian Hecke eigencondition for a family of lines $\mathcal{A}$ on $\text{Pic}(C)$ is

$$h^* \mathcal{A} \simeq E \boxtimes \mathcal{A}, \quad h(x, L) = L(x),$$

with the isomorphisms compatible with successive modifications. Use the coarse Picard space and $C = \mathbb{P}^1$ in this exercise.

- On the spectral sphere with fixed affine parameter z prove $\text{div}(z - a) = (a) - (\infty)$. Deduce that every divisor is equivalent to $d(\infty)$, where d is its degree, and that degree is unchanged by a principal divisor. Prove that multiplication by a meromorphic function realizes the corresponding isomorphism of divisor line bundles. Thus compute the divisor Picard space as $\{\mathcal{O}(d(\infty)) : d \in \mathbb{Z}\}$.
- Use simple connectivity of the sphere to identify every rank-one local system with a constant line E_0 . Set $\mathcal{A}_d = E_0^{\otimes d}$, using dual lines for negative d , and construct the Hecke eigenisomorphism. Prove compatibility for modifications at two different points from the tensor symmetry.
- Fixing $\mathcal{A}_0 = \mathbb{C}$, prove that the eigencondition determines all $\mathcal{A}_d$ and the compatible identifications. Identify the dual torus as $\mathbb{C}^\times$ by interchanging its character and cocharacter lattices. State explicitly the curve, group, and rank for this abelian correspondence.

Exercise 64.2 (Prediction: two-site Gaudin spectroscopy from oper residues). Consider two spin- $\frac{1}{2}$ sites with distinct real z_1, z_2 and physical Hamiltonian $H_{\text{phys}} = J(z_1 - z_2)G_1 = J\mathbf{S}_1 \cdot \mathbf{S}_2$, $J > 0$. The measured energy transfer is $\hbar\omega$.

- For $M = 0$ calculate the oper and its residue at z_1 . For $M = 1$ insert $w = (z_1 + z_2)/2$ and calculate the changed residue. Use $E = J(z_1 - z_2)g_1$ to derive the energies $J/4$ and $-3J/4$.
- Use total spin to determine the multiplicities three and one. Derive a singlet-triplet absorption line at $\hbar\omega = J$ and the partition function $Z = e^{3\beta J/4} + 3e^{-\beta J/4}$. Verify agreement with the two-site quiver prediction after its stated additive energy shift.
- Compute the response to $S_1^z - S_2^z$: it maps the normalized singlet into the zero-magnetization triplet with norm one. Determine the zero-temperature spectral weight of the absorption line. State the isolated-pair, isotropic-exchange, and negligible line-broadening assumptions. Distinguish the derived two-site Bethe-to-oper calculation from the separate abelian Hecke eigencondition on the sphere.

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